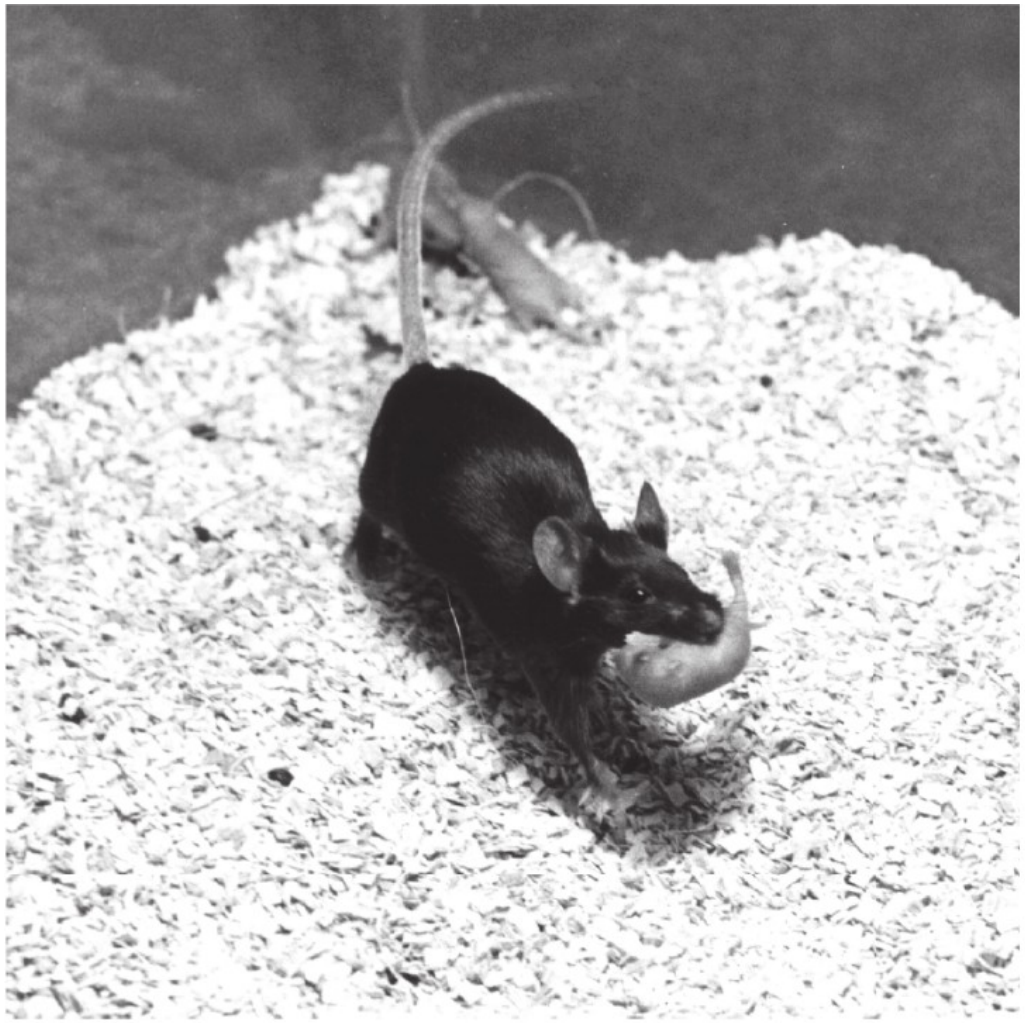


Carlson and Birkett (2022), Fig. 10.22, p. 331.

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Reproductive Development, Mating, Pair Bonds, and Caregiving

From Reproduction to Care: Hormones, Social Circuits, and Parenting

Fall 2026

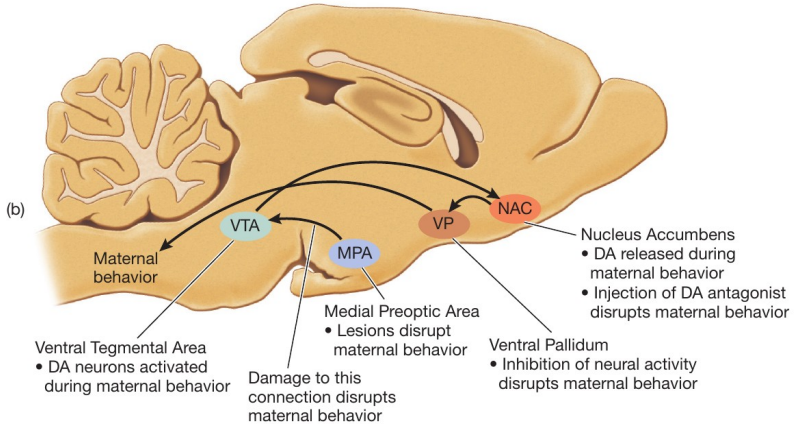
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Reproduction combines development, motivation, and care

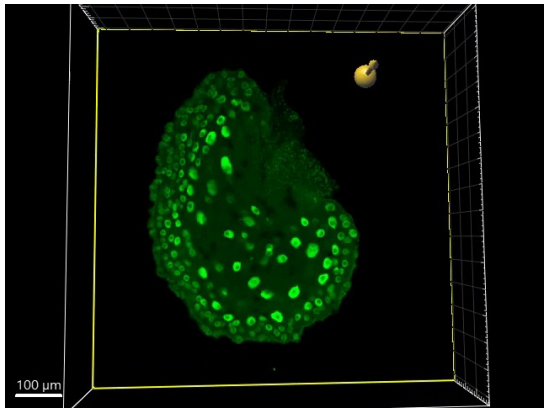
Reproductive behavior includes actions that contribute to finding a partner, mating, and caring for offspring. These actions depend on sensory information, the body's reproductive condition, and previous experience. A hormone can change how strongly a social cue influences behavior without specifying every movement the animal performs.

The **hypothalamus** coordinates endocrine signals and motivated actions through several distinct cell populations. Some neurons control pituitary hormone secretion; others influence mating, social recognition, or caregiving. Their effects depend on their receptors and projection targets, so activating one hypothalamic population does not reproduce every aspect of a reproductive state.

Parental care can occur after pregnancy, after mating, or through experience with another animal's offspring. These routes recruit overlapping mechanisms while differing across species. Human reproduction also involves personal decisions, social relationships, and cultural knowledge, which cannot be inferred directly from a rodent's mating posture or pup-retrieval response.



Carlson and Birkett (2022), Fig. 10.24(b), p. 334.



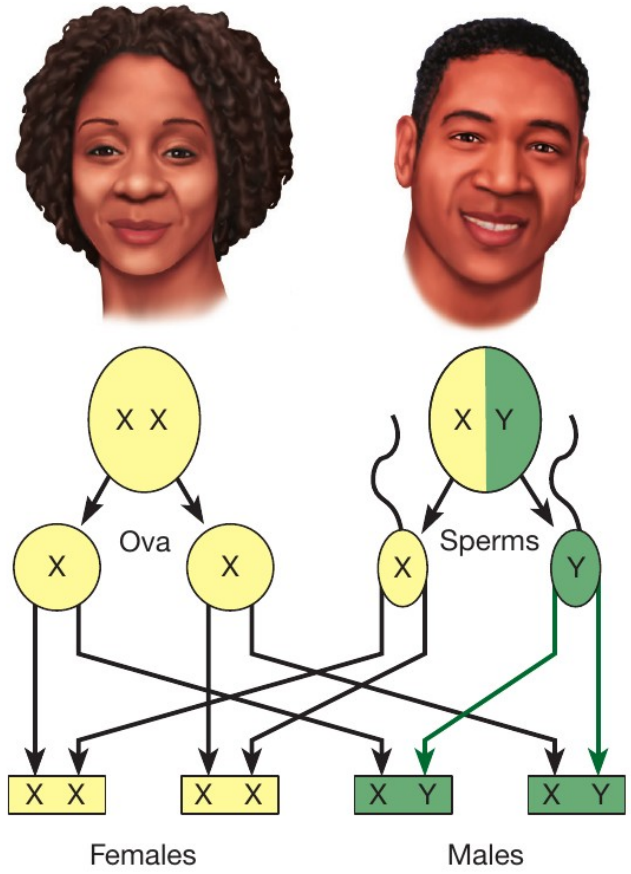
Yin et al. (2025), Video 1.

Gametes contribute one chromosome set to a new embryo

Gametes are reproductive cells: sperm and ova. They are produced through **meiosis**, a form of cell division that reduces the chromosome number. Fertilization combines a chromosome set from each gamete, restoring the paired organization found in most cells of the developing human body.

The usual human chromosome complements include XX and XY, although other complements and mosaic patterns also occur. A chromosome carries many genes, and those genes must be expressed in particular cells at particular times. Chromosome composition therefore begins a developmental process rather than directly specifying an adult's behavior.

Sex traits include chromosomes, gonads, hormone production, internal ducts, and external anatomy. These features usually develop in coordinated patterns but can differ from one another. Describing the particular trait being measured is more informative than assuming that a chromosome label supplies every anatomical, endocrine, and psychological detail.



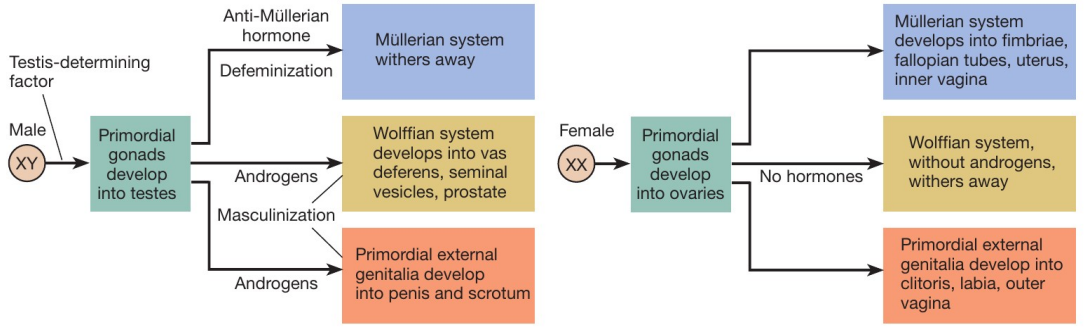
Carlson and Birkett (2022), Fig. 10.1, p. 304.

Gonadal differentiation changes embryonic hormone signals

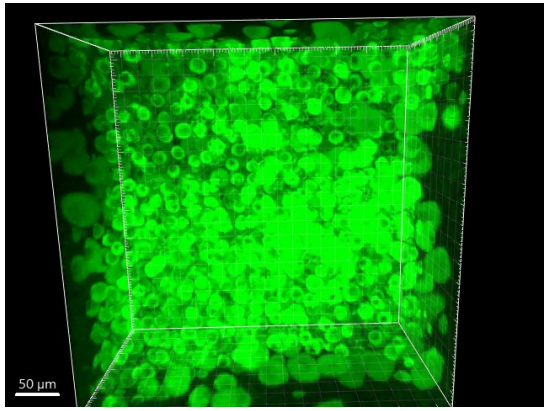
Early gonads contain cells capable of developing along different pathways. In a typical XY embryo, expression of the **SRY** gene helps initiate a testis-development program. SRY acts through other regulatory genes and changes cell differentiation; it is not itself a circulating hormone that travels to the reproductive ducts.

Developing testes contain **Sertoli cells**, which produce anti-Müllerian hormone, and **Leydig cells**, which produce testosterone. These signals act on different targets. Their coordinated actions help organize internal reproductive anatomy, while local hormone conversion and receptor function influence how other tissues respond during the same developmental period.

Ovarian development also requires active gene regulation and maintenance of ovarian cell identity. It should not be understood as the complete absence of a developmental program. Differences in a regulatory gene, a hormone-producing cell, or a target receptor can alter different components of reproductive development independently.



Carlson and Birkett (2022), Fig. 10.4, p. 307.



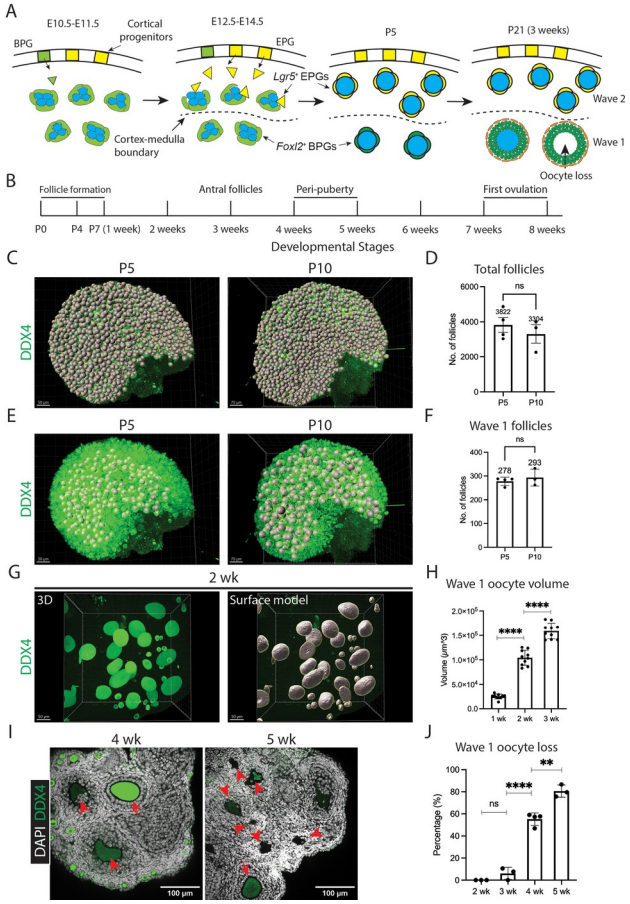
Yin et al. (2025), Video 2.

Early ovarian follicles can follow different fates

An **ovarian follicle** contains an oocyte surrounded by supporting cells. A mouse study used tissue clearing and lineage tracing to follow early follicles through development. **Lineage tracing** permanently marks a cell population and its descendants, allowing later tissue organization to be related to an earlier developmental origin.

The earliest growing follicles occupied the ovarian medulla, the inner region of the ovary. Most of these first-wave follicles lost their oocytes rather than producing the first mature eggs. Their supporting cells did not all disappear at the same time, so loss of an oocyte did not eliminate every contribution of that follicle.

The researchers distinguished these early follicles from later follicles stored nearer the ovarian cortex. Three-dimensional imaging preserved their spatial arrangement, which a single thin section could miss. Position and developmental history helped separate populations that looked superficially similar when classified only by their current size or stage.



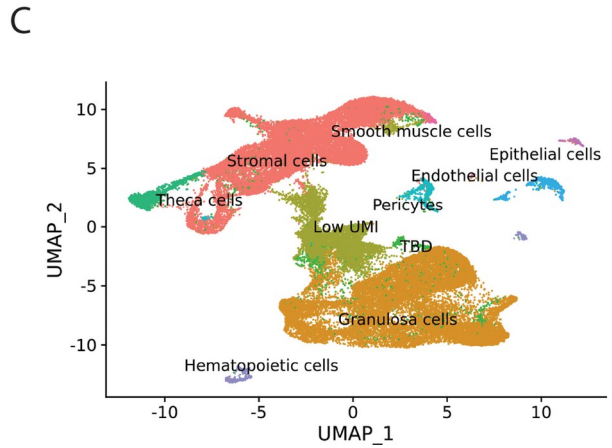
Yin et al. (2025), Fig. 1.

Follicle loss can leave steroid-producing cells behind

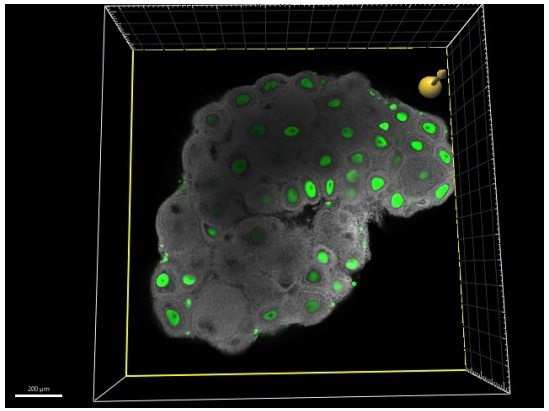
Atresia is the breakdown of a follicle that does not complete maturation. In the mouse lineage-tracing study, many early oocytes degenerated while surrounding **theca cells** persisted. Theca cells contribute to steroid synthesis, allowing the remains of an early follicle to influence endocrine function after its reproductive cell has been lost.

Near puberty, retained theca and interstitial cells expressed genes needed for androgen production. A nearby boundary population of later follicles supplied many of the earliest mature oocytes. The source of steroid-producing tissue therefore differed from the source of most eggs released during the first reproductive period.

The study combined anatomical tracing with single-cell gene-expression measurements. These methods distinguished a cell's location and ancestry from the enzymes it could express. An ovarian structure can consequently contribute to hormone production without surviving as an intact follicle capable of releasing a mature oocyte.



Yin et al. (2025), Fig. 5.



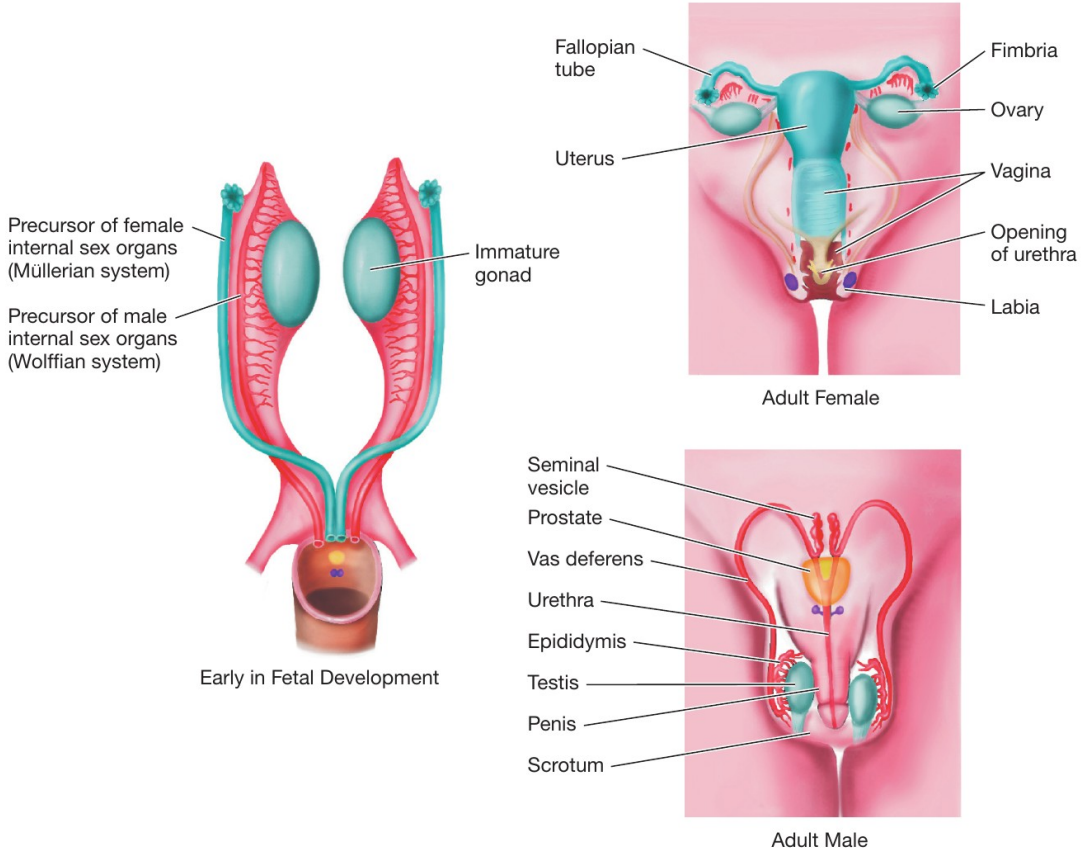
Yin et al. (2025), Video 5.

Different hormones act on the two embryonic duct systems

Early embryos contain **Müllerian** and **Wolffian ducts**, two internal duct systems with different developmental possibilities. Anti-Müllerian hormone acts through its receptor to promote Müllerian regression in typical testicular development. Testosterone supports Wolffian structures, including the epididymis and vas deferens, when their cells can respond to androgens.

These two hormone actions are separable. **Androgen insensitivity** results from reduced or absent androgen-receptor function, so testosterone cannot produce its usual effects in target tissues. Anti-Müllerian signaling can remain active, producing an anatomical pattern that cannot be described simply as a shortage of every testicular hormone.

Conversely, disrupted anti-Müllerian signaling can permit Müllerian structures to persist despite androgen-supported development elsewhere. These developmental variations identify distinct receptor-controlled processes. They concern particular anatomical and endocrine features and do not determine a person's gender identity, relationships, interests, or sexual orientation from anatomy alone.



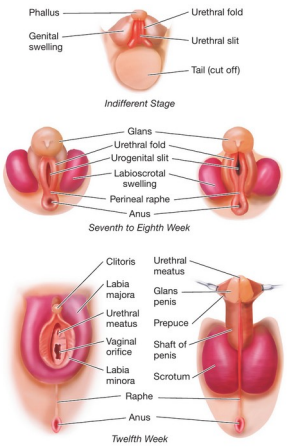
Carlson and Birkett (2022), Fig. 10.2, p. 305.

Local conversion of testosterone affects external anatomy

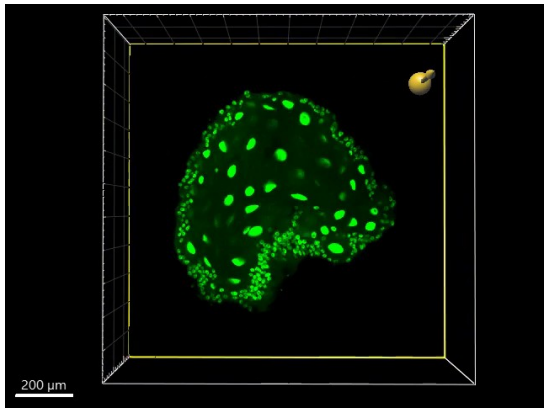
In developing target tissues, the enzyme **5α-reductase** converts testosterone into **dihydrotestosterone**, or DHT. DHT activates androgen receptors and contributes to differentiation of external genital structures. The amount of testosterone in circulation therefore does not, by itself, specify the androgen signal experienced by every tissue.

A change in 5α-reductase can alter external development while leaving other testosterone-dependent processes relatively preserved. A change in the androgen receptor affects the ability to respond to either androgen. Distinguishing hormone synthesis, local conversion, and receptor activation helps explain why developmental variations can produce different anatomical combinations.

Puberty changes the endocrine environment again, and some androgen-dependent features may develop more strongly at that time. Clinical assessment considers the individual's anatomy, hormone responses, and preferences. A developmental mechanism explains a biological process; it does not justify predicting someone's identity from a single enzyme or chromosome result.



Carlson and Birkett (2022), Fig. 10.3, p. 306.



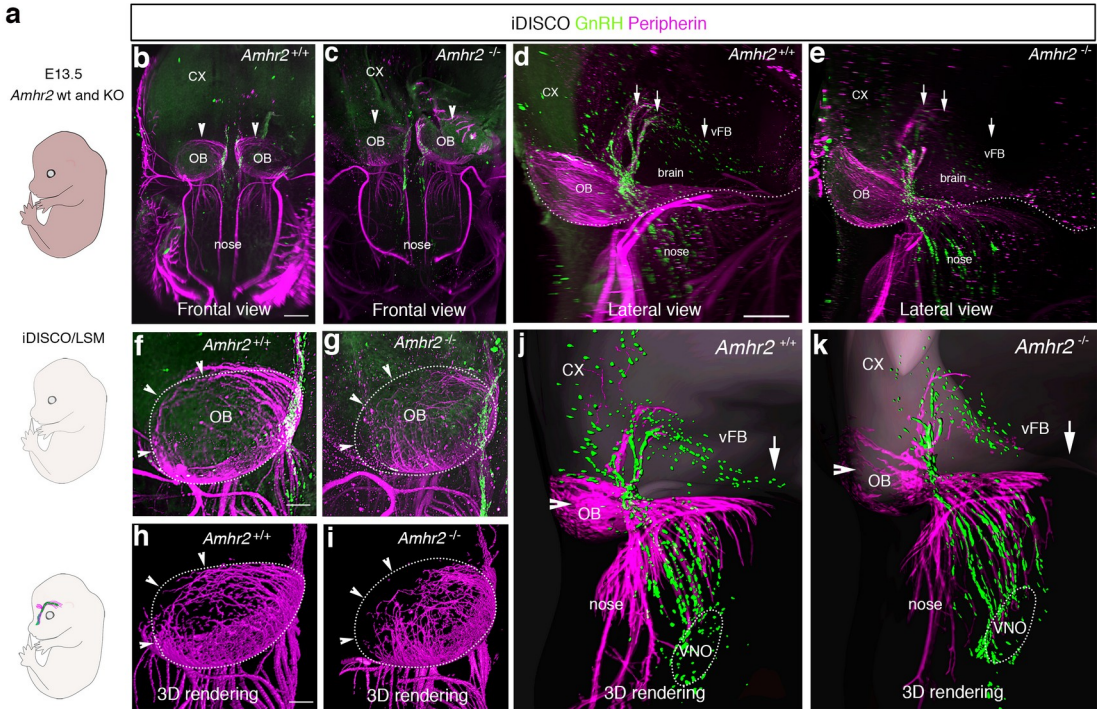
Yin et al. (2025), Video 3.

Reproductive hormone neurons must reach the forebrain

Gonadotropin-releasing hormone, or GnRH, is secreted by neurons that regulate pituitary reproductive output. During development, many of these neurons originate outside the forebrain and migrate inward along nasal-associated pathways. Their final position depends on developmental signals that guide movement and interactions with neighboring axons.

A study of anti-Müllerian hormone signaling found its receptor in developing GnRH-related pathways. Mice lacking normal receptor function showed disturbed olfactory organization and GnRH-neuron migration. The hormone system associated with reproductive duct development therefore also had a separate developmental role in organizing a brain endocrine population.

Cleared-embryo imaging allowed the investigators to examine these pathways in three dimensions. Misplaced or missing GnRH neurons can reduce later reproductive signaling even when gonads are present. This developmental defect lies upstream of the adult pituitary and gonadal responses that would normally follow a hypothalamic secretory signal.



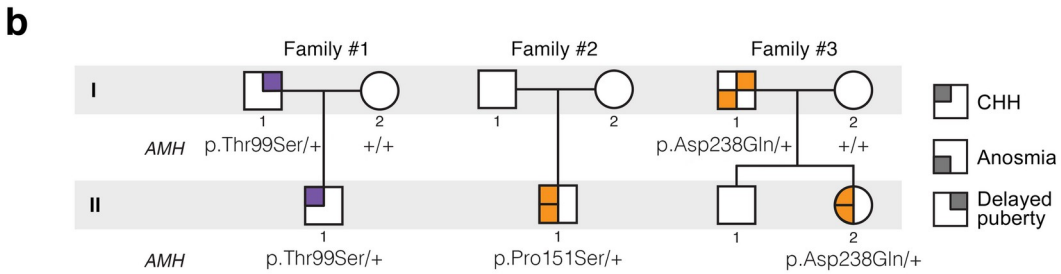
Malone et al. (2019), Fig. 3.

Developmental signals can affect puberty and fertility

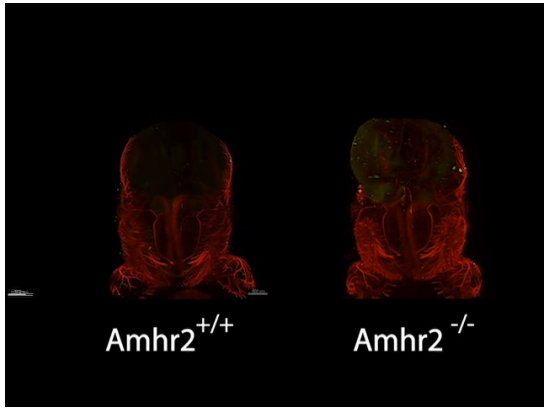
Hypogonadotropic hypogonadism describes reduced gonadal function caused by insufficient stimulation from the brain–pituitary system. The anti-Müllerian signaling study identified variants in affected people and tested related mechanisms in cells and mice. The combined approach asked whether altered signaling could disrupt GnRH development rather than only describing a clinical association.

Receptor-deficient mice had fewer appropriately positioned GnRH neurons and impaired reproductive hormone output and fertility. Cell experiments also examined how altered hormone signaling changed neuronal movement. A developmental migration problem could consequently lead to an adult endocrine deficit, even though the original disturbance occurred long before puberty.

The clinical variants affected only a subset of people with the condition, which has several possible biological causes. A deficient hormone response can arise at the hypothalamus, pituitary, or gonad. Identifying the affected level distinguishes a failure to generate the command from a failure of the downstream tissue to respond.



Malone et al. (2019), Fig. 6.



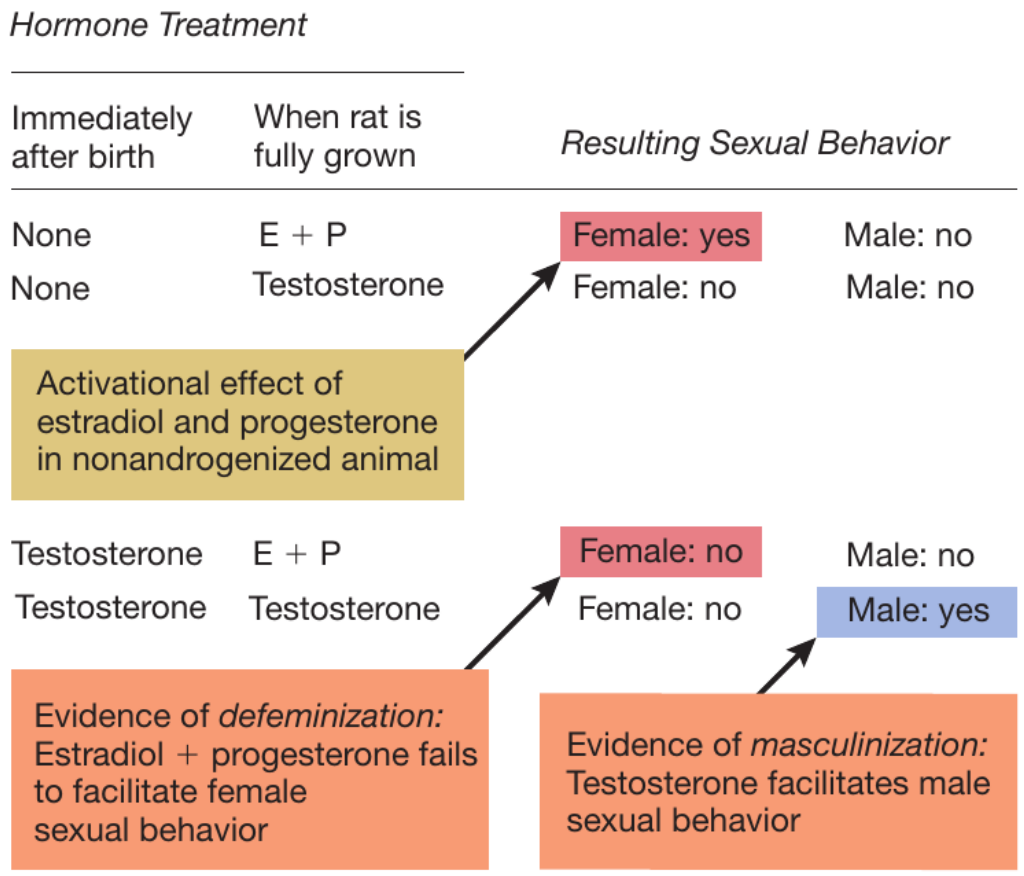
Malone et al. (2019), Fig. 3-Video 1.

Hormones can organize circuits and activate existing ones

Organizational effects change the development or lasting structure of a tissue or circuit during a sensitive period. **Activational effects** change the operation of an existing system while a hormone signal is available. These terms describe the timing and persistence of an effect, rather than two completely unrelated kinds of hormone.

In classic rodent experiments, early androgen exposure altered later hormone-dependent mating responses. In adults, gonadal hormone removal and replacement changed the expression of behavior through already developed circuits. The outcome depended on both developmental history and the hormone conditions present when the animal was tested.

Adolescence can include further circuit maturation, so development is not restricted to prenatal life. Hormones can change transcription, synaptic function, and tissue responsiveness over several timescales. A behavioral change after hormone treatment therefore needs a defined age, treatment duration, and neural target to identify the process being altered.



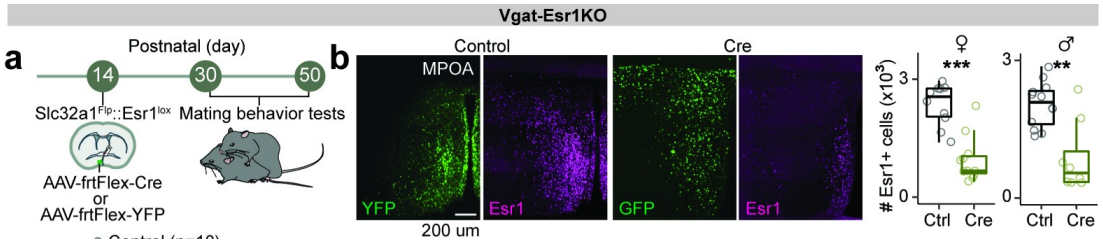
Carlson and Birkett (2022), Fig. 10.8, p. 312.

Estrogen receptors help mating circuits mature

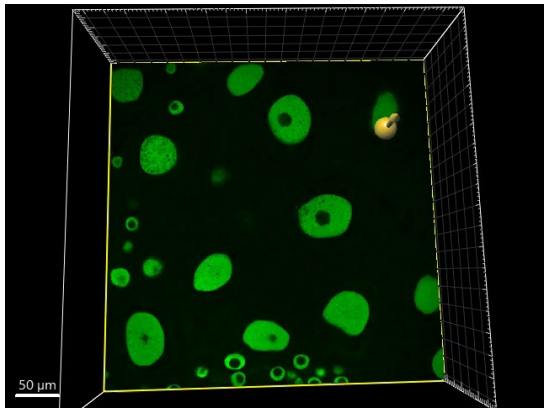
A recent mouse study examined the **medial preoptic area**, or MPOA, across adolescence. The researchers removed **estrogen receptor α**, encoded by Esr1, from defined neuronal populations. This receptor binds an estrogen signal and can regulate gene transcription, changing which proteins a cell produces as it matures.

Removing Esr1 from inhibitory, GABA-releasing neurons disrupted the developmental progression of mating behavior in both sexes. Removing it from the tested glutamate-releasing population did not produce the same behavioral outcome. The relevant hormone effect therefore depended on which cells received the receptor manipulation, rather than on a uniform change throughout the hypothalamus.

The experiment separated a receptor's role in a specific neural population from its effects elsewhere in the body. The MPOA contains several cell types involved in social behavior. A receptor expressed across a region can influence different actions through particular neurons and their targets, rather than through a single regional on–off switch.



Hashikawa et al. (2026), Fig. 1.



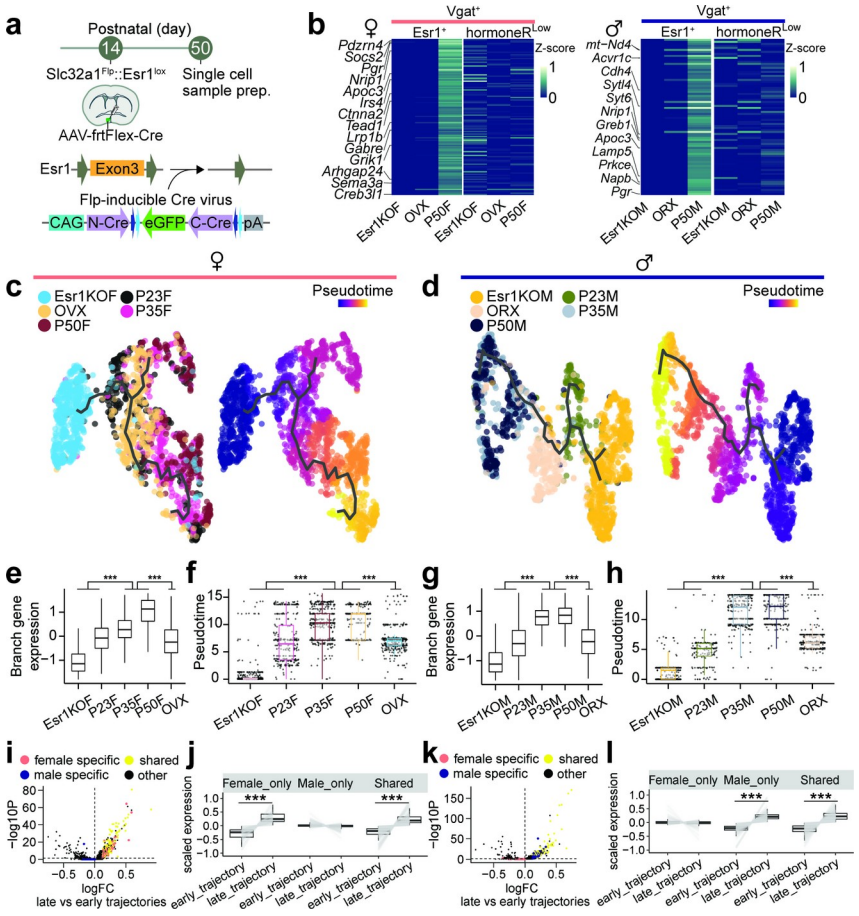
Yin et al. (2025), Video 4.

Hormones change a neuron’s transcriptional development

The adolescent MPOA study measured gene expression in individual cells at successive developmental stages. **Single-cell RNA sequencing** identifies the RNA molecules a cell contains, providing evidence about which genes it is using. Cells were grouped by expression patterns rather than treating every neuron in the preoptic area as equivalent.

GABAergic neurons containing Esr1 followed sex-dependent transcriptional trajectories across adolescence. Deleting the receptor before adolescence interrupted their normal progression through these gene-expression states. The receptor influenced the developmental program of these neurons, including coordinated changes across several genes rather than only one immediate electrical response.

The researchers also used tissue-based RNA detection to locate expression changes within the brain. Sequencing supplied molecular detail, while anatomical labeling retained spatial information. Together with the behavioral manipulation, these measurements related a defined receptor to the maturation of a particular cell population and the later expression of mating behavior.



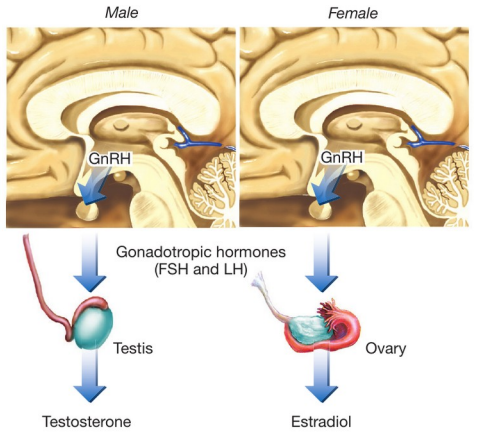
Hashikawa et al. (2026), Fig. 7.

The reproductive axis uses a sequence of endocrine signals

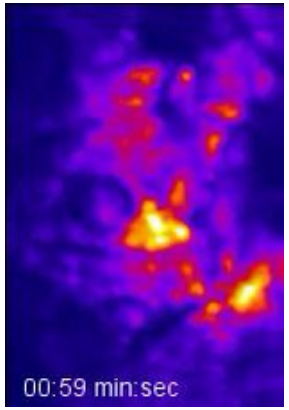
The **hypothalamic–pituitary–gonadal axis** begins with GnRH secretion into blood vessels that supply the anterior pituitary. GnRH activates receptors on pituitary **gonadotrophs**, cells that release luteinizing hormone and follicle-stimulating hormone. These pituitary signals travel through the circulation to reproductive tissues rather than directly moving the muscles used in mating.

Luteinizing hormone, or LH, and **follicle-stimulating hormone**, or FSH, regulate gonadal steroid production and gamete development through different target cells. The resulting gonadal hormones feed information back to the brain and pituitary. Feedback changes subsequent secretion, allowing reproductive output to respond to the current state of the gonads.

The timing of GnRH release matters because repeated pulses and sustained exposure do not have identical effects on pituitary receptors. A blood hormone concentration is therefore only part of the signal. The frequency, duration, and biological target of a secretion pattern help determine what the downstream cells do.



Carlson and Birkett (2022), Fig. 10.5, p. 307.



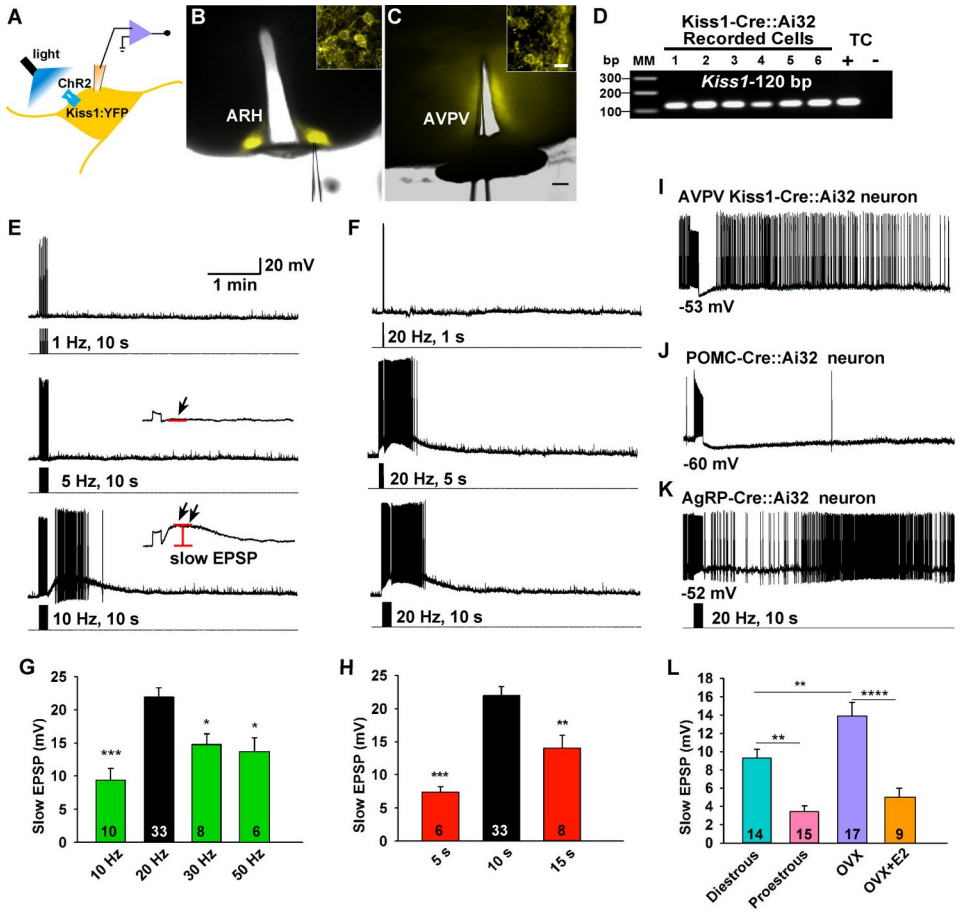
Hollander-Cohen et al. (2024), Fig. 2-Video 2.

Kisspeptin circuits combine excitation and peptide feedback

Kisspeptin is a peptide that strongly excites the reproductive GnRH system. Many arcuate kisspeptin neurons also produce **neurokinin B** and **dynorphin**. A mouse brain-slice study used light-sensitive channels to stimulate these neurons and record the electrical responses produced in their local circuit and downstream targets.

High-frequency stimulation produced a slow excitatory response mediated by neurokinin signaling. Blocking the neurokinin receptor reduced that response, whereas activating a dynorphin-sensitive inhibitory pathway opposed it. Peptide release therefore provided interacting excitatory and inhibitory influences that could coordinate activity across the neuronal population.

The same study traced routes through which arcuate neurons could excite preoptic kisspeptin and GnRH neurons. Fast glutamate signaling and slower peptide signaling acted at different circuit steps. Coordinated endocrine output can consequently emerge from a network whose cells communicate through transmitters with different targets and response durations.



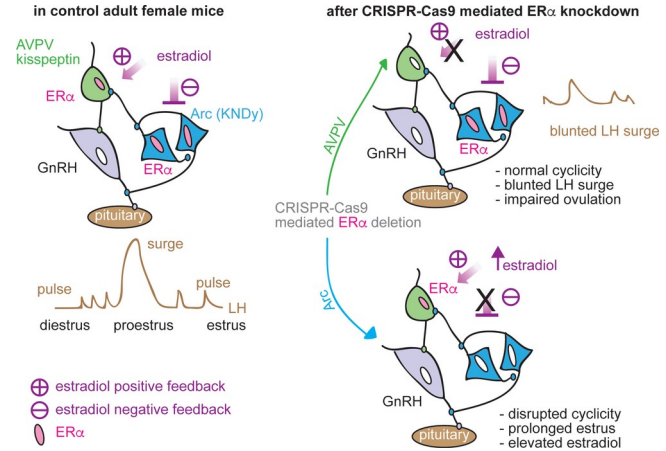
Qiu et al. (2016), Fig. 2.

Estrogen feedback differs across kisspeptin populations

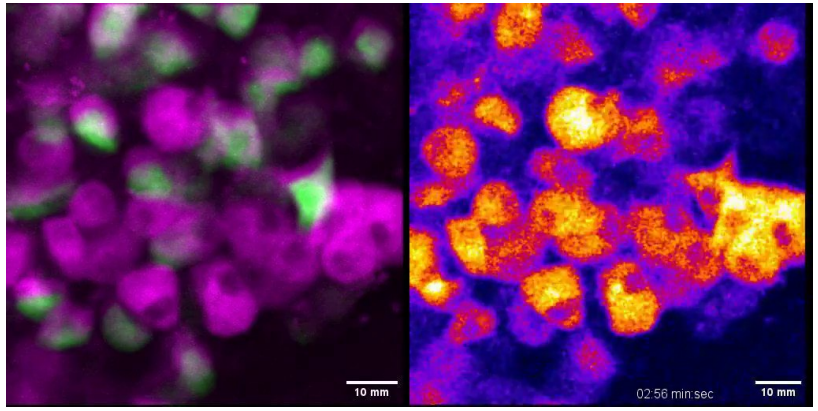
An adult mouse experiment disrupted estrogen receptors separately in preoptic and arcuate kisspeptin populations. Targeting adults helped distinguish ongoing receptor action from an earlier developmental effect. The manipulation changed a hormone-sensing mechanism within an established circuit rather than removing the ovaries or every estrogen-responsive cell.

Reducing estrogen receptor function in preoptic kisspeptin neurons blunted LH surges and reduced neuronal excitability. Disrupting the receptor in arcuate kisspeptin neurons disturbed reproductive cyclicity and changed their synaptic input. The same receptor therefore contributed to different aspects of endocrine timing depending on its location.

Positive feedback strengthens output under the appropriate reproductive conditions, while **negative feedback** limits it. These effects require circuitry capable of interpreting a hormone signal differently across cell populations. A circulating estrogen increase does not produce the same electrical or secretory response in every kisspeptin neuron that encounters it.



Wang et al. (2019), Fig. 7.



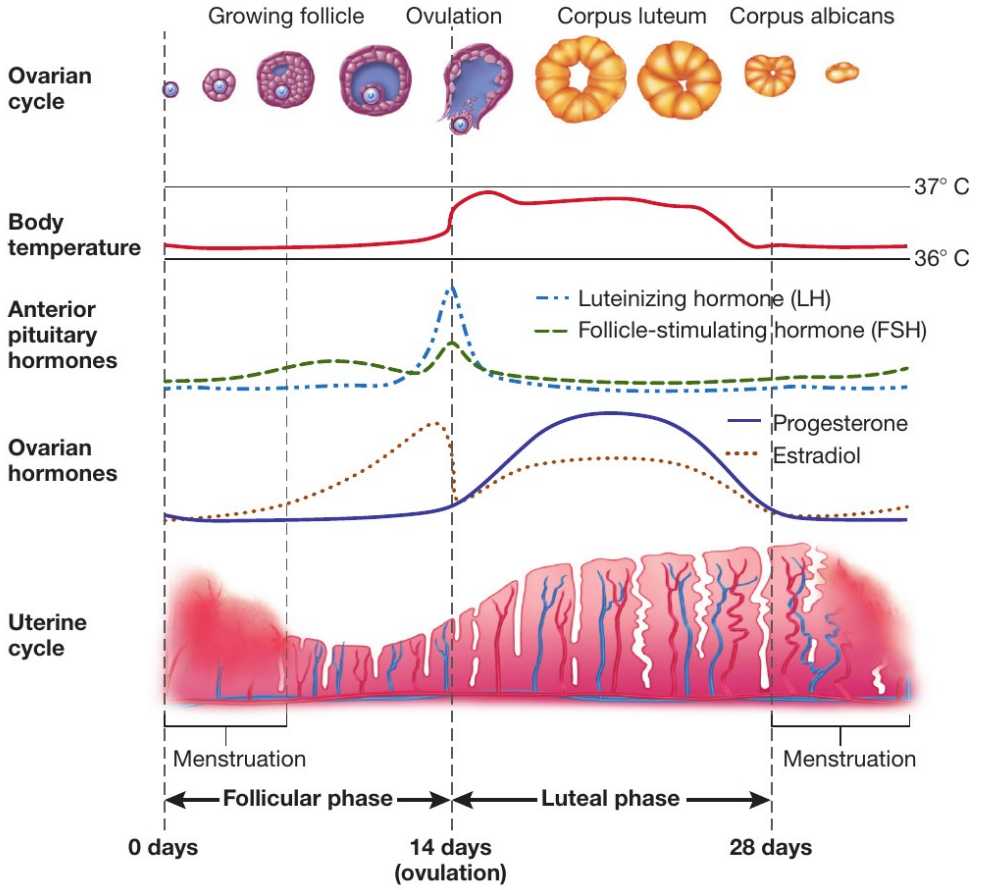
Hollander-Cohen et al. (2024), Fig. 3-Video 3.

Follicles and pituitary signals change across the cycle

During the **follicular phase**, ovarian follicles grow under gonadotropin stimulation and produce increasing estradiol. **Estradiol** is an estrogen that acts in reproductive tissues and the nervous system. Its effects depend on concentration, duration, and cycle stage, including feedback onto the hypothalamic and pituitary cells controlling further secretion.

An LH surge contributes to **ovulation**, the release of an oocyte from a mature follicle. The remaining follicular tissue becomes the **corpus luteum**, which produces progesterone as well as estradiol. The endocrine contribution of the same ovarian structure therefore changes after the oocyte is released.

If pregnancy is not established, corpus-luteum support declines and ovarian steroid levels fall. The uterine lining responds to these endocrine changes, and another cycle can begin. Ovulation, steroid secretion, and uterine changes are coordinated events, but measuring one does not directly reveal the timing of every other component.



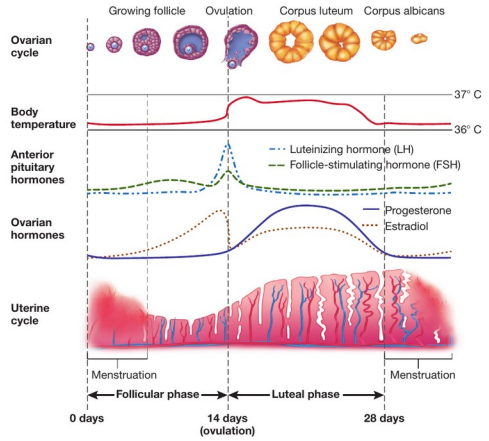
Carlson and Birkett (2022), Fig. 10.6, p. 310.

Progesterone changes reproductive target tissues

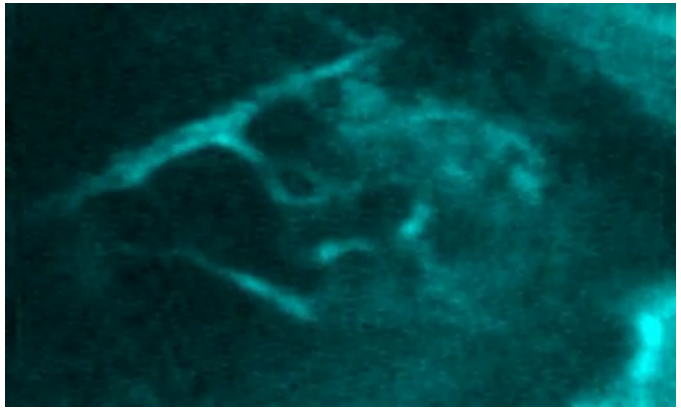
Progesterone is a steroid hormone produced prominently by the corpus luteum after ovulation. It acts through receptors in reproductive tissues and the brain, contributing to changes appropriate to the luteal phase. Its effects depend partly on earlier estrogen exposure, which can change receptor availability and the condition of the target tissue.

In the uterus, ovarian steroids regulate the growth and maintenance of the endometrium, the internal lining. In the nervous system, hormone-sensitive circuits can alter physiological regulation and behavior. A change in circulating progesterone therefore acts on several targets, with different consequences according to each tissue's receptors and ongoing state.

The familiar cycle diagram represents coordinated changes over time rather than a precise calendar shared by every person. Follicular timing and ovulation can vary. Hormone measurements, reproductive anatomy, and observed cycle events provide different evidence and should not be replaced by assuming that ovulation always occurs on one fixed day.



Carlson and Birkett (2022), Fig. 10.6, p. 310.



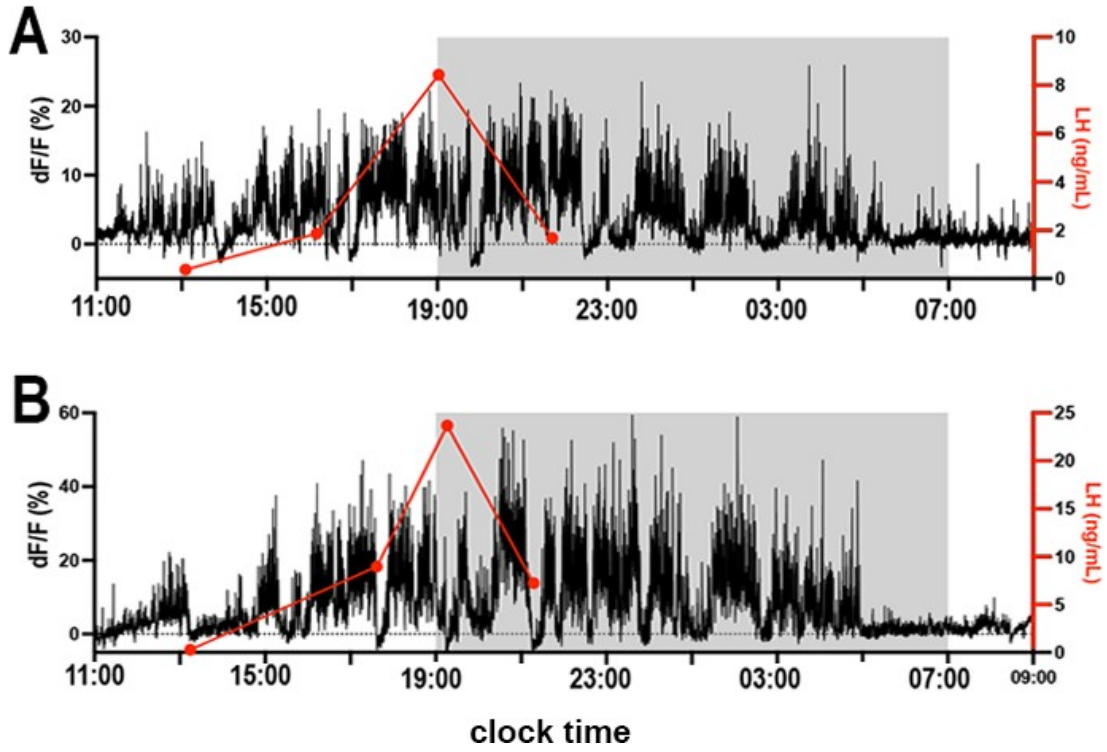
Hollander-Cohen et al. (2024), Fig. 2-Video 1.

Preoptic kisspeptin activity rises before the LH surge

A recent mouse study recorded calcium-dependent fluorescence from preoptic kisspeptin neurons across the estrous cycle. **Fiber photometry** collects light from a labeled neuronal population near an implanted optical fiber. The changing fluorescence reports population calcium activity rather than directly measuring GnRH secretion or the firing of every individual neuron.

On proestrus, the stage preceding ovulation, the researchers observed prolonged slow activity oscillations with faster transients superimposed. These patterns were much less evident at other cycle stages. Concurrent blood sampling showed that the beginning of the population activity preceded the peak of the LH surge rather than occurring only after it.

The onset varied across animals and across repeated cycles in the same mouse. The reproductive signal was therefore a sustained and temporally structured population event, not a single precisely timed spike. Comparing neural fluorescence with blood hormone measurements helped distinguish activity in the control circuit from the endocrine response it accompanies.



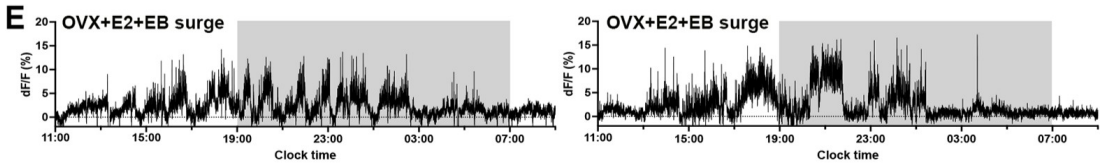
Zhou et al. (2026), Fig. 3.

Estrogen replacement restores a timed kisspeptin pattern

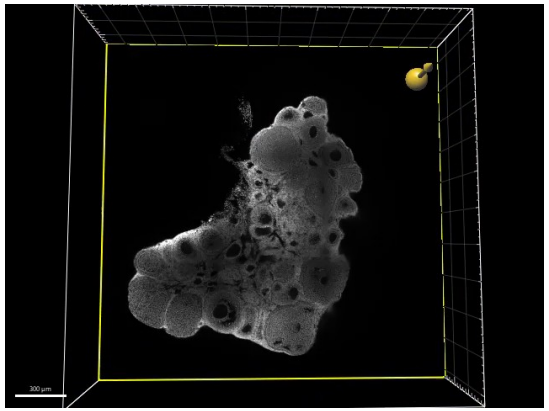
The preoptic kisspeptin study also removed the ovaries and replaced estradiol under controlled conditions. **Ovariectomy** removes an important source of ovarian hormones, allowing replacement experiments to ask which signals are sufficient to restore a particular neural pattern. The surgery also removes normal cyclic ovarian feedback.

Acute estradiol treatments did not simply generate the prolonged oscillatory pattern whenever they were given. The characteristic activity emerged during the expected surge period under the replacement regimen. The response depended on a prepared endocrine and temporal state, rather than on the immediate presence of estrogen alone.

These recordings distinguish a surge-associated preoptic population from the arcuate network involved in pulsatile reproductive output. Both use kisspeptin but have different activity patterns and feedback roles. A shared peptide name therefore does not imply that the two cell populations generate identical signals throughout the reproductive cycle.



Zhou et al. (2026), Fig. 8.



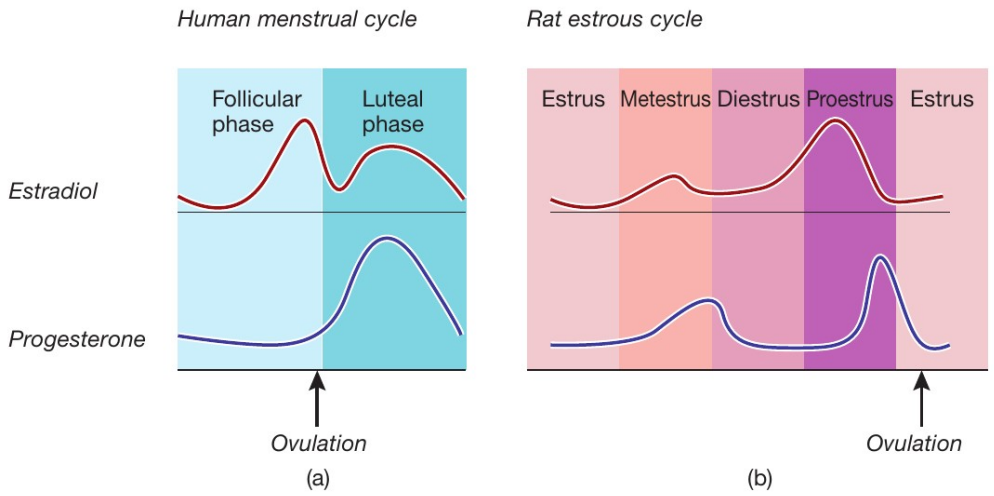
Yin et al. (2025), Video 9.

Human menstrual and rodent estrous cycles differ

A **menstrual cycle** includes cyclic changes in the uterine lining and its shedding when pregnancy is not established. Many laboratory rodents instead have an **estrous cycle**, with a shorter sequence of ovarian and behavioral changes. Their receptive period is more tightly restricted by the cycle than human sexual behavior generally is.

In rats, estradiol followed by progesterone can prepare and promote receptive mating responses. The same simple hormonal sequence does not determine human sexual choices or relationships. Human motivation is influenced by endocrine state together with learned expectations, health, personal meaning, social circumstances, and the presence of a desired partner.

A reproductive cycle is therefore different from a behavioral timetable. Ovulation concerns ovarian physiology, receptivity concerns the response to mating, and desire concerns motivation and experience. These processes can influence one another without occurring as an obligatory sequence whenever a particular hormone reaches a particular concentration.



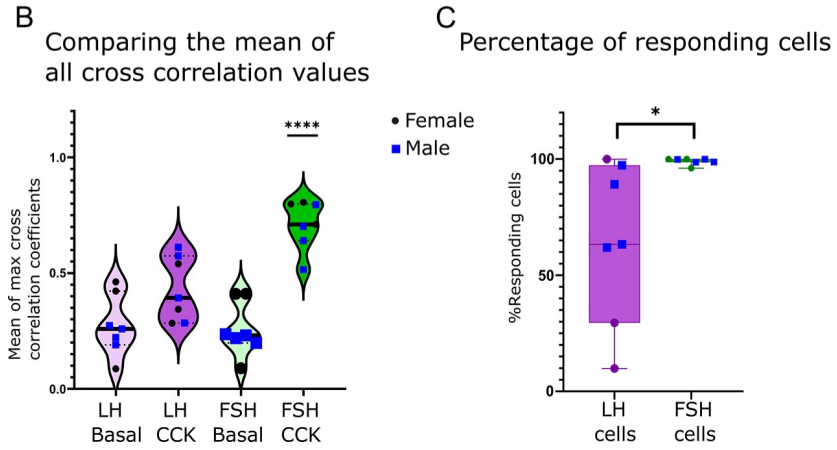
Carlson and Birkett (2022), Fig. 10.7, p. 311.

Pituitary cells can respond to metabolic signals

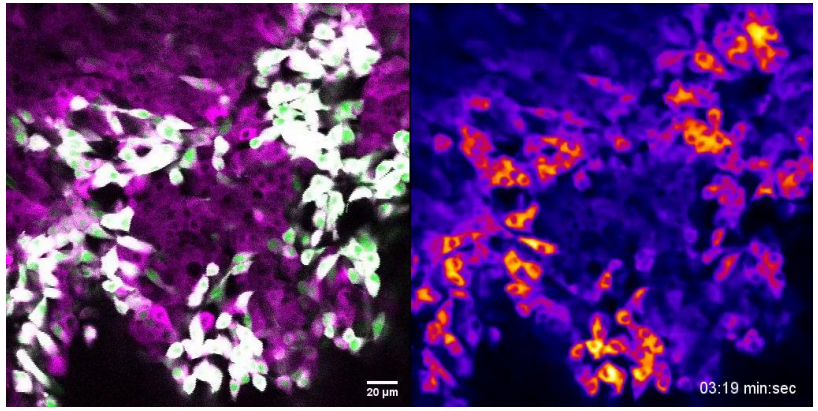
A zebrafish study examined how **cholecystokinin**, or CCK, influences reproductive hormone secretion. CCK is a peptide associated with nutrient signaling and satiety. The researchers found a CCK receptor in pituitary FSH cells, providing a route by which a metabolic signal could act directly on endocrine cells controlling ovarian development.

Calcium imaging showed that FSH cells responded strongly to CCK stimulation, whereas GnRH predominantly activated LH cells under the tested conditions. The response separated two pituitary populations often grouped together as gonadotrophs. Their shared reproductive function did not require identical sensitivity to the same upstream releasing signal.

Mutating the relevant CCK receptor impaired ovarian development and gonadotropin secretion. The physiological effect therefore extended beyond an isolated fluorescence response. This mechanism was established in fish; it should not replace the GnRH-centered organization of the mammalian reproductive axis without corresponding evidence from mammalian pituitary cells.



Hollander-Cohen et al. (2024), Fig. 4.



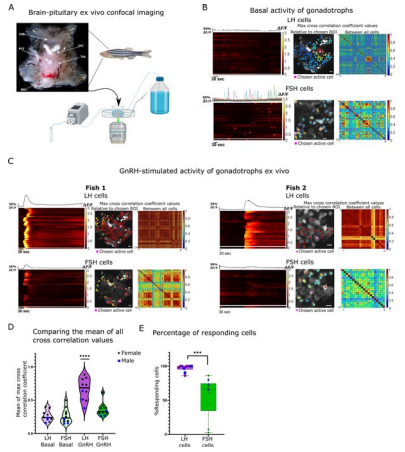
Hollander-Cohen et al. (2024), Fig. 4-Video 1.

Distinct secretory cells do not always share the same input

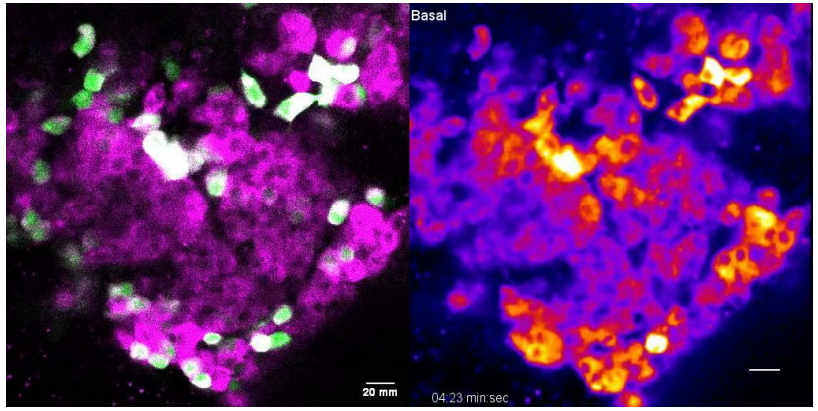
The zebrafish experiment compared LH and FSH cells while preserving their anatomical relationship in the pituitary. Fluorescent labels identified each population, and a calcium indicator tracked responses during stimulation. Nearby endocrine cells could consequently be monitored within the same tissue while their responses were assigned to different cell types.

GnRH evoked prominent activity in LH cells, while CCK recruited the FSH population more effectively. Receptor disruption also disturbed secretion, showing that cellular responsiveness and endocrine output had functional consequences. The investigators tested both the response to an applied signal and the effect of removing the signal's receptor.

In an egg-laying species, growing the ovary and triggering ovulation impose different physiological demands. Separating FSH-related development from LH-related release permits different regulatory inputs to influence those processes. The result concerns the organization of this fish endocrine system rather than a universal rule that one peptide controls all vertebrate reproduction.



Hollander-Cohen et al. (2024), Fig. 3.



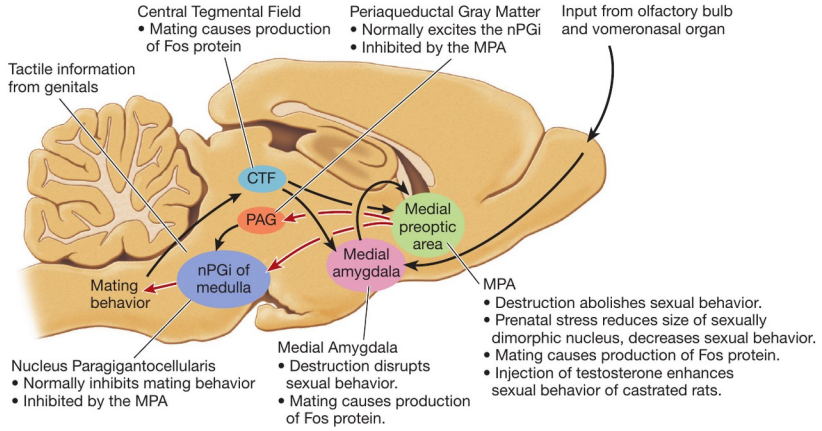
Hollander-Cohen et al. (2024), Fig. 3-Video 1.

Hormone receptors alter rodent mating circuits

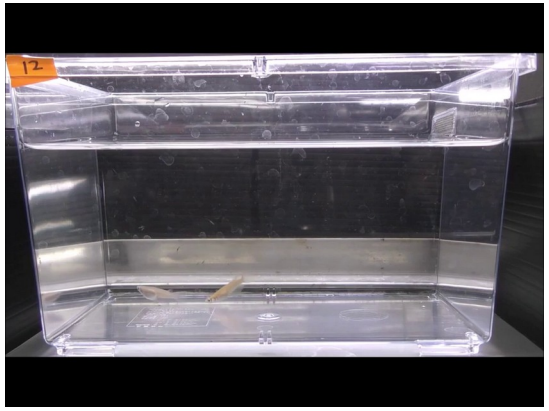
In male rodents, testicular androgens help maintain mating motivation and performance through hormone-sensitive brain regions. Testosterone can act through the androgen receptor or be converted into estradiol by **aromatase**. The importance of each route depends on species, developmental stage, neural population, and the particular behavior being measured.

In female rats, estradiol followed by progesterone facilitates receptive behavior, including **lordosis**, a posture that permits mounting. Estradiol changes the readiness of the relevant circuit, including receptor expression, while later progesterone can promote its activation. Removing either the appropriate hormone signal or its neural target can impair the response.

A hormone-dependent mating response is still produced through sensory and motor pathways. The animal must detect a partner, evaluate contact, and organize posture and movement. Endocrine treatment changes the conditions under which those circuits operate; it does not substitute for every sensory event involved in the interaction.



Carlson and Birkett (2022), Fig. 10.17, p. 321.



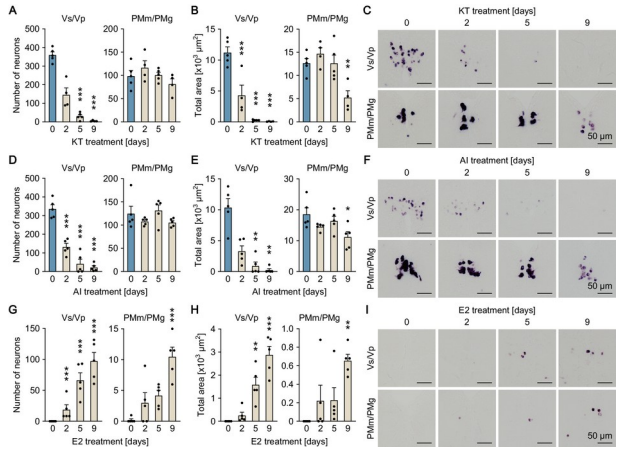
Hiraki-Kajiyama et al. (2019), Video 1.

Estrogen regulates a reversible social peptide signal in fish

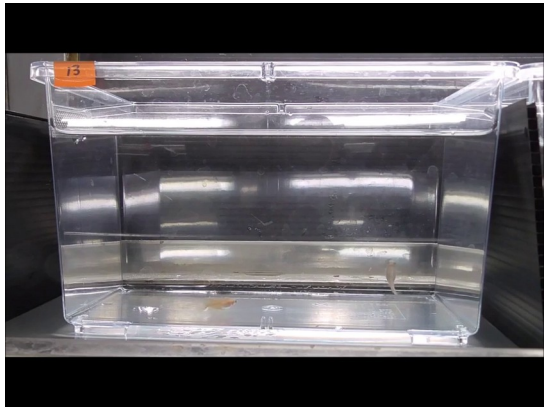
In medaka fish, researchers examined **neuropeptide B**, a signaling molecule expressed strongly in particular female brain regions. Its expression depended on estrogen-responsive DNA regulation. A receptor-bound hormone could therefore change the production of a peptide that subsequently acts on other cells within the social-behavior network.

Changing the adult steroid environment reversed the sex-biased expression pattern. This was an adult regulatory change rather than a permanent consequence of chromosome complement alone. The same brain could alter its peptide expression when the endocrine conditions changed, demonstrating substantial physiological flexibility in this particular fish system.

The investigators identified peptide-related expression in several brain and spinal regions relevant to social behavior. This distribution provided possible routes from estrogen-sensitive gene regulation to a coordinated behavioral response. The mechanism concerns neuropeptide expression and fish mating behavior and does not establish how human identity or orientation develops.



Hiraki-Kajiyama et al. (2019), Fig. 2.



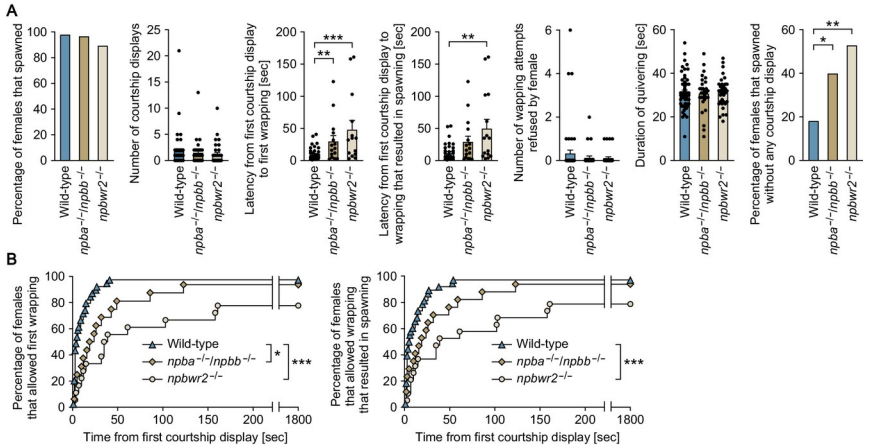
Hiraki-Kajiyama et al. (2019), Video 2.

Removing a peptide pathway changes female receptivity

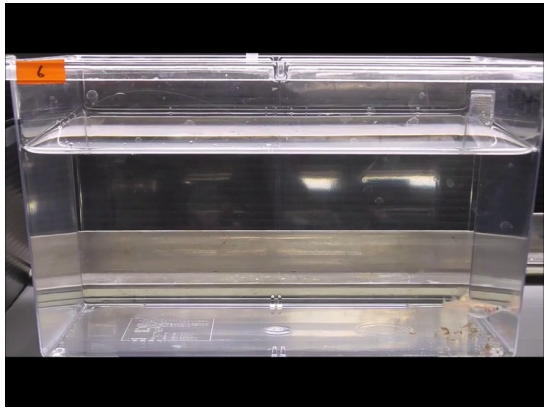
The medaka neuropeptide B study tested behavior after disrupting the peptide genes or their receptor. **A knockout** removes normal gene function, allowing researchers to ask whether an intact pathway contributes to the measured response. They compared female responses to male courtship rather than assuming that altered RNA expression necessarily changed behavior.

Females lacking normal neuropeptide B signaling showed impaired receptivity to male courtship. The receptor manipulation and peptide manipulations affected different steps of the same signaling system. Their converging behavioral effects helped identify a pathway through which estrogen-dependent gene expression could influence a social response.

The original recordings show whole-animal interactions, while gene-expression and anatomical measurements identify candidate biological steps. These measurements answer different questions. A change in mating latency can reflect altered responsiveness within the interaction and should not be described as a direct measurement of a fish's subjective experience or a human psychological category.



Hiraki-Kajiyama et al. (2019), Fig. 7.



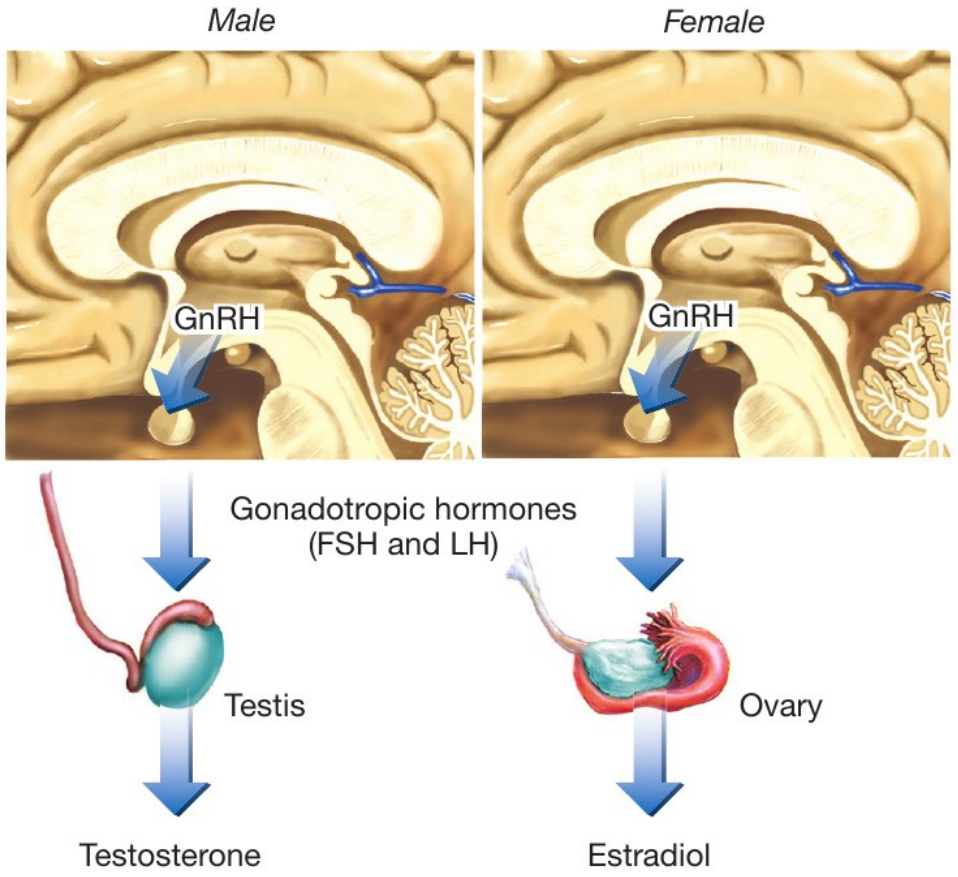
Hiraki-Kajiyama et al. (2019), Video 3.

Human sexual motivation has several interacting influences

Gonadal hormones contribute to human sexual function and motivation, particularly when endocrine function is markedly reduced. They act within a person who also has memories, preferences, relationships, and expectations. The same hormone concentration can therefore accompany different behavior depending on health, medication, personal experience, and social circumstances.

Desire, physiological arousal, and behavior are related but distinguishable processes. A change in genital response does not establish what someone wants, and a reported wish does not guarantee a particular physical response. Measurements need to specify whether they concern motivation, circulation, endocrine status, subjective experience, or an observed action.

Adult hormone treatment can alter the intensity of motivation without determining the sex or gender of a desired partner. **Sexual orientation** concerns patterns of attraction, not simply the amount of sexual interest. Evidence about changing hormone levels should therefore be interpreted at the level actually measured rather than extended automatically to partner preference.



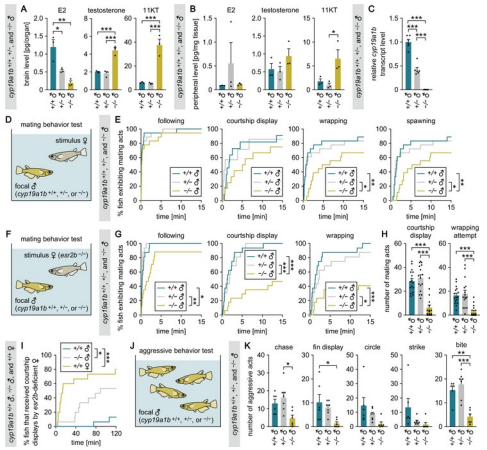
Carlson and Birkett (2022), Fig. 10.5, p. 307.

Brain estrogen strengthens androgen-dependent action

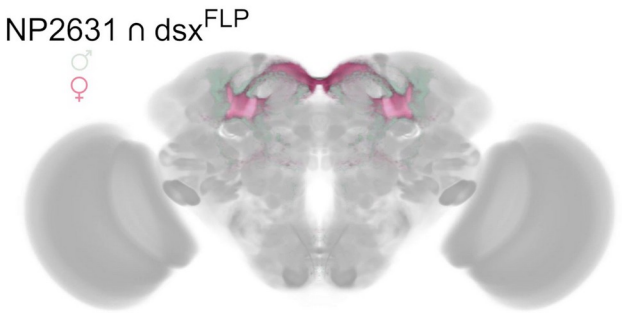
A medaka study removed the brain-enriched aromatase **Cyp19a1b**, which converts androgens into estrogens. Mutant males showed impaired mating and aggressive behavior despite elevated brain androgen levels. The behavioral deficit therefore could not be explained simply by assuming that the brain lacked every form of androgen signal.

Estrogen administration rescued the affected behaviors under the tested conditions. Molecular measurements indicated that brain-derived estrogen helped maintain androgen-receptor expression. Estrogen could consequently facilitate an androgen-dependent behavior by altering the receptor machinery through which another steroid acts, rather than replacing every downstream androgen action.

This mechanism differs from a simple division in which estrogens act only in females and androgens only in males. Both hormone classes can participate in the same male neural pathway. Their effects follow receptor expression, local synthesis, and cellular state rather than the everyday gender labels attached to the hormone names.



Nishiike et al. (2026), Fig. 1.



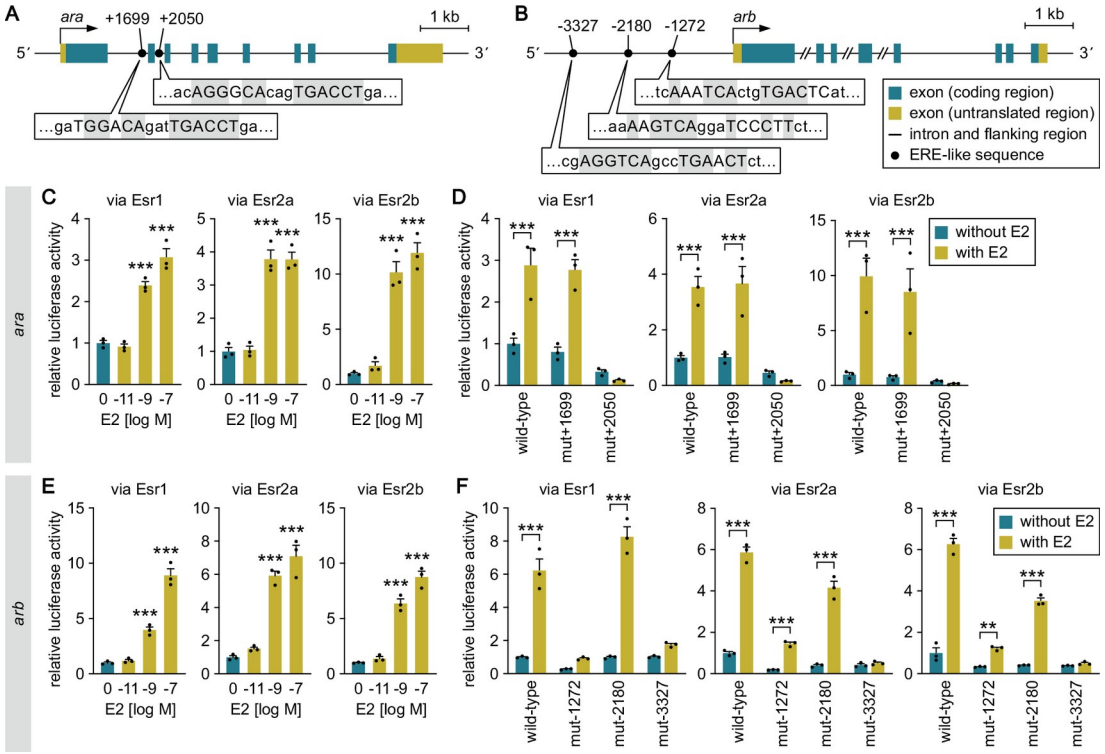
Ishii et al. (2020), Video 3.

One steroid can regulate the receptor for another steroid

The medaka aromatase study examined regulatory DNA associated with androgen-receptor genes. **Transcription** is the production of RNA from a DNA template. When estrogen receptors activate appropriate regulatory sequences, they can increase the RNA available for producing androgen receptors in a hormone-sensitive cell.

Experiments supported direct estrogen-sensitive regulation of androgen-receptor transcription. Different estrogen-receptor subtypes contributed to the expression of the two medaka androgen-receptor genes in relevant brain regions. The signal therefore changed the cell's capacity to detect another hormone rather than merely adding a second independent excitatory input.

The resulting behavior depends on several steps: local estrogen production, estrogen-receptor action, androgen-receptor availability, and downstream androgen signaling. A deficit at the first step can weaken the final response even when androgen remains abundant. Measuring hormone concentration alone would miss the altered receptor availability within the receiving tissue.



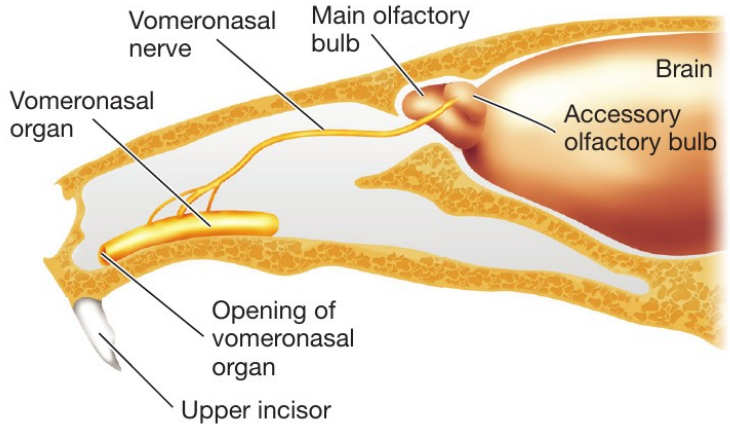
Nishiike et al. (2026), Fig. 4.

Sensory pathways carry social cues to reproductive circuits

Rodents obtain social information through odors, touch, vocalizations, and movements. The **main olfactory system** detects many volatile molecules, while the **vomeronasal organ** detects particular chemical cues through another sensory pathway. These systems send information through different routes that can converge on amygdalar and hypothalamic networks.

The **medial amygdala** receives important social chemosensory input and communicates with reproductive and defensive circuits. Its neurons do not simply label every encountered odor as a mating signal. Responses depend on the chemical pattern, the animal's internal state, and previous experience with the individual producing the cue.

Sensory information must also be translated into a suitable action. Recognizing a conspecific's odor is different from approaching, investigating, mounting, or caring for it. Reproductive circuits combine identity-related sensory input with hormone-sensitive motivation and motor organization before any particular social behavior appears.



Carlson and Birkett (2022), Fig. 10.13, p. 315.



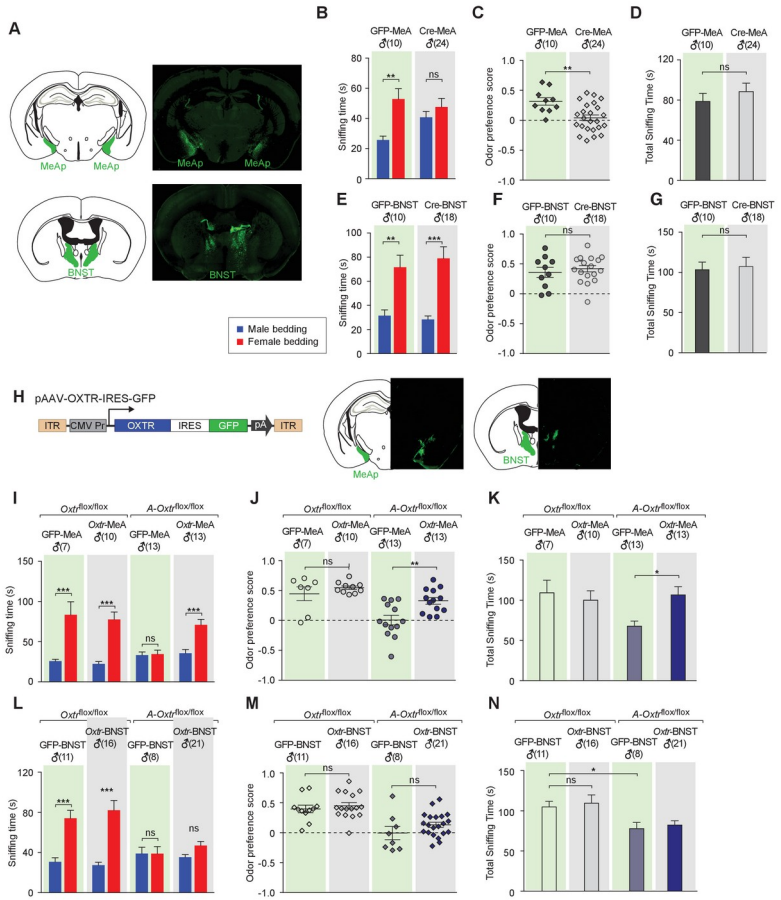
Perrodin et al. (2023), Fig. 2-Video 1.

Oxytocin helps amygdala cells distinguish social cues

A mouse study investigated **oxytocin receptors** in the medial amygdala. Oxytocin is a peptide signal whose effects depend on the cells expressing its receptor. The researchers tested whether this signaling helped animals discriminate social cues from male and female conspecifics rather than treating oxytocin as a general increase in friendliness.

Reducing oxytocin signaling impaired the normal discrimination of these cues and altered social investigation. Receptor manipulation in aromatase-expressing neurons identified a specific cellular target within the medial amygdala. The behavioral effect therefore depended on oxytocin acting in a defined population involved in processing socially relevant sensory information.

Neuronal recordings showed changes in sex-biased responses to conspecific cues when oxytocin signaling was altered. The peptide changed how sensory information was represented by the circuit. It did not simply instruct the animal to perform one complete motor sequence irrespective of what social stimulus was present.



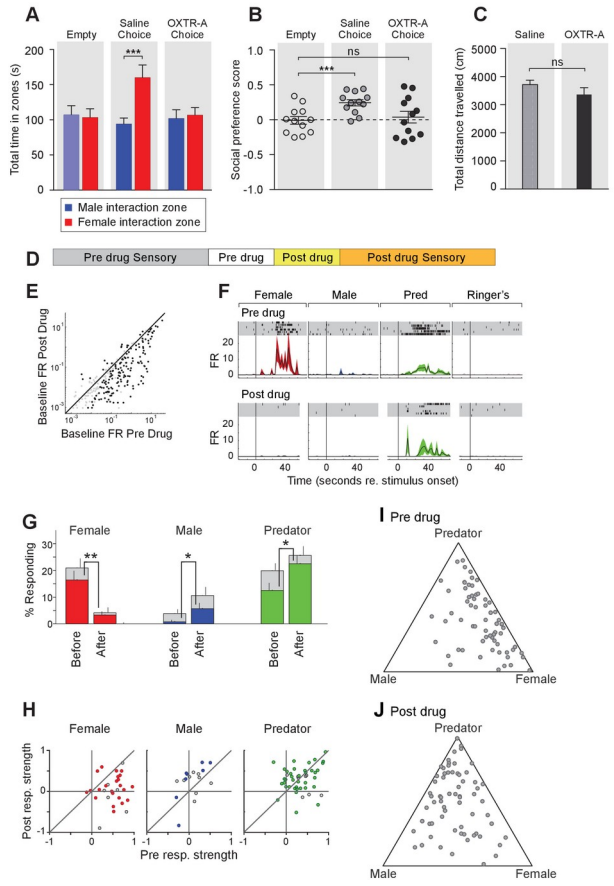
Yao et al. (2017), Fig. 3.

Acute receptor blockade alters a sensory response

The medial-amygdala study combined developmental manipulations with acute inhibition of oxytocin signaling. **Acute inhibition** changes signaling during a relatively short experimental period. This approach asks whether ongoing peptide action contributes to a response after the sensory circuit has already developed, rather than only shaping its early construction.

Blocking oxytocin receptors altered social interaction preference and the neuronal response profile. Manipulating the oxytocin source also changed the discrimination pattern. These interventions acted at different points in the pathway, linking peptide availability and receptor activation to the operation of an established social-sensory network.

A behavioral preference in this assay describes investigation of particular mouse stimuli under controlled conditions. It is not a direct model of every aspect of pair bonding or human attraction. The neural result is that oxytocin can adjust the representation and behavioral influence of socially relevant cues within a defined amygdala circuit.



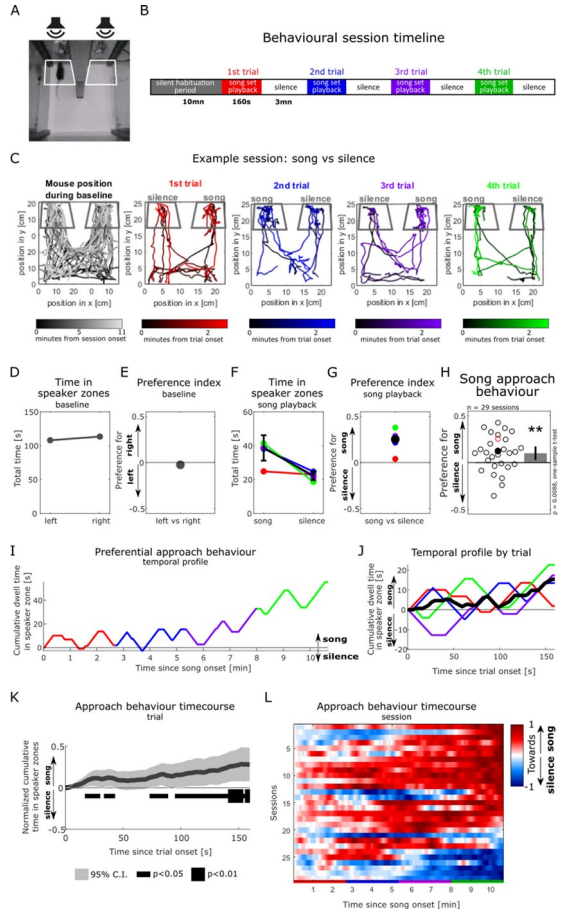
Yao et al. (2017), Fig. 5.

Female mice use song timing when choosing an approach

Male mice produce sequences of ultrasonic vocalizations during courtship. A playback study asked which acoustic features female listeners used when approaching these sounds. **Playback** presents a controlled recording through a speaker, allowing investigators to change one feature while preserving much of the remaining sound rather than changing the live male.

Females preferentially approached intact songs over versions whose temporal regularity had been disrupted. The relevant information concerned how vocal events were distributed in time. Altering the rhythm changed the receiver's behavior even though the sound still contained the kinds of syllables associated with the original courtship recording.

The experiment measured a movement toward a sound source, not mating success or a report of attraction. Speaker location and the animal's tracked trajectory made the response observable. The finding identifies an acoustic feature that influences a social decision while leaving later partner interaction and reproductive outcome unmeasured.



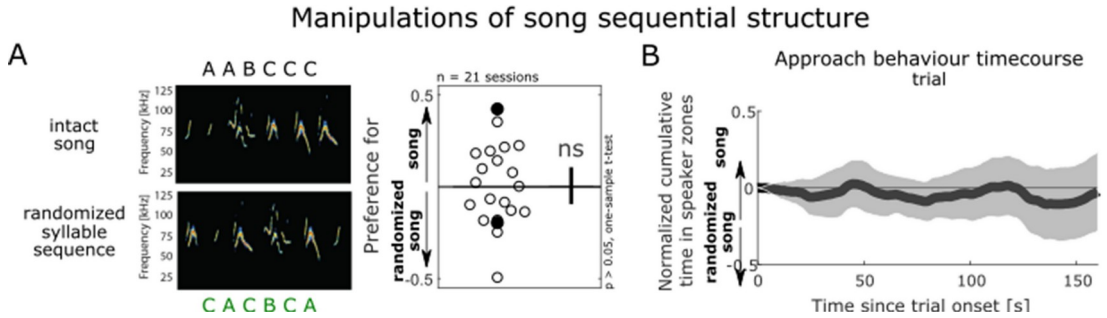
Perrodin et al. (2023), Fig. 2.

A signal can preserve syllables but lose its rhythm

The courtship-song study separately manipulated temporal regularity, syllable sequence, and the spectrotemporal structure of individual syllables. **Spectrotemporal structure** describes how sound frequencies change through time. These controlled changes tested whether the female response depended on a general distortion or on one particular property of the vocal sequence.

Approach behavior was more sensitive to disrupted temporal regularity than to the tested changes in syllable order or fine syllable structure. The female auditory system did not need to preserve every physical detail to treat the recording similarly. Different acoustic dimensions therefore had different behavioral importance in this particular social task.

Tracking the animal during repeated sound presentations also separated an initial position from a subsequent approach. The receiver extracted information from a time-varying signal and converted it into a movement decision. That transformation is one mechanism by which a partner's behavior can influence reproductive interactions before direct physical contact occurs.



Perrodin et al. (2023), Fig. 3.



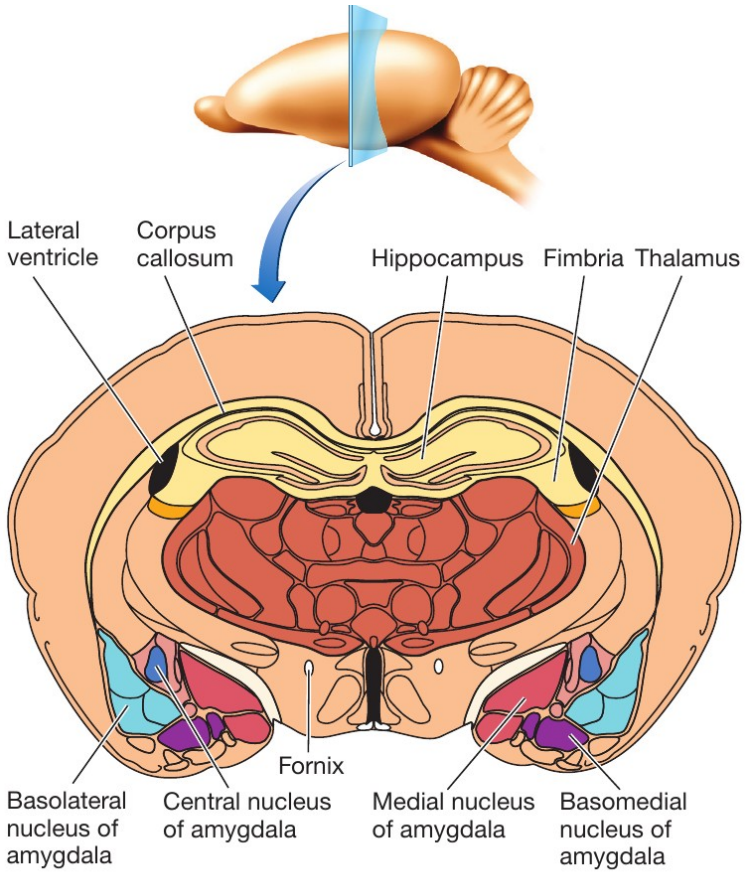
Perrodin et al. (2023), Fig. 2-Video 2.

Human chemical communication requires specific evidence

A **pheromone** is a chemical signal that produces a defined effect in another member of the same species under an established signaling system. Many rodent pheromone mechanisms are well characterized. Human odor communication is studied through different methods, and a laboratory response to an odor does not automatically establish a natural human pheromone.

Human studies may measure odor detection, mood, physiological responses, or judgments of social information. Each endpoint tests a different process. Demonstrating that an applied compound changes a judgment requires controls for smell intensity and expectations, but it still does not show that the compound governs natural partner choice.

The vomeronasal pathways central to many rodent experiments should not be assumed to operate identically in adult humans. Human social chemosensation can involve ordinary olfactory processing. Claims about endocrine modulation of an odor task must identify the actual stimulus, dose, population, and response measured rather than borrowing a rodent mating mechanism unchanged.



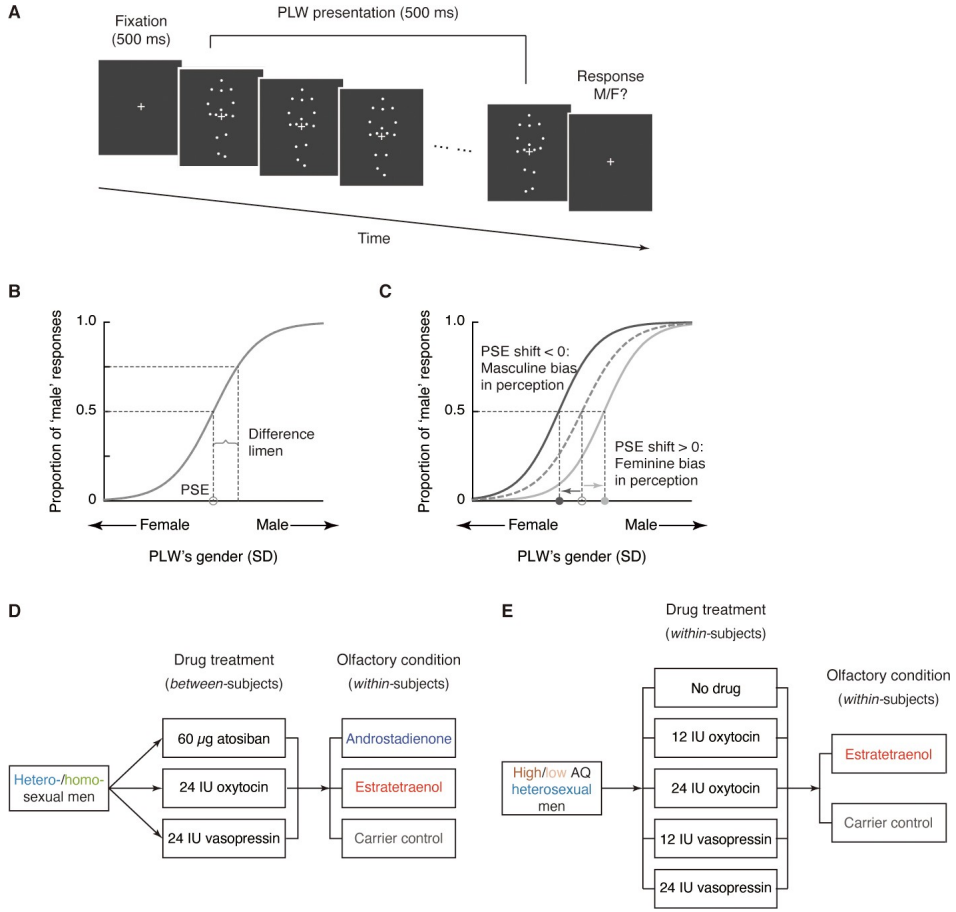
Carlson and Birkett (2022), Fig. 10.14, p. 316.

A human odor study measured judgments of body motion

One human experiment tested whether steroid-like odor compounds changed judgments of **point-light walkers**, displays made from dots representing body joints. Participants classified the apparent sex-related characteristics of the moving figure. The task measured a perceptual decision about an ambiguous stimulus rather than choosing a real-life romantic or sexual partner.

The investigators combined odor presentation with intranasal oxytocin, vasopressin, or a receptor antagonist. Changes in the point where participants switched between response categories indicated altered perceptual judgments. Comparing treatments tested whether the peptide-signaling condition influenced the effect of the chemical stimulus on that specific task.

The study examined defined groups of men and controlled laboratory conditions. Its outcome concerns chemosensory modulation of a visual judgment, not a demonstrated universal human pheromone. A shift in an ambiguous classification task cannot establish that an odor changes someone's sexual orientation or determines whom they desire.



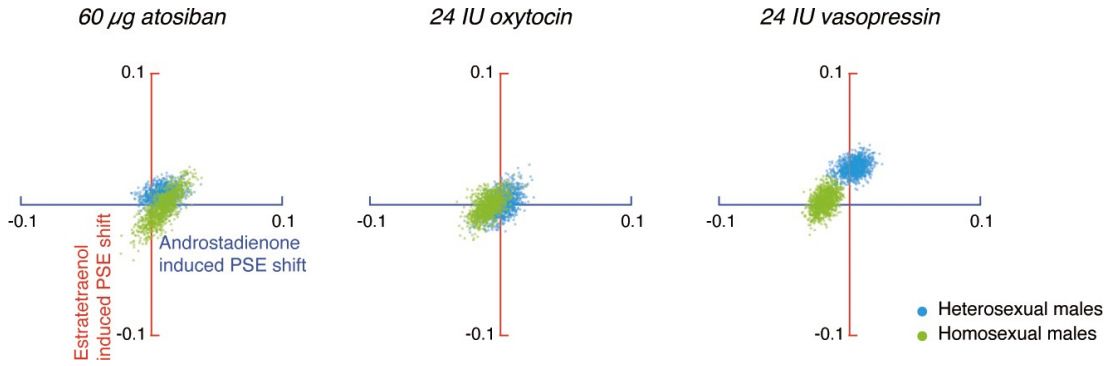
Chen et al. (2021), Fig. 1.

Oxytocin effects can depend on dose and the measured task

The human chemosensory experiment found that oxytocin did not act as a simple dose-dependent gain knob. The perceptual effect varied nonlinearly with dose and differed across participant characteristics. **A nonmonotonic response** changes direction or relative strength across doses rather than increasing steadily whenever more of a compound is administered.

The antagonist **atosiban**, which can block oxytocin and vasopressin receptors, reduced the tested odor-related perceptual effects. Because the drug acts on more than one receptor system, its result does not isolate a single receptor subtype by itself. The comparison still identifies receptor-dependent modulation within the conditions of the experiment.

Intranasal administration, blood hormone concentration, and peptide action at a particular synapse are different measurements or interventions. This study manipulated administration and measured perceptual choices. It did not directly record peptide release in the human medial amygdala or demonstrate the rodent receptor-specific mechanism at the corresponding human cellular site.



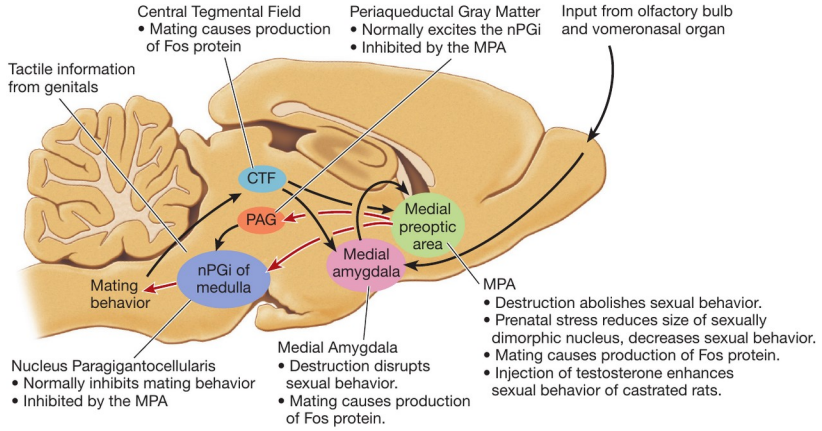
Chen et al. (2021), Fig. 3.

Preoptic and spinal circuits organize male mating responses

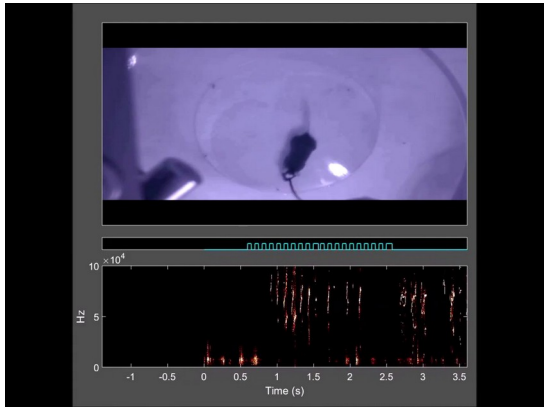
The **medial preoptic area** contributes to the sensory integration and motivation involved in male rodent mating. Its connections reach hypothalamic and brainstem regions that influence motor and autonomic output. Hormone receptors within this network alter its responsiveness to a partner and to genital sensory feedback during an interaction.

Spinal and brainstem circuits organize patterned components of copulation and ejaculation. Sensory input from the genitals can change the state of these circuits, while descending influences can facilitate or inhibit their activity. A coordinated mating response therefore requires communication across several levels of the nervous system rather than activity in the preoptic area alone.

Lesions, hormone replacement, and local stimulation answer different questions about the network. A lesion can establish that intact tissue contributes to normal performance, while stimulation can reveal actions the manipulated circuit can produce. Neither method by itself identifies every neuron normally active at each stage of the behavior.



Carlson and Birkett (2022), Fig. 10.17, p. 321.



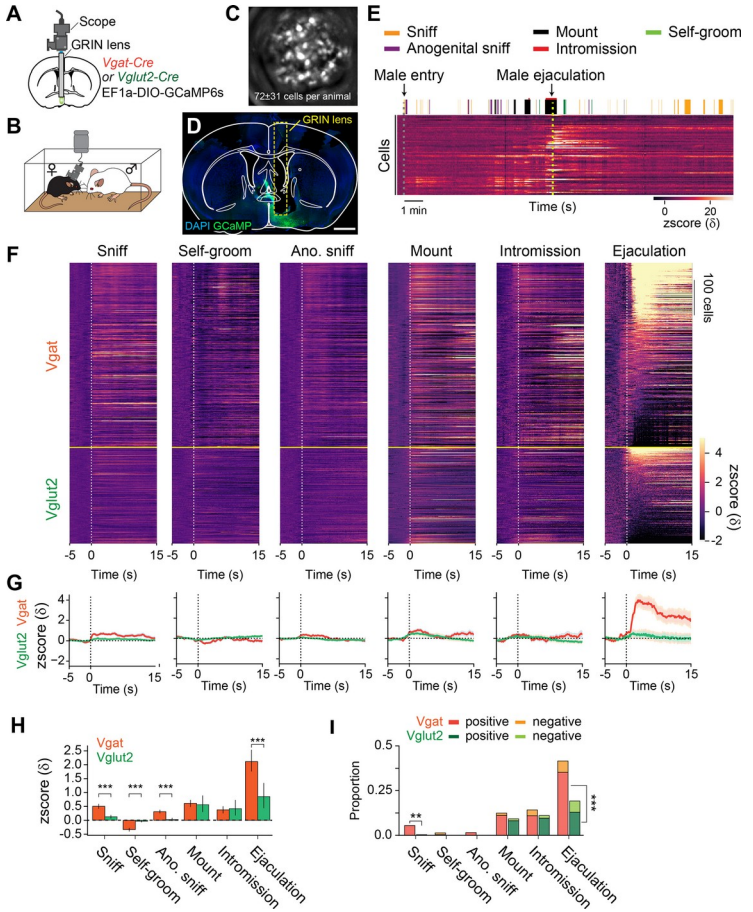
Michael et al. (2020), Video 1.

Female preoptic cells respond to events within mating

A mouse study followed female behavior and preoptic neuronal activity during a self-paced mating task. **Self-paced mating** allows the female to control aspects of access to the male. This arrangement distinguishes changes in her approach or contact behavior from a sequence imposed entirely by the experimenter or the male's location.

The researchers observed heterogeneous neural responses to mounting, intromission, and male ejaculation. Some preoptic cells showed prolonged activity after ejaculation, when subsequent female sexual motivation was reduced. Calcium imaging associated particular cellular activity patterns with events in the interaction rather than assuming one signal represented the entire mating episode.

Excitatory and inhibitory neurons did not all follow the same response profile. The identity and timing of the responding population helped distinguish the sensory event from the longer behavioral state that followed. A brief interaction can thus leave a neural state that affects later willingness to re-enter the mating situation.



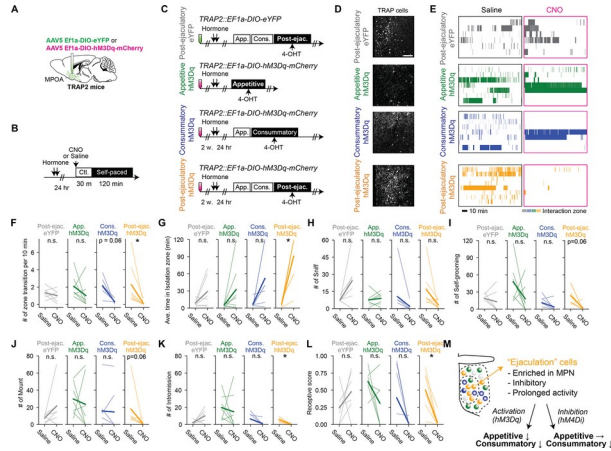
Ishii et al. (2025), Fig. 4.

Activating a mating-tagged ensemble suppresses behavior

The female preoptic study used activity-dependent labeling to identify neurons recruited during mating-related events. **An ensemble** is a group of neurons whose activity is considered together because they participate in a common event or state. Labeling provided a way to manipulate cells selected by their previous activity rather than only by anatomical position.

Activating a labeled preoptic subpopulation suppressed subsequent female sexual behavior under the tested conditions. The intervention moved beyond observing a correlation between activity and reduced motivation. It showed that driving part of the event-associated population could alter the later behavioral response in an established social circuit.

This result does not mean that all medial preoptic neurons suppress mating. Other populations contribute to motivation, performance, and parental behavior. The effect depended on which cells were labeled and activated, illustrating how anatomically neighboring neurons can promote different phases or opposing components of the same broad behavioral domain.



Ishii et al. (2025), Fig. 7.



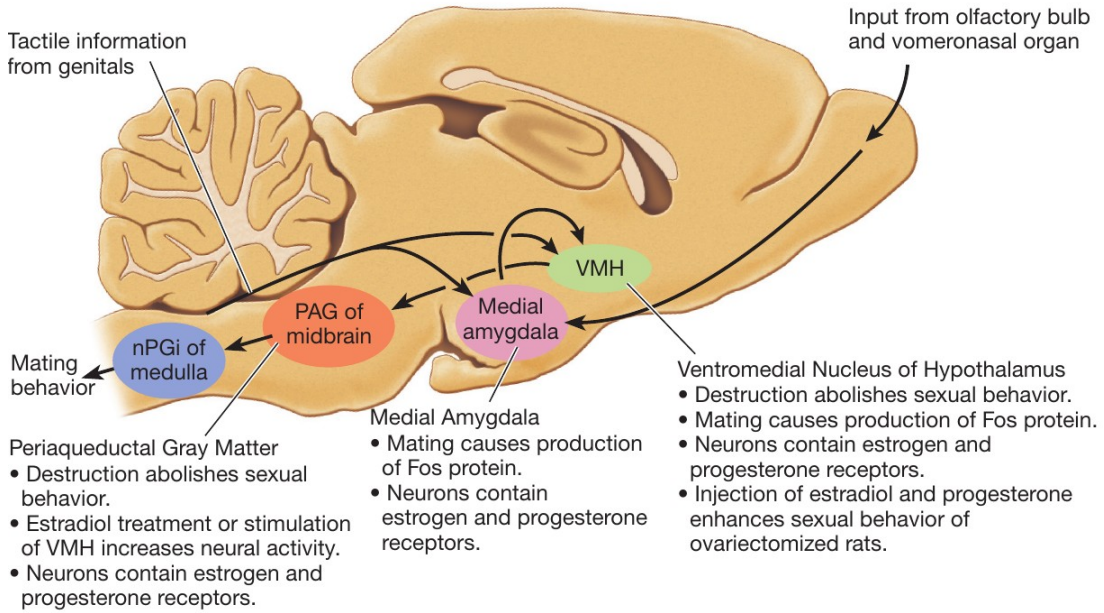
Michael et al. (2020), Video 2.

Ventromedial hypothalamic circuits help regulate receptivity

The **ventromedial hypothalamus**, or VMH, contains hormone-sensitive populations important for female rodent mating responses. Estradiol and progesterone change the responsiveness of this network. Sensory information from social investigation and physical contact can then influence the descending pathways that organize receptive posture and related actions.

In classic rat experiments, VMH damage prevented normal lordosis even with ovarian hormone treatment, while stimulation could facilitate the response. These findings distinguished an available hormone signal from the intact neural tissue needed to translate it into behavior. Hormone administration could not simply bypass the loss of the relevant circuit.

Descending VMH influences interact with midbrain and spinal motor pathways. Genital and flank stimulation also provide feedback during the interaction. A receptive posture therefore reflects hormone-dependent readiness combined with the appropriate sensory event and motor circuitry, rather than estrogen acting directly on every muscle participating in the response.



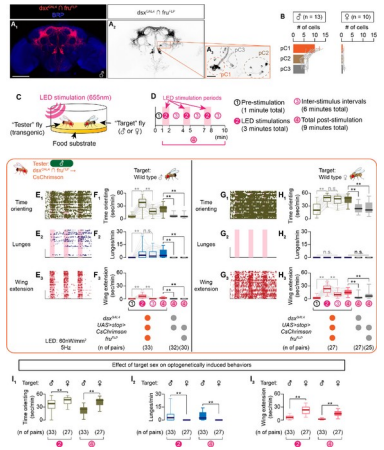
Carlson and Birkett (2022), Fig. 10.18, p. 322.

Sex-determining genes can specify parts of a social circuit

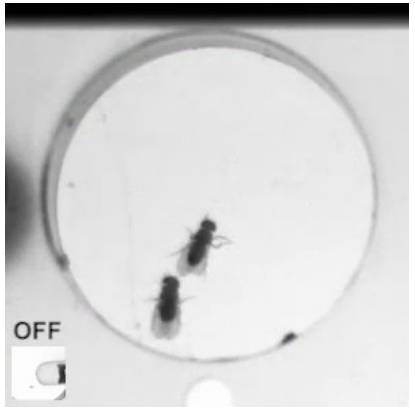
In fruit flies, **doublesex** and **fruitless** regulate development and function of sex-biased neural populations. These genes influence transcription rather than circulating as steroid hormones. A study used genetic intersections to select neurons expressing particular combinations of these regulators and then tested the behavior produced when those cells were activated.

The investigators distinguished neurons contributing to the ability to execute courtship from mechanisms that enhance courtship toward females. Activating defined populations could produce courtship actions toward different targets. The execution of a motor pattern and its direction toward a particular kind of social stimulus were therefore partly separable properties of the circuit.

Fly genetic tools allow precise tests of neurons whose developmental identity is controlled by sex-determining genes. These experiments concern a species-specific reproductive circuit. The labels used for a fly's target-directed courtship should not be treated as direct equivalents of human gender identity, attraction, or the social meaning of a relationship.



Ishii et al. (2020), Fig. 1.



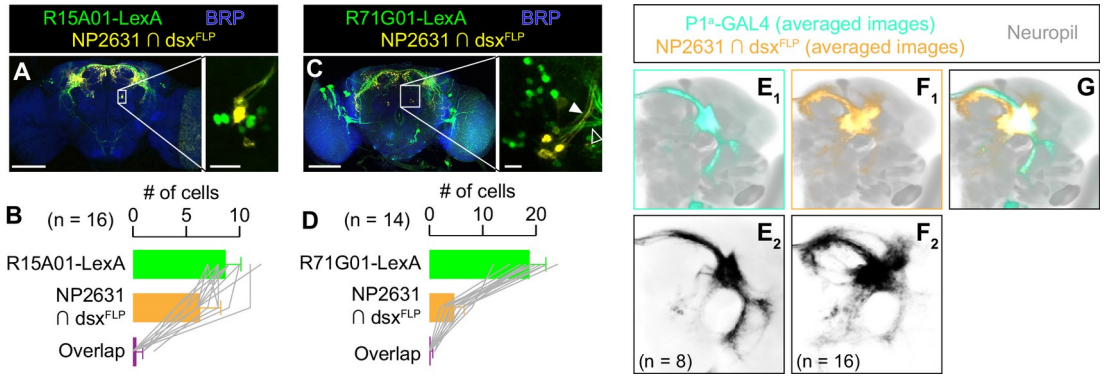
Ishii et al. (2020), Video 1.

Courtship execution and target selection are separable

The fruit-fly study altered fruitless function while preserving the doublesex-dependent component of a selected neural population. The manipulation tested whether an animal retained the machinery needed to perform courtship when the usual enhancement toward females was changed. It separated a developmental regulator from the complete disappearance of every courtship-related neuron.

The researchers found distinct contributions to courtship capability and female-directed enhancement, with cooperative regulation in other neuronal subclasses. Related populations could also influence aggression toward males under different genetic and stimulation conditions. A genetically specified social circuit therefore contained interacting components rather than one neuron group encoding every aspect of biological sex.

An observed action can be decomposed into the sensory target, the internal drive, and the available motor program. Changing one component need not abolish all the others. This circuit logic explains why a genetic manipulation can preserve an action while changing when or toward which stimulus that action is expressed.



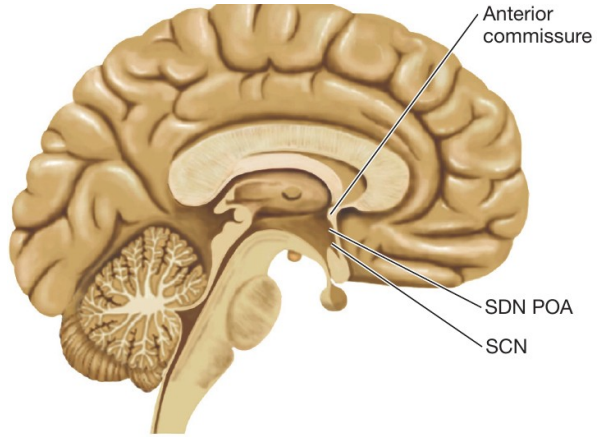
Ishii et al. (2020), Fig. 5.

Sex-related brain variation depends on the feature measured

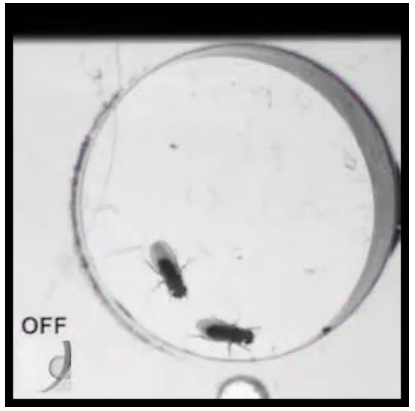
Sexual dimorphism describes an average difference between groups in a biological trait. It does not mean every individual falls into two nonoverlapping categories for that trait. Brain volume, receptor expression, cell number, and gene transcription are different measurements and can show different patterns within the same species.

A larger structure is not automatically more active or better at a psychological function. Regional volume can reflect several cellular and anatomical contributors. Establishing a behavioral mechanism requires evidence about the relevant cells, their signals, and their effects rather than inferring function directly from a group-average anatomical difference.

Individual developmental histories also differ across hormonal, genetic, and environmental conditions. Consequently, an anatomical measurement does not establish a person's gender identity or sexual orientation. Such psychological characteristics require their own descriptions and measurements rather than being assigned from an average difference observed in a brain image.



Carlson and Birkett (2022), Fig. 10.19, p. 326.



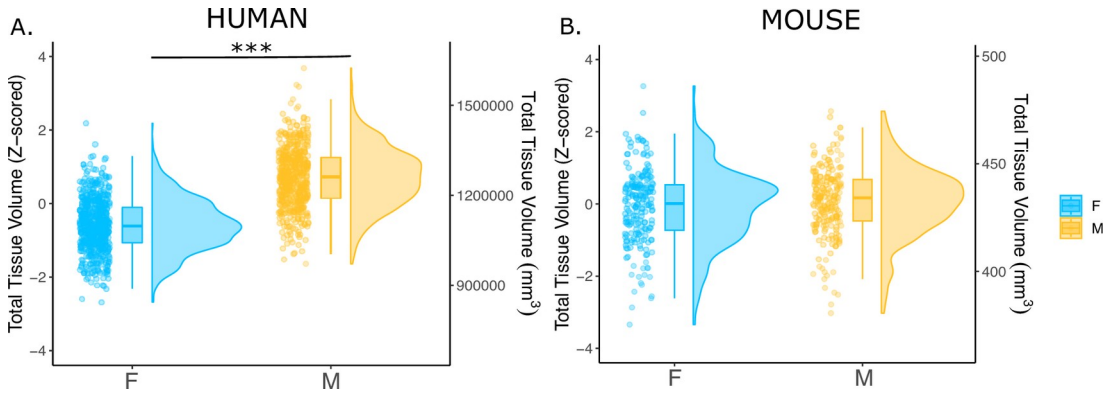
Ishii et al. (2020), Video 2.

Human and mouse brain differences do not match uniformly

A comparative MRI study examined sex-related variation in human and mouse brain anatomy. **Magnetic resonance imaging**, or MRI, measures tissue-related signals that can be used to estimate regional volumes. The researchers compared homologous regions, structures judged to correspond across species, instead of assuming that any similarly named region had identical sex effects.

Human males had larger and more variable total brain volume on average, but the same pattern was not mirrored in mice. Adjusting regional measurements for total volume changed the question from overall size to regional organization. The comparison revealed only modest agreement across corresponding structures rather than a uniform cross-species anatomical rule.

This matters when selecting an animal model for a particular developmental mechanism. A mouse region can help test cellular processes, but its group difference may be retained, reduced, or reversed in humans. The appropriate comparison depends on the actual anatomical feature being studied rather than on a general claim that one species reproduces every human sex difference.



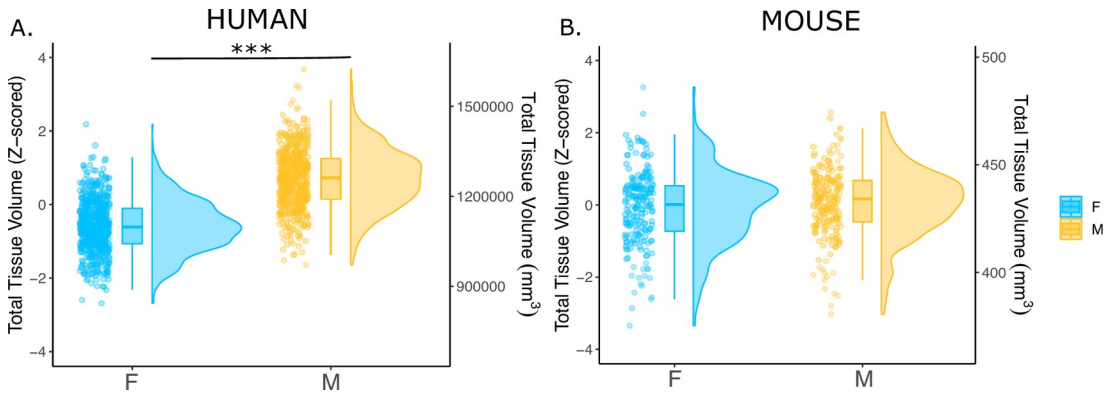
Guma et al. (2024), Fig. 1.

Group averages differ from individual variation

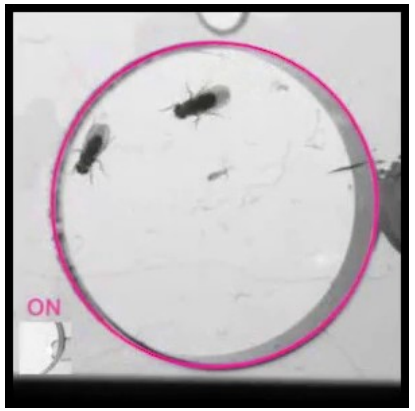
The comparative MRI study also examined variability within groups and patterns of regional anatomy. An average difference summarizes the centers of two distributions, while variability describes their spread. Two groups can differ in their average measurement while many individuals have overlapping values, making a single person difficult to characterize from that feature.

Cross-species agreement was stronger in some cortical regions than in other brain territories. Regional gene-expression comparisons provided another level of analysis, but anatomical agreement and molecular similarity were not interchangeable. The researchers measured whether related patterns occurred rather than demonstrating that one gene directly caused a particular human behavior.

Anatomical observations are therefore most informative when the species, region, scaling method, and measured trait are specified. They can guide experiments on development without functioning as psychological classification tests. The study did not provide a method for identifying a person's orientation or gender from a structural brain scan.



Guma et al. (2024), Fig. 1.



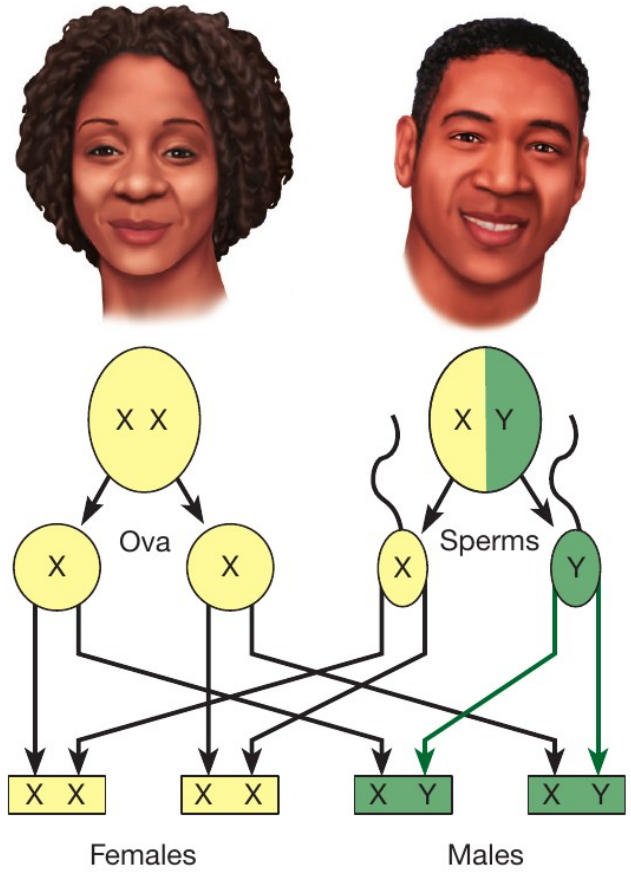
Ishii et al. (2020), Video 6.

Sex traits, gender identity, and orientation are distinct

Gender identity concerns a person's experienced gender, while **sexual orientation** concerns patterns of attraction. Neither is identical to chromosomes, genital anatomy, hormone concentration, or a particular sexual action. These characteristics can be related in someone's life while still describing different biological, psychological, and social dimensions.

Sexual behavior records what someone does, whereas attraction describes whom they desire and identity describes how they understand or describe themselves. Studies sometimes measure only one of these dimensions. A result obtained from a question about past behavior should not automatically be reported as a complete explanation of every dimension of orientation.

Development includes interacting genetic, endocrine, experiential, and social processes whose contributions vary across traits. Human diversity cannot be reduced to a single brain region or hormone level. Describing the measured characteristic precisely preserves the difference between a population-level developmental finding and an unsupported prediction about an individual person.



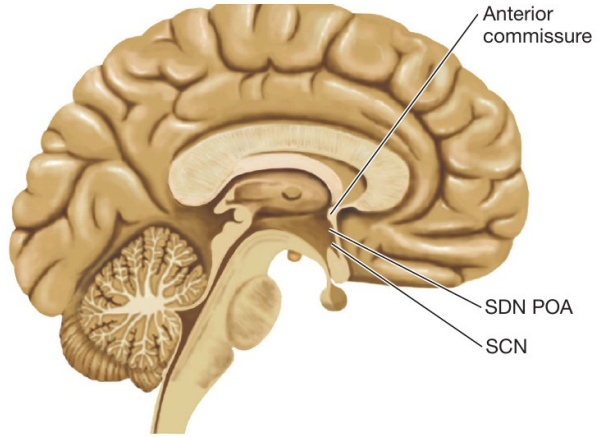
Carlson and Birkett (2022), Fig. 10.1, p. 304.

Orientation develops through multiple interacting influences

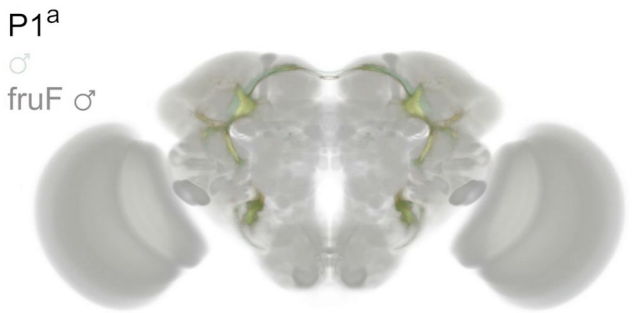
Research on sexual orientation examines genetic variation, prenatal environments, developmental history, and reported attraction or behavior. **Heritability** describes the contribution of genetic differences to variation within a studied population under particular conditions. It does not specify the fraction of an individual's orientation that is genetic or establish a single determining gene.

Clinical variations in prenatal hormone signaling have contributed evidence about development, but outcomes differ among individuals. Such findings are informative about particular pathways and exposures rather than being deterministic tests. A hormone-sensitive developmental process can influence a probability without fixing every later preference, identity, or life experience.

Twin comparisons, endocrine observations, and anatomical studies also measure different levels of biology. Agreement among relatives can be informative without making their experiences identical, and group-average anatomy does not identify an individual's attraction. A multifactorial account preserves these distinct observations rather than forcing them into one universal developmental pathway.



Carlson and Birkett (2022), Fig. 10.19, p. 326.



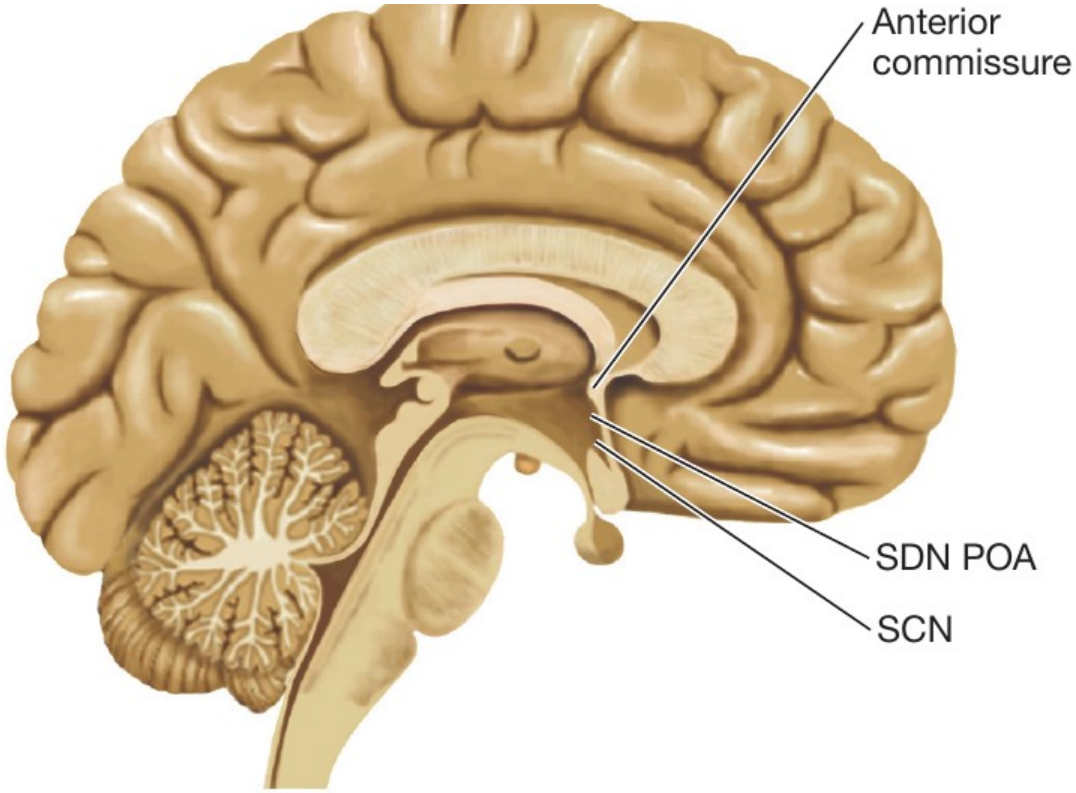
Ishii et al. (2020), Video 8.

Adult hormones do not diagnose sexual orientation

An adult blood sample describes a current endocrine state, not the complete hormonal environment experienced during brain development. Hormone levels also vary with time of day, health, medication, age, and reproductive condition. A present-day concentration therefore cannot reconstruct a person's prenatal neural development or establish the origin of their attraction.

Likewise, an experimental change in sexual motivation is not the same as changing orientation. Hormones can affect the strength of interest or aspects of physiological function while attraction remains directed toward the same partners. The dependent variable must remain the one actually measured rather than being expanded into a different psychological claim.

Human sexual and gender diversity are not disease categories requiring a neural explanation to justify their existence. Neuroscience can investigate development and variation without treating a normal identity as a defect. Biological findings explain processes and distributions; they do not supply a hierarchy of legitimate identities or desirable patterns of attraction.



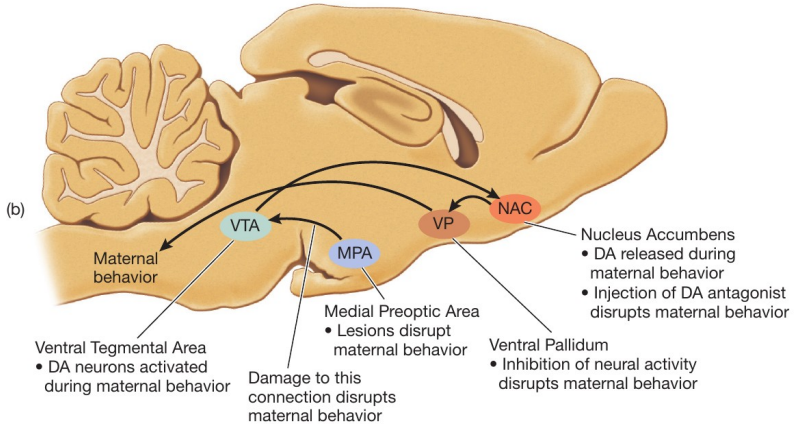
Carlson and Birkett (2022), Fig. 10.19, p. 326.

Pair bonding requires recognizing and valuing a partner

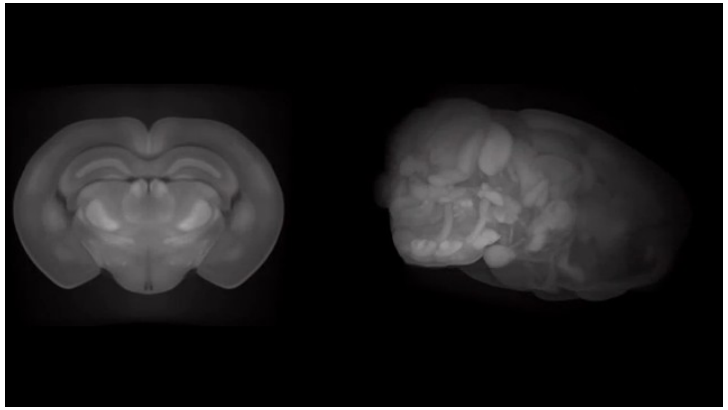
A **pair bond** is a selective, enduring social attachment to a particular partner. It requires more than recognizing that another animal is present. Sensory information must identify the individual, while memory and motivation give interactions with that individual a different value from interactions with an unfamiliar conspecific.

Prairie voles provide one laboratory model because paired animals often prefer familiar partners in controlled contact tests. These assays measure selective affiliation rather than every aspect of human attachment. Related vole species differ in social organization, helping researchers examine which neural and receptor differences accompany particular patterns of bonding.

Oxytocin, vasopressin, dopamine, and other signals can influence bonding-related networks through particular receptors and brain regions. Their effects depend on experience and experimental conditions. No single peptide should be described as containing a complete representation of the partner or mechanically producing every feature of a lasting relationship.



Carlson and Birkett (2022), Fig. 10.24(b), p. 334.



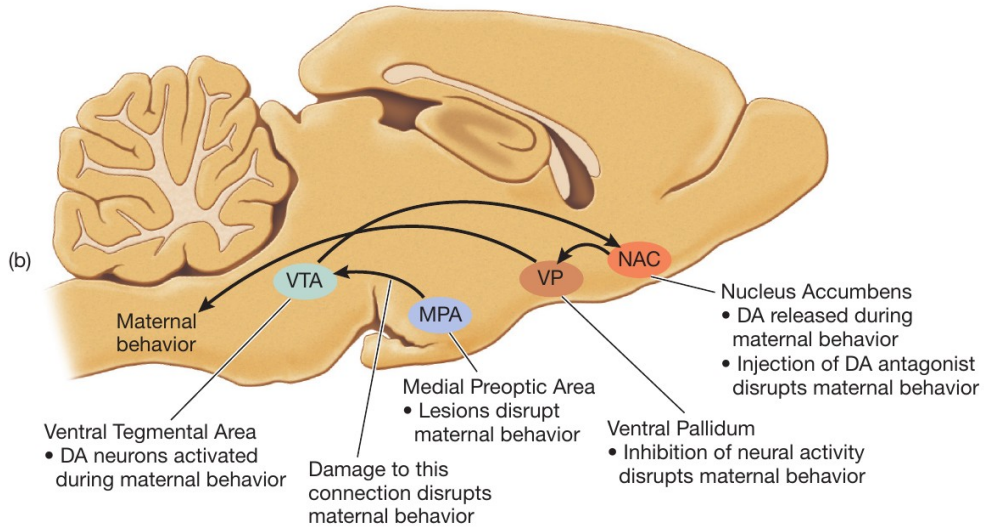
Gustison et al. (2024), Video 1.

Reward and memory contribute to partner preference

The **nucleus accumbens** and **ventral pallidum** participate in motivated behavior and the value of social interactions. In vole experiments, neuropeptide and dopamine manipulations in these networks can alter the development or expression of partner preference. These effects connect individual recognition to the motivation to seek or maintain contact.

Vasopressin and **oxytocin** act through different receptors whose distribution varies across species and brain regions. Changing receptor availability can alter the effect of a social encounter. A circulating peptide concentration consequently does not describe how strongly every bonding-related target is responding at the synaptic level.

Bonding also has phases: initial attraction, repeated interaction, selective affiliation, and responses to separation. A manipulation can affect one phase more than another. Measuring partner contact after a defined pairing period tests a specific part of this process rather than demonstrating that an animal has acquired every component of an enduring social bond.



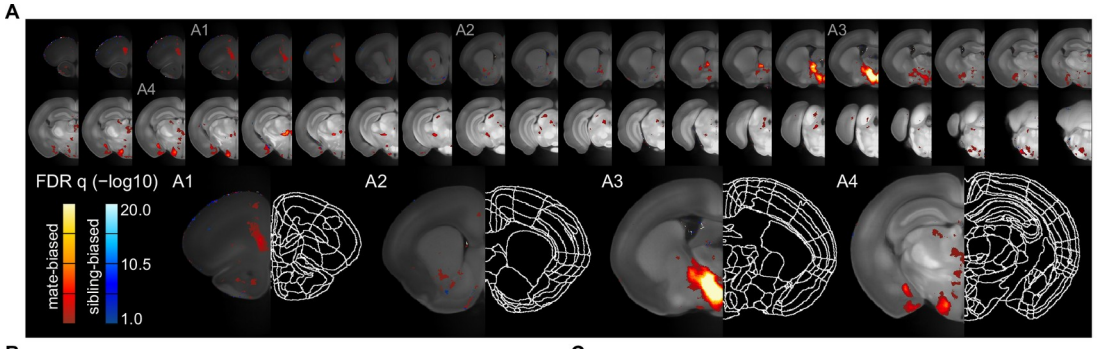
Carlson and Birkett (2022), Fig. 10.24(b), p. 334.

Pair formation recruits coordinated brain-wide activity

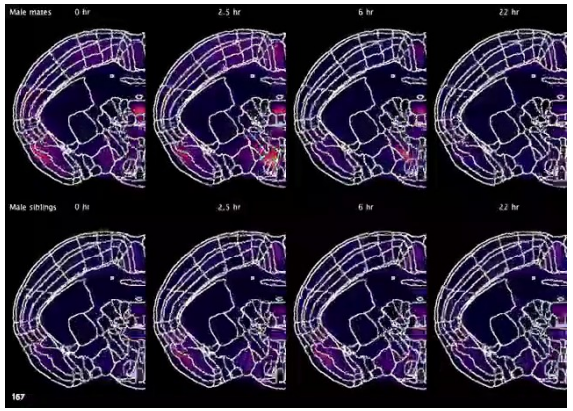
A prairie-vole study used whole-brain imaging to map **c-Fos**, a protein whose expression often increases after neuronal activation. The investigators compared pairing conditions and time points to identify patterns associated with mating and bond formation. c-Fos mapping integrates activity over time rather than reporting the moment-by-moment electrical firing of each cell.

Bonding-related activity involved networks spanning hypothalamic, amygdalar, reward, and cortical regions. Patterns were coordinated across the members of a pair and were strongly related to their mating interactions. The findings extended beyond one isolated bonding nucleus to circuits involved in social sensation, reproductive behavior, reward, and territorial responses.

Males and females showed substantial similarity in these coordinated patterns rather than entirely separate bonding systems. The atlas made it possible to compare the same anatomical locations across animals. Activity mapping identified candidate networks for later causal experiments without establishing that every c-Fos-positive cell was individually necessary for forming a bond.



Gustison et al. (2024), Fig. 3.



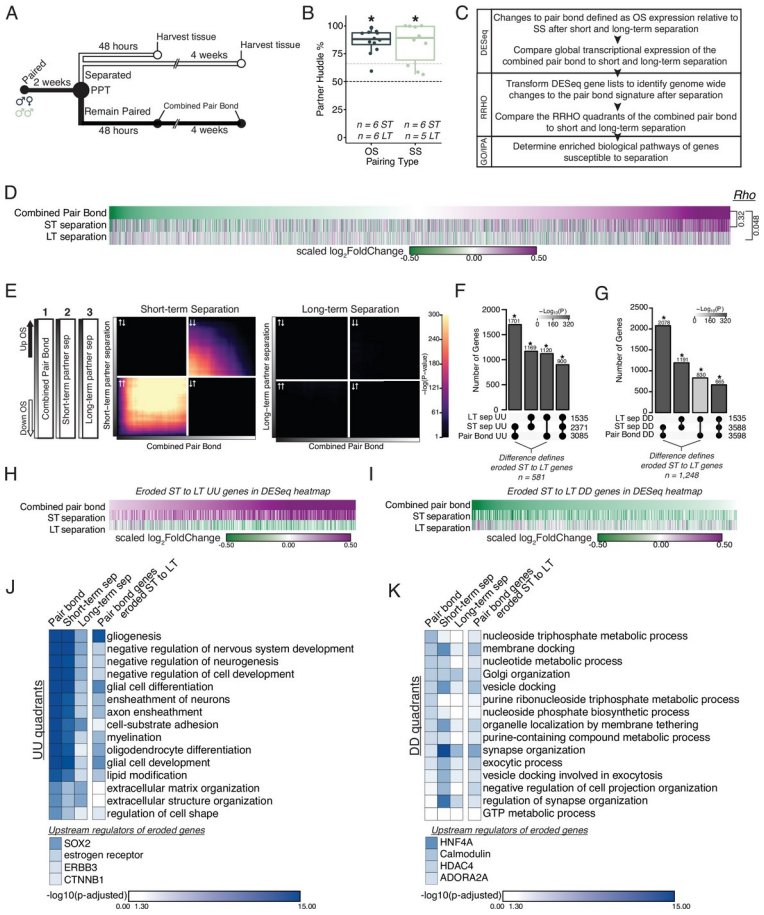
Gustison et al. (2024), Video 2.

Separation changes molecules before preference fades

A prairie-vole study compared behavior and nucleus-accumbens gene expression after partner separation. **RNA sequencing** measures the abundance of gene transcripts in a tissue sample. It can reveal changes in molecular state without directly measuring whether an individual neuron fires more often or whether the animal still prefers its familiar partner.

After prolonged separation, some pair-bond-associated transcriptional patterns weakened even though partner-directed affiliation remained detectable. Molecular adaptation and the measured social preference therefore followed different time courses. The absence of one transcript pattern did not mean that every behavioral feature of the bond had already disappeared.

The researchers also enriched RNA from neurons to distinguish neuronal changes from patterns in the mixed tissue. Different analyses implicated cellular maintenance, signaling, and glial-related processes. The result concerns a vole's adaptation to separation and provides possible biological targets without equating that molecular process with the complete human experience of grief.



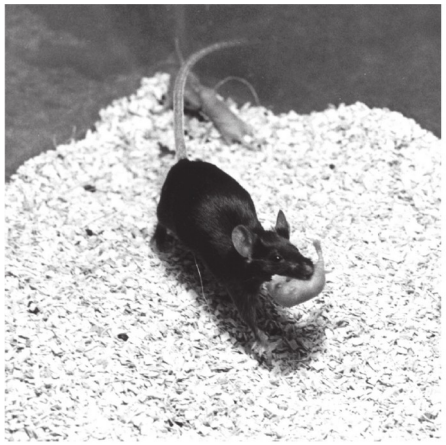
Sadino et al. (2023), Fig. 3.

Parental care solves several different problems for offspring

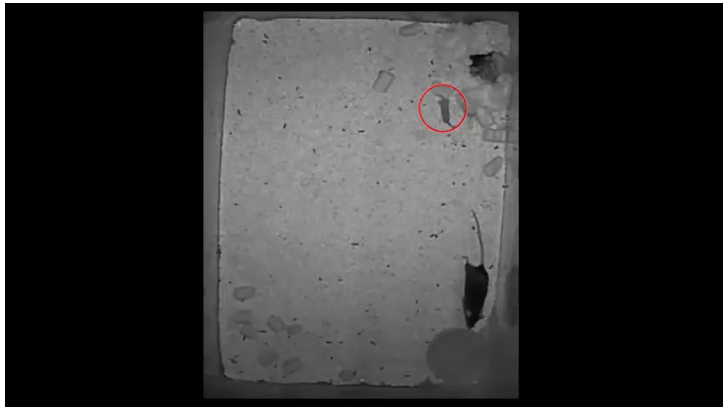
Newborn rodents depend on adults for warmth, nutrition, grooming, and protection. **Pup retrieval** returns a displaced infant to the nest, while nursing supplies milk and huddling helps conserve heat. These behaviors solve different physiological problems, so one successful retrieval does not establish that every component of care is functioning normally.

An adult must detect infant cues, approach without attacking, handle the pup appropriately, and return it to a suitable location. The sequence combines sensory recognition, motivation, and coordinated movement. A deficit can arise at any of these steps rather than reflecting a complete loss of every response to an infant.

Caregiving is not restricted to the biological mother in every species or situation. Fathers and other adults can care for offspring under appropriate conditions. The neural mechanisms must therefore support flexible responses to infant needs rather than requiring pregnancy as an unavoidable prerequisite for every form of parental action.



Carlson and Birkett (2022), Fig. 10.22, p. 331.



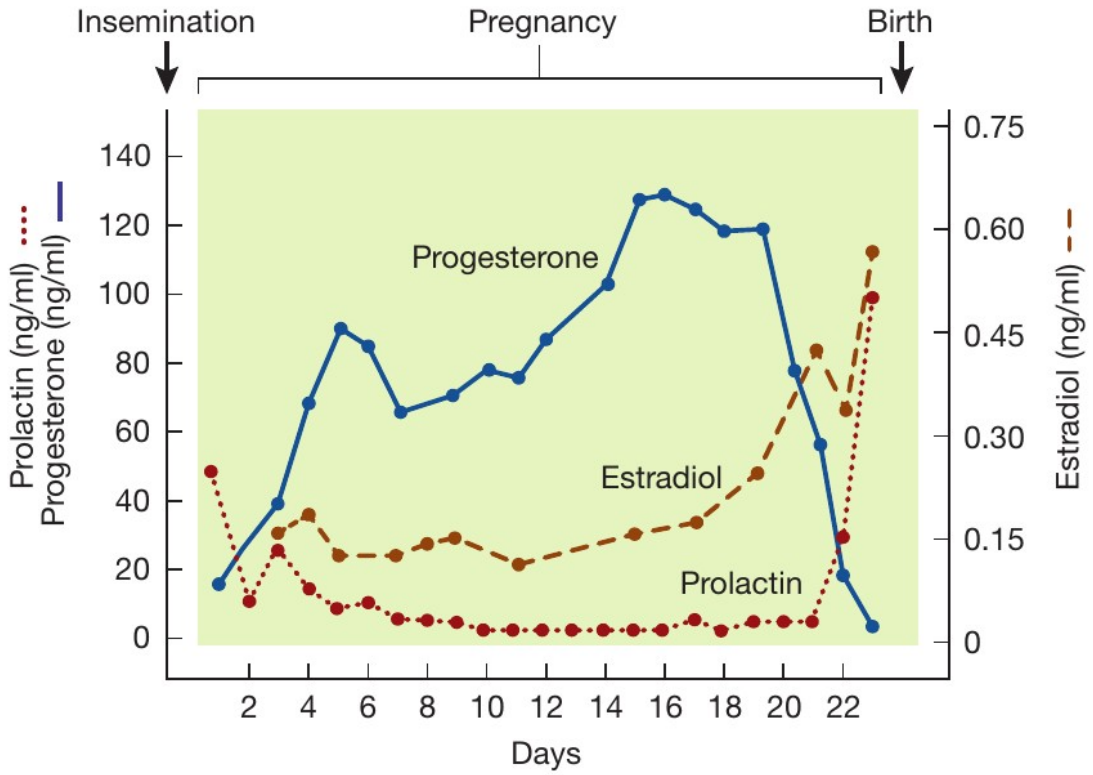
Carcea et al. (2021), Video 1.

Pregnancy and experience both prepare caregiving

During pregnancy and birth, changes in progesterone, estradiol, prolactin, and oxytocin alter reproductive and caregiving systems. **Prolactin** contributes to milk production, while **oxytocin** promotes milk ejection and acts within social circuits. These are distinct functions: producing milk and releasing it to the infant require different biological steps.

In rodent experiments, hormonal conditions around birth facilitate the onset of maternal behavior. Repeated exposure to pups can also promote care in adults that have not undergone pregnancy. Endocrine preparation and experience therefore provide partially different routes for making infant cues effective at recruiting caregiving responses.

Once care has been learned, the hormonal requirements for its expression can differ from those for its initial acquisition. The relevant experiment must specify whether it tests the first appearance, later maintenance, or a particular component of behavior. A hormone that accelerates learning need not be required identically for every subsequent retrieval.



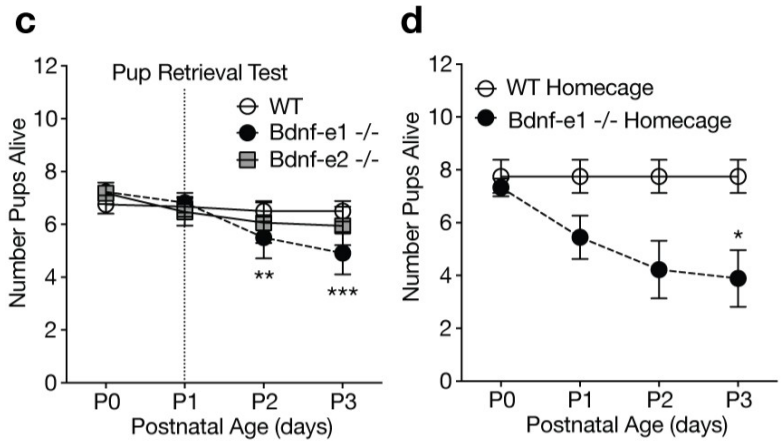
Carlson and Birkett (2022), Fig. 10.23, p. 332.

A growth-factor signal contributes to maternal behavior

Brain-derived neurotrophic factor, or BDNF, is a protein signal that regulates neuronal development and plasticity. Different gene promoters can produce BDNF transcripts encoding the same protein. A mouse study disrupted particular promoter-driven expression patterns to ask whether the source and developmental regulation of this signal affected female social behavior.

Females with the relevant BDNF deficits showed impaired maternal care and reduced sexual receptivity. Their hypothalamic oxytocin expression was also reduced during development. The changes linked a growth-factor pathway to a neuropeptide system, while leaving open which cellular effects contributed to each distinct behavioral deficit.

The study measured specific caregiving actions rather than assigning a single global parenting score as the entire mechanism. A molecular alteration can affect how an adult approaches or handles pups without changing every movement or sensory ability. Such component measurements help locate the behavioral consequences of a developmental signaling disturbance.



Maynard et al. (2018), Fig. 1.



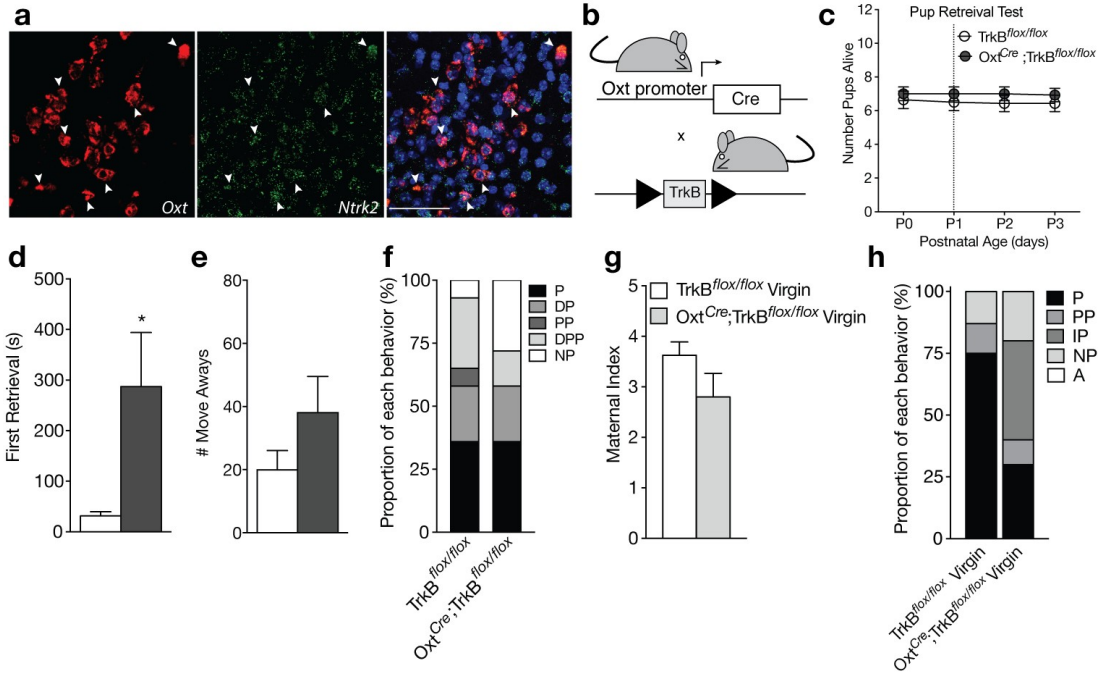
Carcea et al. (2021), Video 2.

Oxytocin neurons also depend on the signals they receive

BDNF acts through **TrkB**, a receptor that changes intracellular signaling after binding the growth factor. The maternal-behavior study removed TrkB selectively from oxytocin neurons. This manipulation tested the receiving side of the pathway rather than reducing BDNF everywhere in the brain or eliminating every oxytocin-producing cell.

The receptor manipulation reproduced part of the maternal-care impairment seen with broader BDNF disruption. The result indicated that BDNF signaling within oxytocin neurons contributed to normal caregiving responses. It did not imply that oxytocin neurons were the only BDNF targets involved in every reproductive or social effect.

Cell-enriched RNA measurements identified molecular pathways related to neuronal structure and plasticity in the oxytocin population. A cell that releases a social peptide is therefore also regulated by developmental and trophic inputs. Changes in those inputs can alter how that neuron develops and functions before its peptide is released to downstream targets.



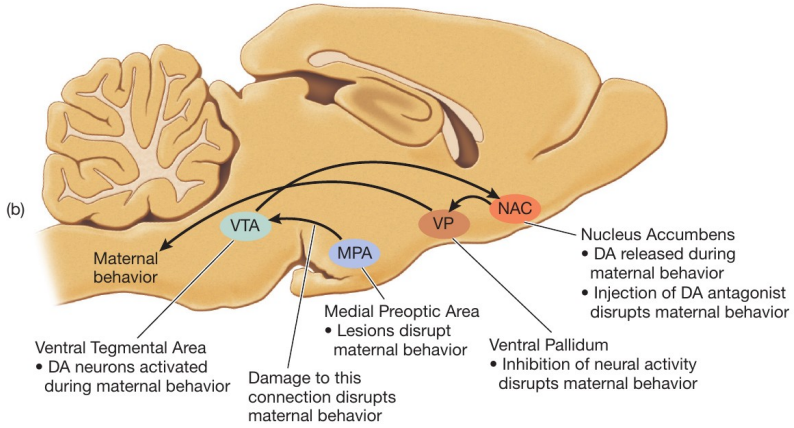
Maynard et al. (2018), Fig. 3.

Preoptic caregiving circuits compete with infant avoidance

The medial preoptic area helps coordinate parental responses to infant cues. In rodent experiments, damage to this region can disrupt retrieval and other aspects of care. Its projections interact with motivational and motor systems, allowing a pup's sensory features to recruit an organized approach and handling response.

Other pathways can promote avoidance or attack, particularly in animals whose current social and reproductive state does not favor care. Parental behavior therefore involves both facilitating appropriate actions and reducing competing responses. An infant cue does not carry an invariant command that produces the same behavior in every adult encountering it.

Cell-specific research has identified galanin-expressing preoptic neurons that participate in parenting in both sexes. Their manipulation can change pup-directed actions, while nearby populations contribute to other social behaviors. The preoptic area is a collection of interacting populations rather than a single center whose activation always produces one complete caregiving sequence.



Carlson and Birkett (2022), Fig. 10.24(b), p. 334.



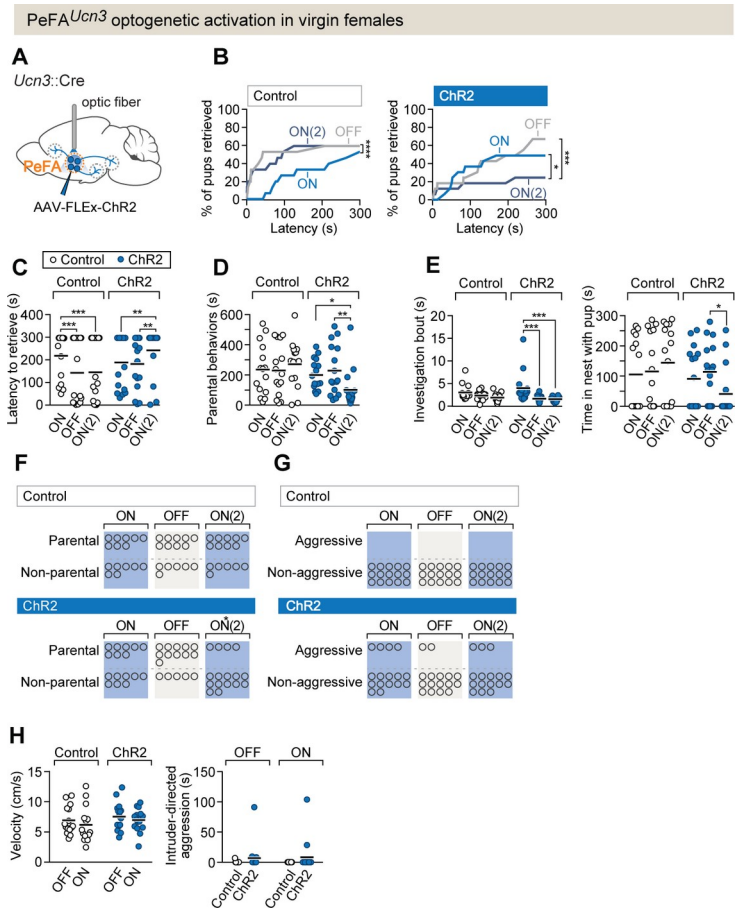
Carcea et al. (2021), Video 3.

A hypothalamic population can suppress infant care

Urocortin-3, or Ucn3, marks a population of neurons in the hypothalamic perifornical area. Researchers found these cells active during infant-directed attack and tested their function with selective manipulations. The experiment examined the negative control of parenting rather than assuming that absent care always reflected weak activity in a positive caregiving pathway.

Activating the Ucn3 population in virgin female mice reduced normal pup-directed care and could produce avoidance-related or aggressive responses. In males, silencing the population reduced infant-directed aggression. The same broad behavioral outcome can therefore be altered by suppressing an opponent pathway as well as by strengthening a care-promoting circuit.

The effects were evaluated alongside other social behaviors and anatomical connections. Infant-directed responses were not treated as identical to every form of adult aggression. Identifying a cell population involved in one target-specific behavior allows a more precise account than labeling the entire hypothalamus as generally aggressive or generally parental.



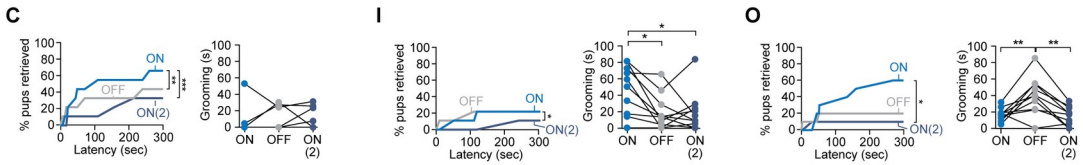
Autry et al. (2021), Fig. 3.

Different projections alter different pup-directed actions

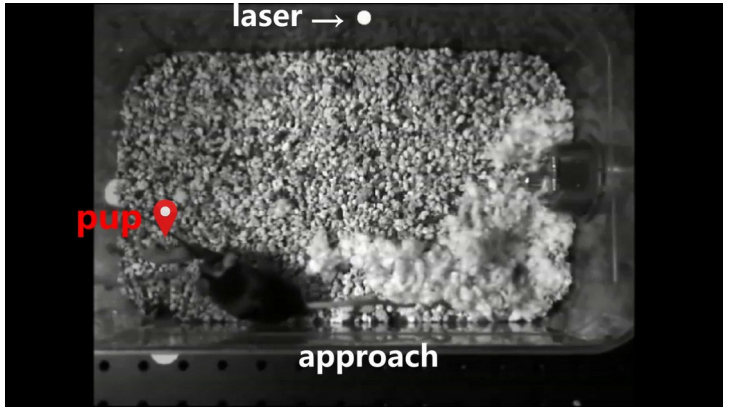
The Ucn3 study traced axons to several forebrain targets and stimulated their endings separately. **Projection-specific stimulation** changes activity at an identified output route, helping test which behavioral effects can be produced through that branch. It does not require assuming that every axon from the source population serves the same function.

Activating projections to different targets produced distinguishable effects, including reduced interaction, pup avoidance, and more aggressive handling. The pattern separated components of infant-directed response that might look similar if recorded only as a failure to retrieve. Target location changed the action produced by stimulating the source population's output.

Some source neurons sent branches to more than one target, so the anatomy was not simply a set of completely independent channels. Functional stimulation and tracing supplied complementary evidence. The result was a distributed opponent system capable of biasing different stages of the approach, contact, and care sequence.



Autry et al. (2021), Fig. 9.



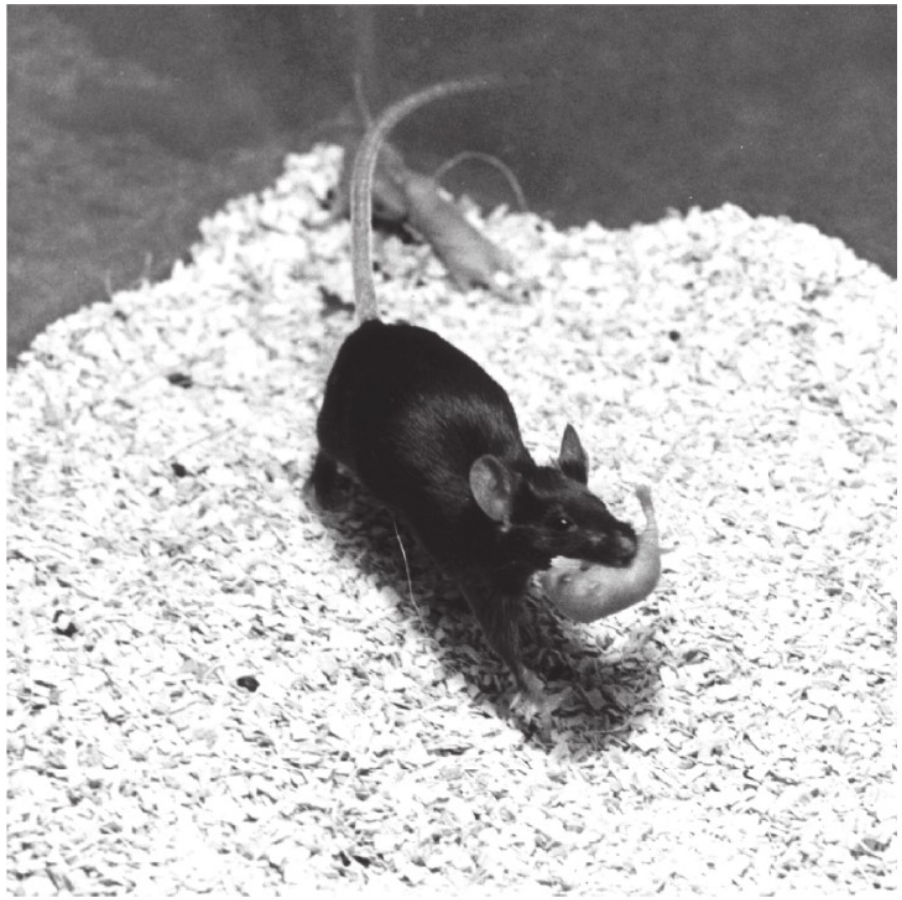
Autry et al. (2021), Video 2.

Experience changes the meaning of infant sounds

Rodent pups emit ultrasonic distress calls when displaced or separated from the nest. Adult detection of these sounds depends on auditory pathways, while the caregiving response depends on their biological meaning. Hearing a call and treating it as a reason to retrieve a pup are therefore different neural operations.

Experience with infants can change cortical responses to pup calls and the behavioral actions they recruit. Oxytocin contributes to this plasticity in experimentally defined mouse conditions. The peptide affects the receiving circuit's responsiveness to a sound rather than physically carrying the acoustic information from the pup to the adult's ear.

The adult must also combine the call with location, odor, and contact cues to complete retrieval. A stronger auditory response can facilitate recognition without supplying the complete motor sequence. Sensory plasticity provides one component of a caregiving system whose other circuits determine approach, gentle handling, and return to the nest.



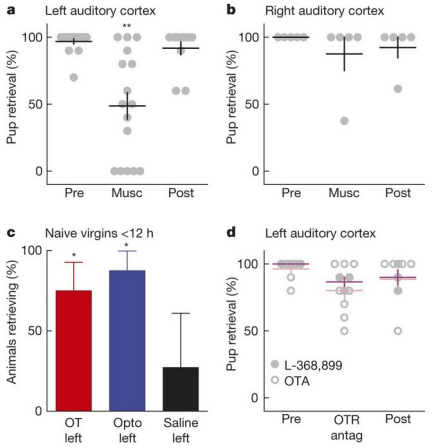
Carlson and Birkett (2022), Fig. 10.22, p. 331.

Oxytocin in auditory cortex helps adults learn retrieval

A mouse study tested whether oxytocin in auditory cortex contributed to pup retrieval. **Auditory cortex** contains neurons that process sound patterns after information has passed through earlier auditory stations. The researchers combined behavioral tests, local drug delivery, receptor labeling, and recordings to locate a specific site where a social peptide could alter infant-cue processing.

Blocking or inactivating the left auditory cortex impaired retrieval under the tested conditions, while local oxytocin could accelerate retrieval in inexperienced females. The right cortex did not show an identical contribution in these experiments. The result identified a lateralized circuit effect rather than a uniform increase in caregiving caused by exposing the whole brain to a hormone.

Oxytocin-receptor involvement also differed between acquiring a response and expressing an established one. Once an adult had learned to retrieve, the same receptor blockade did not reproduce every initial learning deficit. A signal that opens a window for plasticity need not serve as an obligatory moment-by-moment command for each later action.



Marlin et al. (2015), Fig. 3.



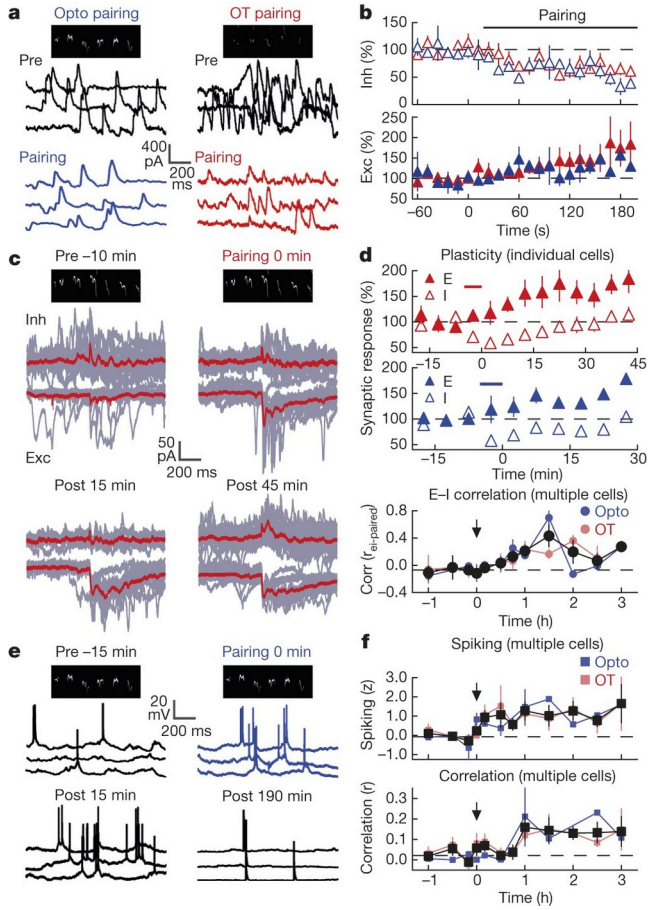
Marlin et al. (2015), Video 3.

Oxytocin changes cortical excitation and inhibition

The auditory-cortex study recorded responses to pup calls in experienced and inexperienced females. **Excitation** increases a neuron's likelihood of firing, while **inhibition** opposes or shapes that response. The timing of these inputs matters because a sound-evoked excitatory event can be strengthened, limited, or delayed by the inhibition arriving around it.

Experienced females showed more reliable pup-call responses and better coordinated excitatory and inhibitory input patterns in the tested left cortical circuit. Pairing calls with oxytocin changed the balance and timing of synaptic responses. This provided a cellular route through which experience could make an infant vocalization more effective at driving cortical output.

The mechanism was not a permanent removal of every inhibitory connection. Inhibitory responses were reorganized alongside excitation as the circuit became more selective and temporally reliable. Oxytocin-dependent plasticity therefore changed how an existing sensory pattern was processed, contributing to recognition of a behaviorally important call rather than simply amplifying all sounds equally.



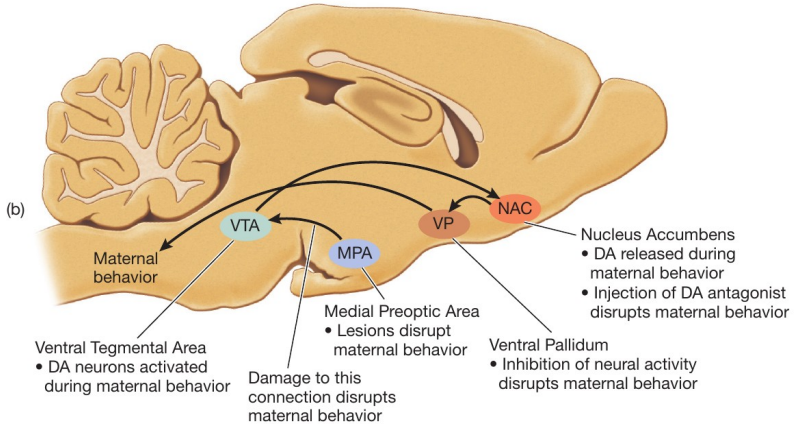
Marlin et al. (2015), Fig. 6.

Milk production and milk ejection require different signals

Lactation includes making milk and delivering it to the infant. Prolactin supports milk synthesis in mammary tissue, while oxytocin contracts cells around the milk-producing structures and promotes ejection. These endocrine functions are coordinated with nursing but are not equivalent to the adult's motivation to approach or retrieve an infant.

Suckling activates sensory pathways that influence hypothalamic and pituitary output. **A neuroendocrine reflex** translates a neural sensory signal into hormone secretion, which then acts on a distant tissue. The infant's behavior therefore helps regulate a physiological response in the caregiver through a sequence of sensory, neural, endocrine, and muscular events.

Oxytocin can also be released within the brain, where its effects depend on receptor location and current circuit activity. Peripheral milk ejection and central sensory plasticity involve the same peptide but different target tissues. A blood concentration does not establish the strength or timing of oxytocin action at every central synapse.



Carlson and Birkett (2022), Fig. 10.24(b), p. 334.



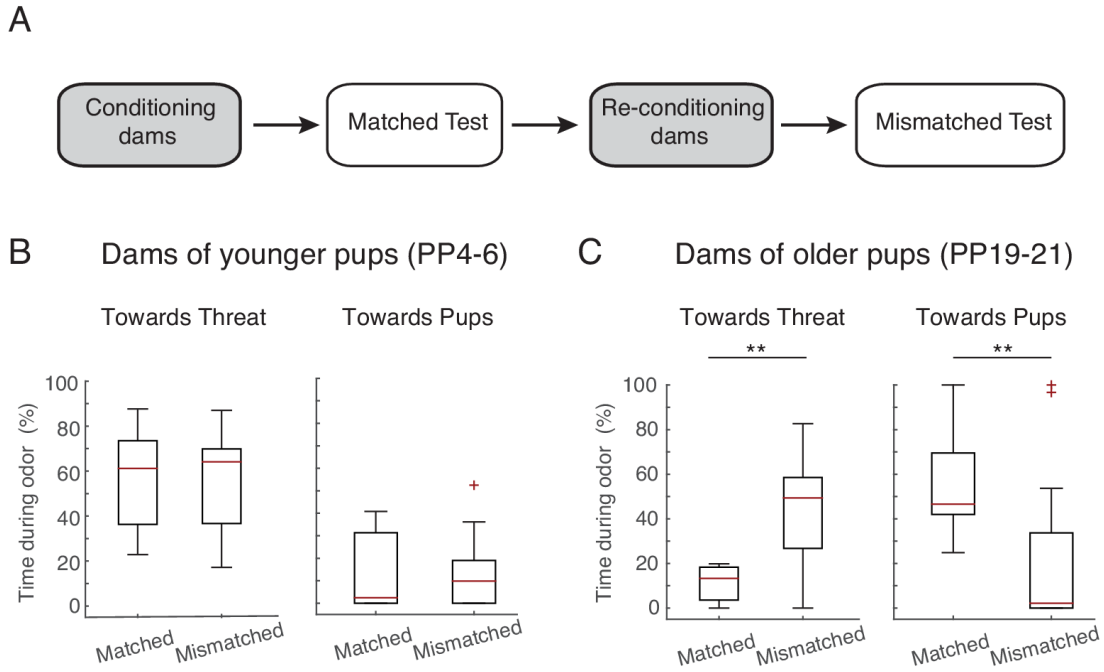
Carcea et al. (2021), Video 4.

Oxytocin can shift maternal defense from freezing to action

A rat study compared defensive responses of conditioned mothers tested alone or with pups. **Conditioned freezing** is the suppression of movement in response to a learned threat cue. Alone, the mothers froze to the conditioned odor; with pups present, they instead maintained contact with offspring or directed behavior toward the threat.

Blocking oxytocin signaling in the central amygdala prevented the normal suppression of maternal freezing. The peptide helped alter the selected defensive action rather than merely eliminating detection of the threat. Remaining mobile allowed the mother to perform responses that would be impossible during sustained immobility.

Pups also acquired information about the odor through interactions with the conditioned mother. Disrupting maternal oxytocin signaling changed the pups' later defensive learning. A manipulation in the adult therefore influenced offspring behavior through the caregiving and threat-learning situation, rather than by directly administering the antagonist to the infant's brain.



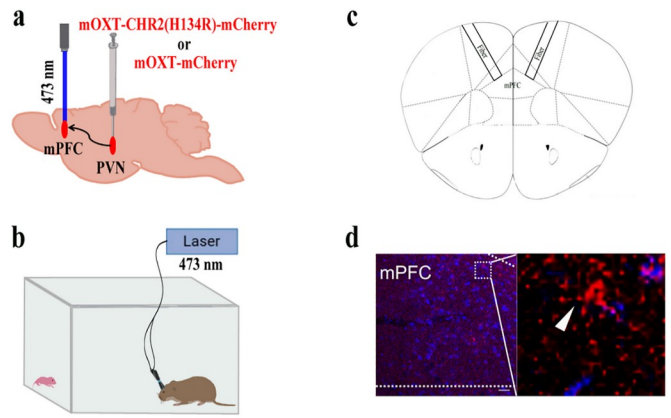
Rickenbacher et al. (2017), Fig. 2.

A hypothalamic–prefrontal route alters pup-directed action

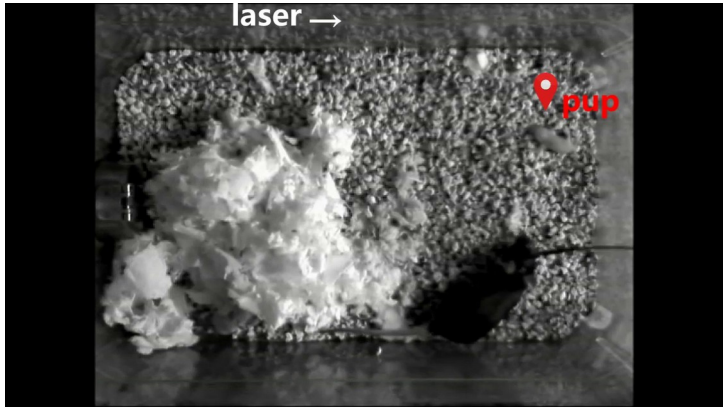
A mandarin-vole study examined oxytocin neurons in the **paraventricular nucleus**, or PVN, and their projections to medial prefrontal cortex. The researchers compared adults that cared for pups with adults that attacked them. This design treated pup-directed behavior as a state-dependent response rather than assuming every virgin adult was uniformly parental.

Activating PVN oxytocin neurons promoted care and delayed attacks in the tested animals. Stimulating the PVN-to-prefrontal projection shortened approach and retrieval latencies in males under the relevant conditions, while the same effect was not found identically in caregiving females. The behavioral influence depended on sex and initial response state.

Inhibiting the pathway facilitated infant-directed attack in animals prone to that response. These opposing manipulations identified an output route through which the peptide system could bias action selection. The result concerns a defined vole circuit and does not establish that administering oxytocin is a general treatment for human caregiving difficulties.



Li et al. (2024), Fig. 6.



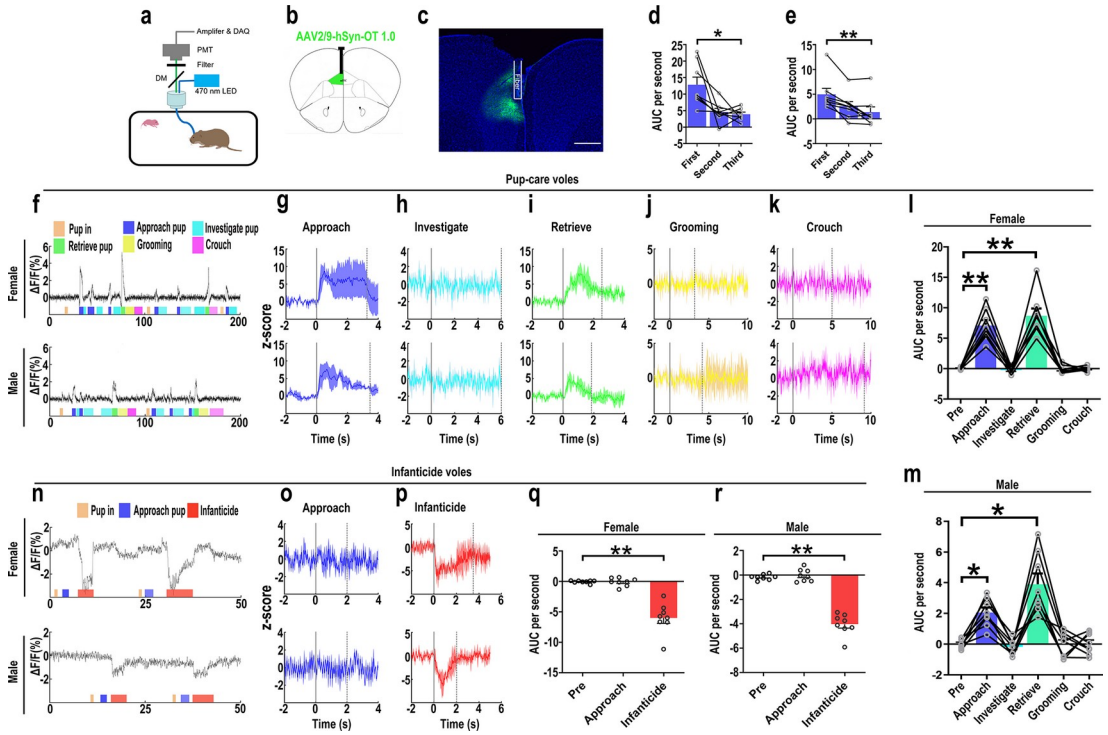
Autry et al. (2021), Video 3.

Local peptide release varies with the action being performed

The mandarin-vole study monitored oxytocin-related signals in medial prefrontal cortex during pup interactions. Measuring local release addresses a different question from measuring peptide-producing cell activity. A neuron can have several projection targets, and activity at its cell body does not by itself specify the peptide signal reaching each target.

The prefrontal oxytocin signal rose during approach and retrieval and decreased during infant-directed attack in the tested animals. These changes aligned the chemical signal with particular stages of behavior. They provided temporal information beyond a single hormone measurement taken before or after the entire social encounter.

Combining release measurements with projection activation and inhibition strengthened the circuit interpretation. The recordings described when a signal changed, while the interventions tested how changing a pathway affected behavior. Neither measurement implied that oxytocin alone contained all the sensory information needed to identify the infant and organize a complete retrieval.



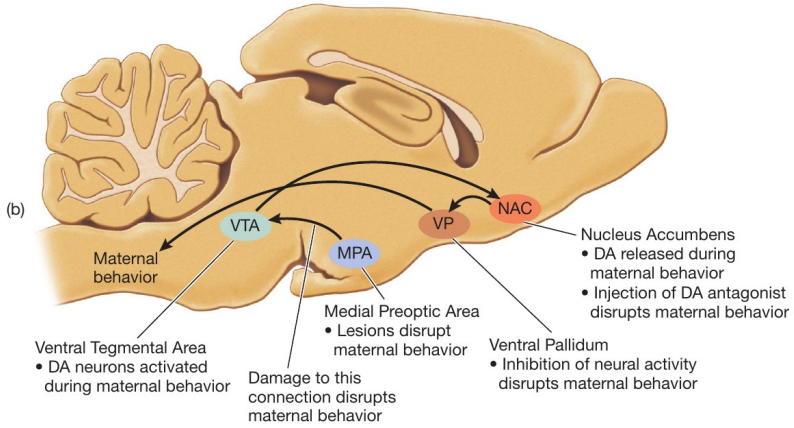
Li et al. (2024), Fig. 4.

Caregiving combines incentive with coordinated action

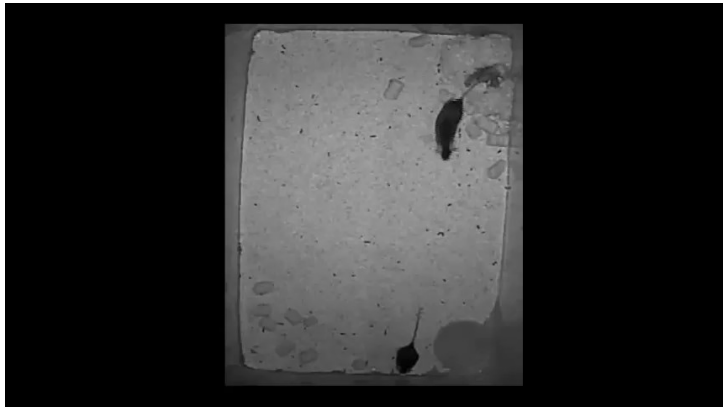
Infant cues must recruit enough motivation for an adult to approach and continue care despite competing demands. **Motivational value** refers to the capacity of a stimulus or outcome to guide effort and choice. Preoptic pathways interact with reward-related regions so that locating and handling offspring can become a prioritized activity.

This motivational contribution is separable from basic movement ability. An adult may be capable of carrying an object but not approach a pup, or approach repeatedly without organizing suitable handling. Assessing several behavioral components distinguishes a deficit in incentive, sensory recognition, or action sequencing from a general inability to move.

Care also involves repeated choices over time: remaining in the nest, retrieving a displaced pup, feeding, and responding to a disturbance. Hormonal state and prior experience alter these choices through particular circuits. The outcome is a changing allocation of behavior to offspring needs rather than a single uninterrupted motor program called parenting.



Carlson and Birkett (2022), Fig. 10.24(b), p. 334.



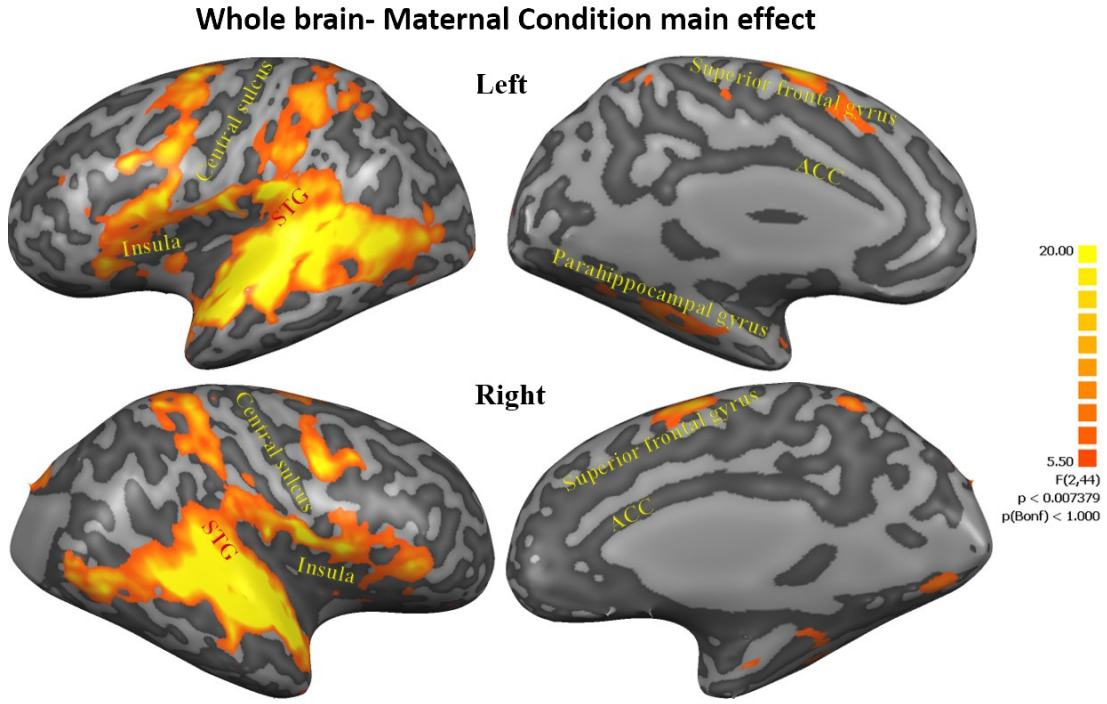
Carcea et al. (2021), Video 5.

Maternal networks respond to social interaction context

A human fMRI study presented mothers with naturalistic mother–infant interactions recorded at home. Conditions included socially synchronized play and situations in which the adult was unavailable or unresponsive. The researchers asked how the caregiving network responded to different interaction patterns rather than contrasting infants with every unrelated visual stimulus.

Socially synchronized interactions produced stronger responses across regions including amygdala, insula, anterior cingulate cortex, and reward-related midbrain tissue. **The fMRI signal** reflects blood-oxygen changes associated with local activity, not a direct count of neuronal spikes or a measurement of one neurotransmitter released at a synapse.

The videos preserved features of real social exchanges, including timing and reciprocity. Comparing conditions showed that maternal neural responses depended on the interaction being viewed. The result did not imply that one region alone represented maternal affection or that the measured response supplied a complete assessment of caregiving quality.



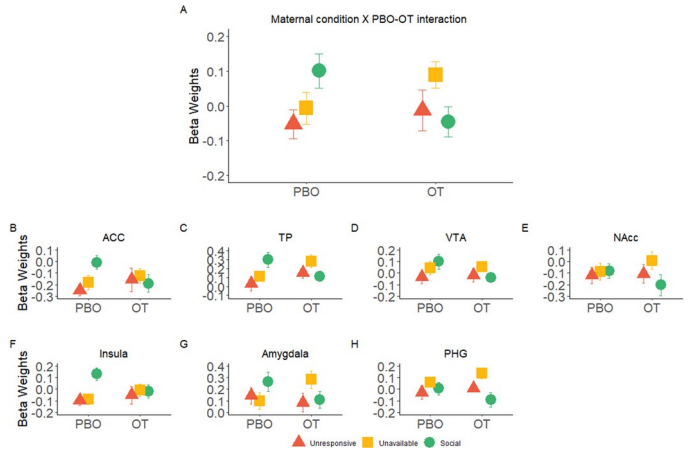
Shimon-Raz et al. (2021), Fig. 2.

Oxytocin effects in humans depend on social context

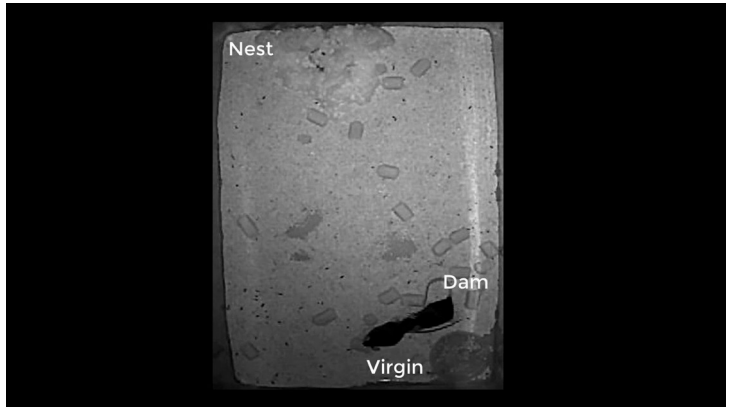
The maternal fMRI study used a within-person comparison of intranasal oxytocin and placebo. **A crossover design** exposes the same participant to different conditions at separate sessions, helping compare treatment responses against that person's own baseline. The investigators examined how administration altered responses to the different mother–infant interaction contexts.

Oxytocin affected responses particularly in the social condition and reduced some differences between social and nonsocial interaction conditions. The direction of change was therefore not a uniform increase in every caregiving region. Its effect depended on the presented social context and the neural measure being analyzed.

Temporal consistency across sessions also differed among interaction conditions and regions. The study measured responses to recorded exchanges, not the participant's ongoing behavior with an infant during the scan. An administration effect on a blood-oxygen signal should consequently be distinguished from demonstrated improvement in daily caregiving or a direct measurement of endogenous peptide release.



Shimon-Raz et al. (2021), Fig. 4.



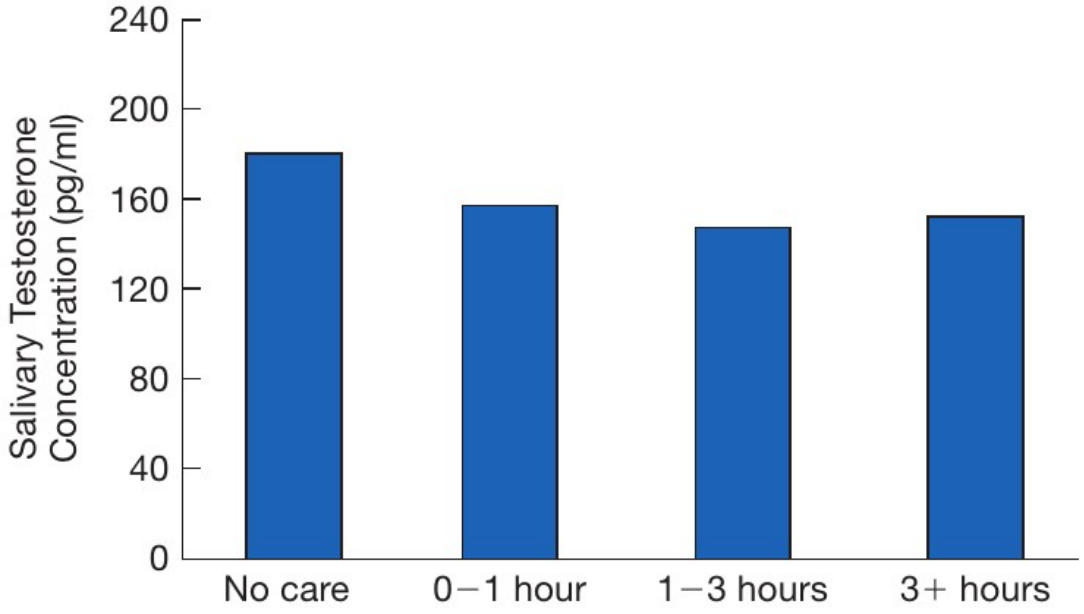
Carcea et al. (2021), Video 6.

Paternal care involves flexible hormone and neural states

Fathers provide care in humans and in several animal species, but its expression varies with ecology and social organization. **Paternal behavior** includes the same kinds of offspring-directed actions that can be measured in other caregivers, such as contact, protection, grooming, and retrieval. Pregnancy is therefore not a necessary experience for every caregiving circuit to function.

Hormonal changes associated with fatherhood can include altered testosterone, prolactin, and oxytocin measures, depending on species and conditions. Such changes may reflect both preparation for care and experience with offspring. An association between a hormone concentration and time spent caring does not by itself establish which process caused the other.

Preoptic and motivational circuits participate in male caregiving as well as female caregiving. Prior mating, cohabitation, infant exposure, and sensory processing can change the adult's response to a pup. Studying the transition into care is therefore different from assuming that a fixed sex category permanently determines whether the animal will approach or attack.



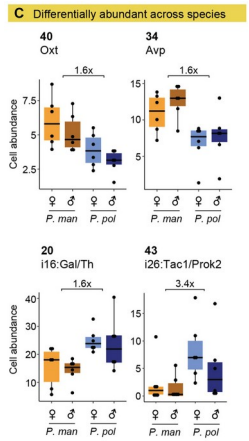
Carlson and Birkett (2022), Fig. 10.25, p. 335.

Related mouse species differ in preoptic cell populations

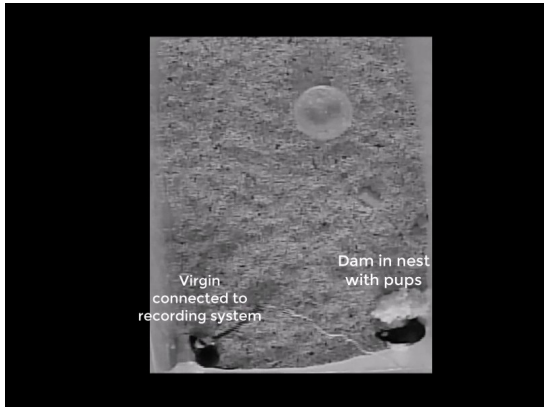
A comparative study examined the preoptic area of two related deer-mouse species with different mating and caregiving patterns. The researchers used **single-nucleus RNA sequencing**, which profiles RNA from individual cell nuclei. This allowed them to compare molecularly defined cell populations across species and sexes while retaining a common anatomical focus.

Several cell types differed in abundance across the species. Some differences occurred in populations whose known functions made them candidates for contributing to caregiving variation. The comparison identified possible cellular substrates of social-system differences rather than proving that every abundance difference caused the observed behavior.

The species also differed in the extent of sex-biased expression and caregiving. A shared anatomical region can therefore contain different proportions or molecular states of neuronal populations across closely related animals. Similar regional names do not guarantee identical cellular organization or identical behavioral effects when the entire area is manipulated.



Chen et al. (2025), Fig. 2.



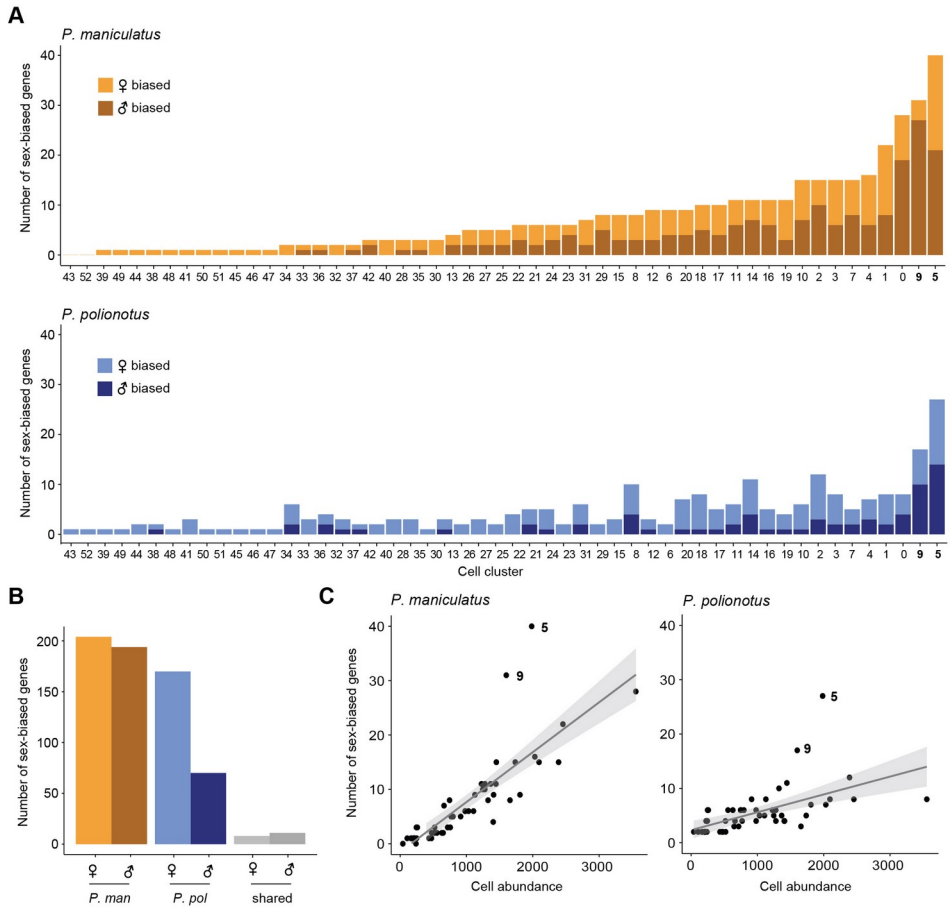
Carcea et al. (2021), Video 7.

Behavioral evolution changes cells within a shared region

The deer-mouse atlas compared gene expression within matched preoptic cell types as well as their relative abundance. These are different kinds of change. A population can retain a similar number of cells while changing the receptors, peptides, or signaling machinery those cells produce, potentially altering their responses to the same social input.

The more socially monogamous species showed reduced sex-biased gene expression in the studied region alongside reduced sex differences in caregiving. These observations motivated hypotheses about circuit evolution without establishing a simple one-gene explanation. Expression profiles describe molecular state, and experimental manipulation is needed to test the contribution of a candidate difference.

The atlas makes those later tests more specific by identifying cell types and molecular markers that can be targeted. It also prevents averaging together populations with opposing responses. A species-level behavioral difference may involve changes in selected cells within a shared circuit rather than the appearance of an entirely new brain structure.



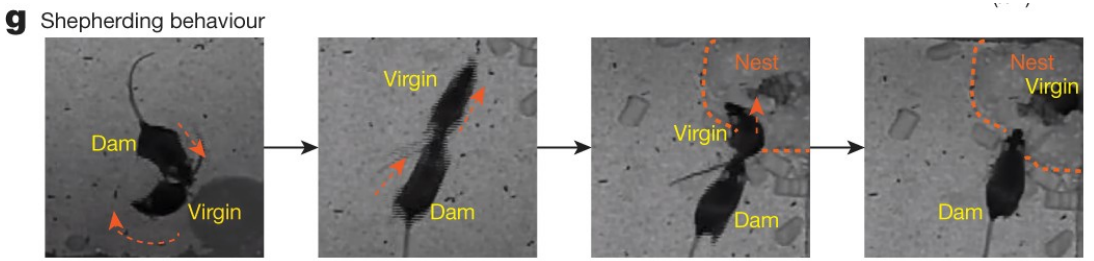
Chen et al. (2025), Fig. 5.

Experienced mothers can recruit other adults into care

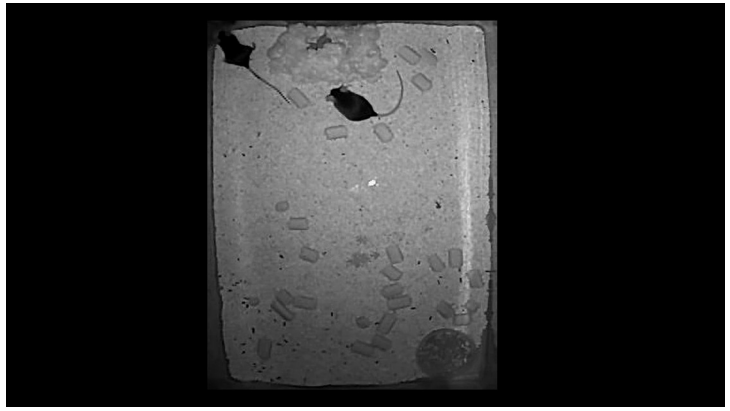
A mouse study continuously monitored inexperienced females housed with an experienced mother and pups. The recordings showed that mothers sometimes **shepherded** the inexperienced adult back toward the nest. These interactions altered the caregiver's exposure to infant cues rather than leaving experience entirely dependent on spontaneous exploration by the inexperienced animal.

Adults housed with a mother and pups acquired reliable retrieval more readily than adults experiencing pups without the same maternal partner. The researchers recorded behavior across days, allowing changes in nest participation and retrieval to be related to repeated social encounters. Care was acquired through a particular experience rather than appearing immediately in every adult.

The original footage preserves sequences of maternal retrieval, nest entry, and interaction between the two adults. These events provide different learning opportunities, including observation of an effective action and direct exposure to infants. The experimental question was how those encounters recruited neural plasticity in the initially inexperienced caregiver.



Carcea et al. (2021), Fig. 1.



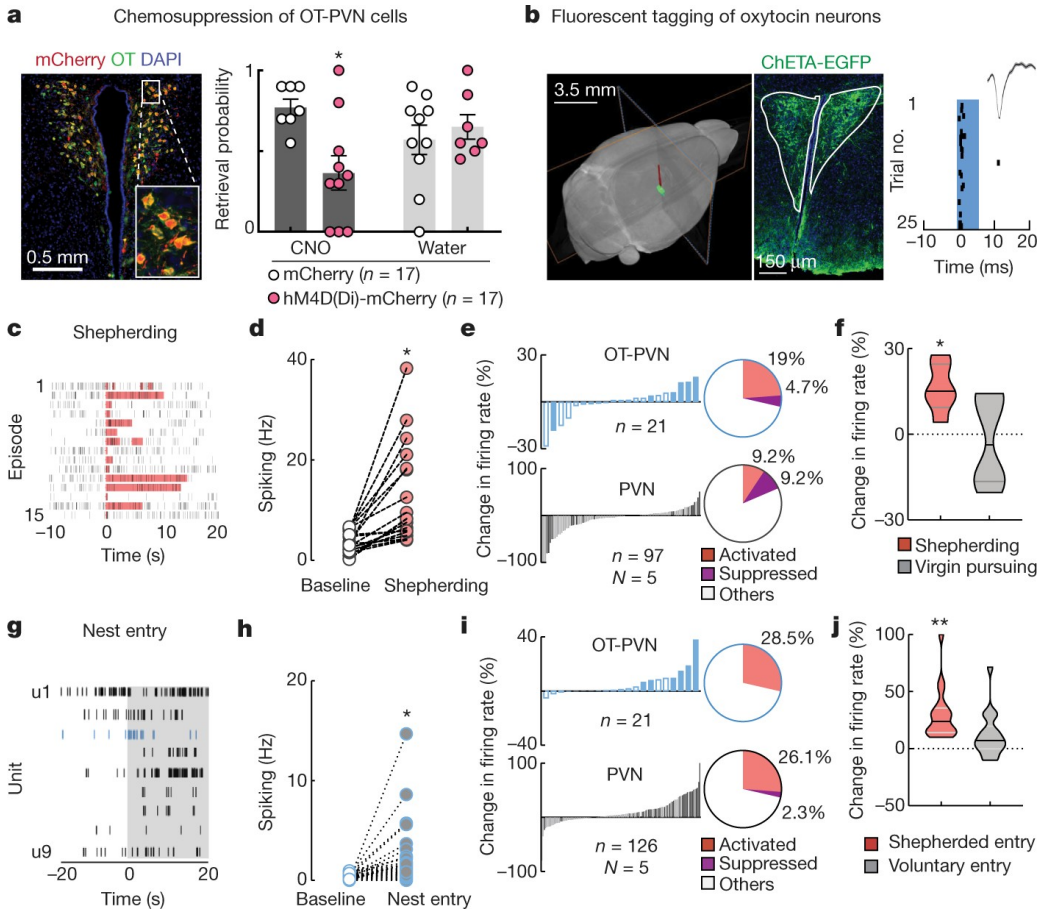
Carcea et al. (2021), Video 8.

Social encounters activate oxytocin neurons during learning

The social-transmission study recorded neurons in the inexperienced female's hypothalamic PVN while behavior unfolded in the home cage. Some cells were identified as oxytocin neurons through selective labeling. This combination related the activity of a defined population to specific interactions rather than inferring its activation only from a hormone sample collected later.

PVN and oxytocin-neuron activity increased around particular maternal interactions, including shepherding and nest entry. Disrupting the oxytocin system during the learning period impaired acquisition of retrieval. The peptide system therefore contributed to the transition into caregiving, while its role after learning could be tested separately.

The strongest neural response did not require every infant sound to activate every recorded PVN neuron directly. Interaction with the experienced mother supplied another route for recruiting the hormone-sensitive learning process. Adult social behavior could consequently prepare a sensory circuit to respond more effectively to the needs of a younger animal.



Carcea et al. (2021), Fig. 2.

Repeated care changes what infant cues can recruit

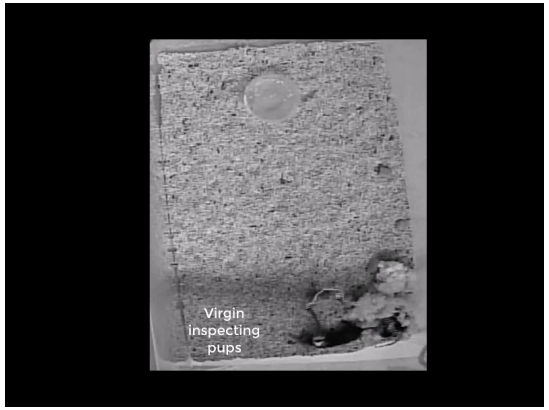
Alloparenting is care provided by an individual other than the biological parent. Repeated infant exposure can help establish such behavior by changing sensory recognition, motivation, and action organization. The adult learns which cues predict a displaced or distressed infant and which actions return it to a safer condition.

Learning can also change the balance between initial avoidance and sustained approach. A cue that once prompted investigation or withdrawal can later recruit retrieval and nest attendance. The neural transition is therefore a change in how existing sensory information influences behavior rather than a requirement for a new infant signal to appear.

Experience operates within a particular social and physiological setting. Learning alongside a skilled caregiver, handling pups directly, and hearing calls from a nest provide different information. Comparing these conditions helps identify which experiences make care possible rather than attributing every successful response to a single circulating hormone.



Carlson and Birkett (2022), Fig. 10.21, p. 330.



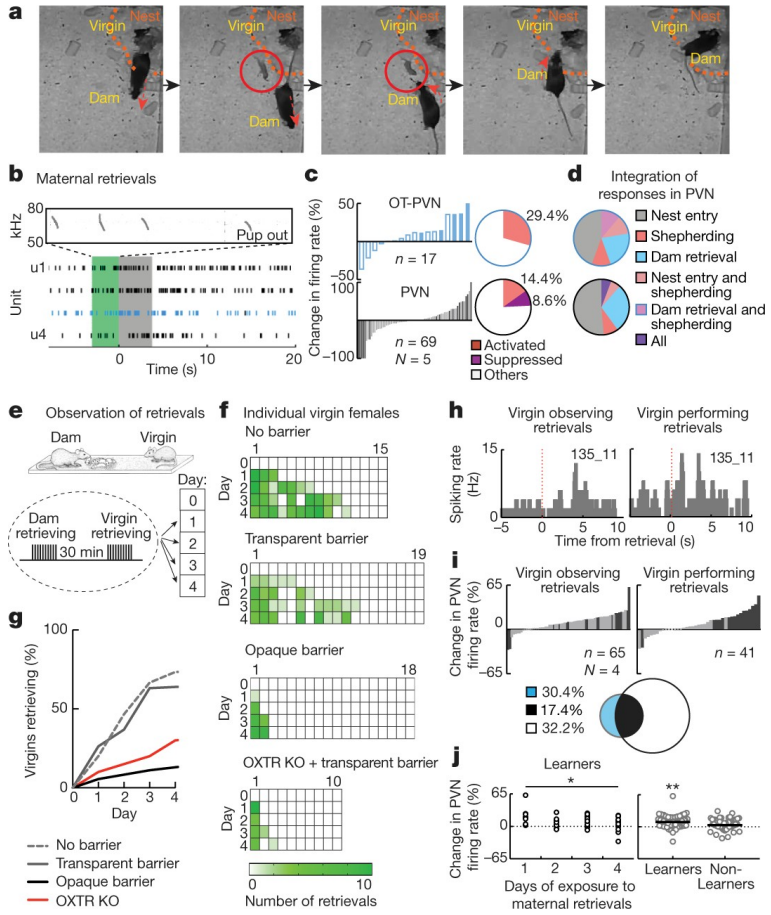
Carcea et al. (2021), Video 9.

Observing retrieval can promote a new caregiver's response

The mouse social-transmission experiment separated inexperienced females from a mother and pups using barriers. A **transparent barrier** preserved visual access while restricting direct interaction; an opaque barrier removed that view. The comparison tested whether observing maternal behavior contributed beyond merely spending time in the same general experimental environment.

Females that could observe maternal retrieval acquired retrieval more effectively than those behind the opaque barrier under the tested conditions. The experiment therefore identified visual access as a contributor to learning. Observation did not mean that the adult instantly copied every behavior, because acquisition developed across repeated opportunities and testing sessions.

The investigators also examined pathways carrying visual information toward oxytocin-related circuitry. Observing an action could recruit a modulatory system that changes later responsiveness to infant cues. This route links social information to learning without requiring the inexperienced adult to undergo pregnancy before the caregiving response can develop.



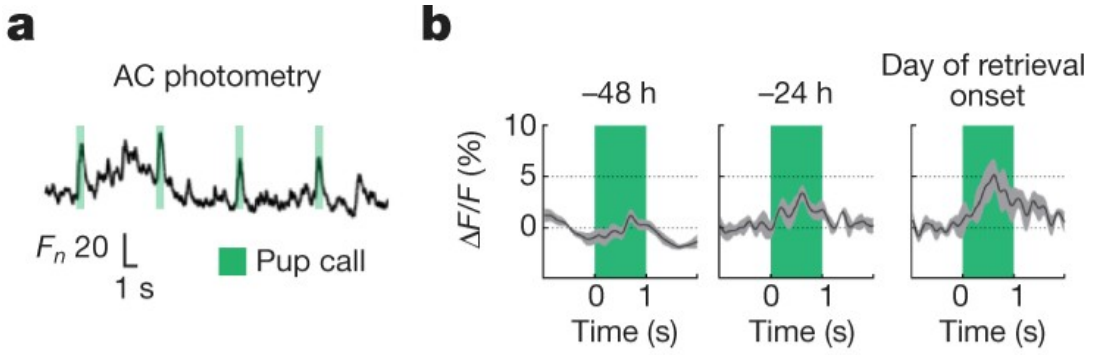
Carcea et al. (2021), Fig. 3.

Care learning changes responses to infant calls

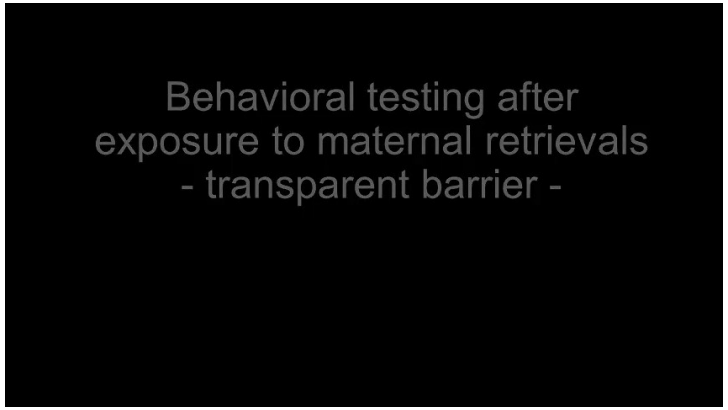
The social-transmission study followed responses in the inexperienced female's left auditory cortex as caregiving emerged. Neural responses to pup calls changed during the acquisition period. These measurements related experience-dependent sensory processing to the appearance of retrieval rather than treating the cortex as a fixed detector whose response could never change in adulthood.

Oxytocin-related activity and sensory plasticity occurred within a broader sequence of interactions with the mother and pups. The visual encounter could help recruit the modulatory signal, while later auditory processing made infant calls more effective. Different sensory channels therefore contributed at different stages of establishing the caregiving response.

Perturbing the relevant signaling during learning altered the outcome, while established behavior did not depend identically on every initial learning condition. The experiment distinguished acquiring a new response from maintaining it. Adult caregiving can consequently emerge through regulated plasticity in an existing social and sensory network.



Carcea et al. (2021), Fig. 4.



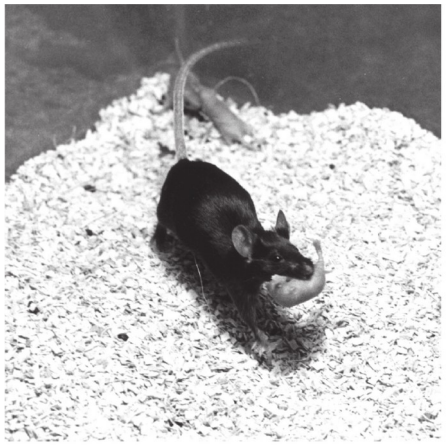
Carcea et al. (2021), Video 11.

Human caregiving uses biology and experience

Animal experiments identify receptors, neurons, and pathways that can alter particular caregiving actions. Human parenting additionally involves language, planning, relationships, resources, and shared care across adults. Translating a circuit mechanism therefore requires matching the measured behavior and population rather than treating a mouse retrieval latency as a complete model of human parenting.

Endocrine transitions can facilitate responsiveness, while repeated caregiving experience can modify perception and motivation in adults with different reproductive histories. Biological and non-biological caregivers can both develop reliable responses to infant needs. The existence of several routes into care is consistent with a nervous system capable of learning across adulthood.

A useful mechanistic explanation identifies what a signal changes: a sensory response, a motivational bias, a competing action, or the plasticity needed for learning. Those distinctions preserve the biology without turning one hormone or brain region into an explanation for every individual difference in a family's caregiving practices.



Carlson and Birkett (2022), Fig. 10.22, p. 331.



Carcea et al. (2021), Video 12.

Key takeaways

Development has several distinct biological targets. Chromosomes, gonads, hormone synthesis, local conversion, and receptors contribute through separable steps; no single trait specifies every later characteristic.

Hormone timing changes what a circuit can do. Organizational and activational effects interact; pulsatile and surge output depend on distinct populations and feedback mechanisms.

Social actions require sensory information and internal state. Chemosensory, acoustic, and contact cues reach reproductive circuits whose responses depend on hormones, experience, and cell-specific connectivity.

Human identity and attraction need their own measurements. Sex traits, gender identity, sexual orientation, desire, and behavior describe different dimensions; animal assays or group anatomy do not diagnose an individual.

Bonding and parenting involve distributed systems. Recognition, reward, approach, competing avoidance, and coordinated action contribute through different pathways rather than one universal social-hormone switch.

Experience can establish caregiving through adult plasticity. Infant exposure and interaction with experienced caregivers can recruit oxytocin-dependent sensory learning; acquisition and maintenance need not have identical requirements.